Microsoft Certified: Fabric Data Engineer Associate

124 Real Exam Question and Answers

DP-700

Microsoft CERTIFIED FABRIC DATA ENGINEER ASSOCIATE

ShapingPixel

PDF

Case Study 1:

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Overview:

Litware, Inc. is a publishing company that has an online bookstore and several retail bookstores worldwide. Litware also manages an online advertising business for the authors it represents.

Existing Environment. Fabric Environment
Litware has a Fabric workspace named Workspace1. High concurrency is enabled for Workspace1.
The company has a data engineering team that uses Python for data processing.

Existing Environment. Data Processing
The retail bookstores send sales data at the end of each business day, while the online bookstore constantly provides logs and sales data to a central enterprise resource planning (ERP) system.
Litware implements a medallion architecture by using the following three layers: bronze, silver, and gold.
The sales data is ingested from the ERP system as Parquet files that land in the Files folder in a lakehouse.
Notebooks are used to transform the files in a Delta table for the bronze and silver layers. The gold layer is in a warehouse that has V-Order disabled.
Litware has image files of book covers in Azure Blob Storage. The files are loaded into the Files folder.

Existing Environment. Sales Data
Month-end sales data is processed on the first calendar day of each month. Data that is older than one month never changes.
In the source system, the sales data refreshes every six hours starting at midnight each day.
The sales data is captured in a Dataflow Gen2 dataflow.
When the dataflow runs, new and historical data is captured. The dataflow captures the following fields of the source:
Sales Date
Author
Price
Units
SKU
A table named AuthorSales stores the sales data that relates to each author. The table contains a column named AuthorEmail. Authors authenticate to a guest Fabric tenant by using their email address.
Existing Environment. Security Groups
Litware has the following security groups:
Sales
Fabric Admins
Streaming Admins
Existing Environment. Performance Issues
Business users perform ad-hoc queries against the warehouse. The business users indicate that reports against the warehouse sometimes run for two hours and fail to load as expected. Upon further investigation, the data engineering team receives the following error message when the reports fail to load: "The SQL query failed while running."

The data engineering team wants to debug the issue and find queries that cause more than one failure.
When the authors have new book releases, there is often an increase in sales activity. This increase slows the data ingestion process.
The company's sales team reports that during the last month, the sales data has NOT been up-to-date when they arrive at work in the morning.

Requirements. Planned Changes
Litware recently signed a contract to receive book reviews. The provider of the reviews exposes the data in Amazon Simple Storage Service (Amazon S3) buckets.
Litware plans to manage Search Engine Optimization (SEO) for the authors. The SEO data will be streamed from a REST API.

Requirements. Version Control
Litware plans to implement a version control solution in Fabric that will use GitHub integration and follow the principle of least privilege.

Requirements. Governance Requirements
To control data platform costs, the data platform must use only Fabric services and items. Additional Azure resources must NOT be provisioned.

Requirements. Data Requirements
Litware identifies the following data requirements:
Process the SEO data in near-real-time (NRT).
Make the book reviews available in the lakehouse without making a copy of the data.
When a new book cover image arrives in the Files folder, process the image as soon as possible.

Case Study 2:

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Overview:
- Company Overview
Contoso, Ltd. is an online retail company that wants to modernize its analytics platform by moving to Fabric. The company plans to begin using Fabric for marketing analytics.

Overview:
- IT Structure
The company's IT department has a team of data analysts and a team of data engineers that use analytics systems.
The data engineers perform the ingestion, transformation, and loading of data. They prefer to use Python or SQL to transform the data.
The data analysts query data and create semantic models and reports. They are qualified to write queries in Power Query and T-SQL.

Existing Environment. Fabric
Contoso has an F64 capacity named Cap1. All Fabric users are allowed to create items.
Contoso has two workspaces named WorkspaceA and WorkspaceB that currently use Pro license mode.

Existing Environment. Source Systems
Contoso has a point of sale (POS) system named POS1 that uses an instance of SQL Server on Azure Virtual Machines in the same Microsoft Entra tenant as Fabric. The host virtual machine is on a private virtual network that has public access blocked. POS1 contains all the sales transactions that were processed on the company's website.
The company has a software as a service (SaaS) online marketing app named MAR1. MAR1 has seven entities. The entities contain data that relates to email open rates and interaction rates, as well as website interactions. The data can be exported from MAR1 by calling REST APIs. Each entity has a different endpoint.
Contoso has been using MAR1 for one year. Data from prior years is stored in Parquet files in an Amazon Simple Storage Service (Amazon S3) bucket. There are 12 files that range in size from 300 MB to 900 MB and relate to email interactions.

Existing Environment. Product Data
POS1 contains a product list and related data. The data comes from the following three tables:
Products
ProductCategories
ProductSubcategories
In the data, products are related to product subcategories, and subcategories are related to product categories.

Existing Environment. Azure
Contoso has a Microsoft Entra tenant that has the following mail-enabled security groups:
DataAnalysts: Contains the data analysts
DataEngineers: Contains the data engineers
Contoso has an Azure subscription.
The company has an existing Azure DevOps organization and creates a new project for repositories that relate to Fabric.

Existing Environment. User Problems
The VP of marketing at Contoso requires analysis on the effectiveness of different types of email content. It typically takes a week to manually compile and analyze the data. Contoso wants to reduce the time to less than one day by using Fabric.
The data engineering team has successfully exported data from MAR1. The team experiences transient connectivity errors, which causes the data exports to fail.

Requirements. Planned Changes
Contoso plans to create the following two lakehouses:
Lakehouse1: Will store both raw and cleansed data from the sources
Lakehouse2: Will serve data in a dimensional model to users for analytical queries
Additional items will be added to facilitate data ingestion and transformation.
Contoso plans to use Azure Repos for source control in Fabric.

Requirements. Technical Requirements
The new lakehouses must follow a medallion architecture by using the following three layers: bronze, silver, and gold. There will be extensive data cleansing required to populate the MAR1 data in the silver layer, including deduplication, the handling of missing values, and the standardizing of capitalization.
Each layer must be fully populated before moving on to the next layer. If any step in populating the lakehouses fails, an email must be sent to the data engineers.
Data imports must run simultaneously, when possible.
The use of email data from the Amazon S3 bucket must meet the following requirements:
Minimize egress costs associated with cross-cloud data access.
Prevent saving a copy of the raw data in the lakehouses.
Items that relate to data ingestion must meet the following requirements:
The items must be source controlled alongside other workspace items.
Ingested data must land in the bronze layer of Lakehouse1 in the Delta format.
No changes other than changes to the file formats must be implemented before the data lands in the bronze layer.
Development effort must be minimized and a built-in connection must be used to import the source data.
In the event of a connectivity error, the ingestion processes must attempt the connection again.
Lakehouses, data pipelines, and notebooks must be stored in WorkspaceA. Semantic models, reports, and dataflows must be stored in WorkspaceB.
Once a week, old files that are no longer referenced by a Delta table log must be removed.

Requirements. Data Transformation
In the POS1 product data, ProductID values are unique. The product dimension in the gold layer must include only active products from product list. Active products are identified by an IsActive value of 1. Some product categories and subcategories are NOT assigned to any product. They are NOT analytically relevant and must be omitted from the product dimension in the gold layer.

Requirements. Data Security
Security in Fabric must meet the following requirements:
The data engineers must have read and write access to all the lakehouses, including the underlying files.
The data analysts must only have read access to the Delta tables in the gold layer.
The data analysts must NOT have access to the data in the bronze and silver layers.
The data engineers must be able to commit changes to source control in WorkspaceA.

Case Study 3:

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Existing Environment. Data Processing
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A table named AuthorSales stores the sales data that relates to each author. The table contains a column named AuthorEmail. Authors authenticate to a guest Fabric tenant by using their email address.

Existing Environment. Security Groups
Litware has the following security groups:
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Existing Environment. Performance Issues
Business users perform ad-hoc queries against the warehouse. The business users indicate that reports against the warehouse sometimes run for two hours and fail to load as expected. Upon further investigation, the data engineering team receives the following error message when the reports fail to load: "The SQL query failed while running."
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Requirements. Planned Changes
Litware recently signed a contract to receive book reviews. The provider of the reviews exposes the data in Amazon Simple Storage Service (Amazon S3) buckets.
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Requirements. Version Control
Litware plans to implement a version control solution in Fabric that will use GitHub integration and follow the principle of least privilege.

Requirements. Governance Requirements
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Requirements. Data Requirements
Litware identifies the following data requirements:
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Make the book reviews available in the lakehouse without making a copy of the data.
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The data engineers perform the ingestion, transformation, and loading of data. They prefer to use Python or SQL to transform the data.
The data analysts query data and create semantic models and reports. They are qualified to write queries in Power Query and T-SQL.

Existing Environment. Fabric
Contoso has an F64 capacity named Cap1. All Fabric users are allowed to create items.
Contoso has two workspaces named WorkspaceA and WorkspaceB that currently use Pro license mode.

Existing Environment. Source Systems

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Contoso has been using MAR1 for one year. Data from prior years is stored in Parquet files in an Amazon Simple Storage Service (Amazon S3) bucket. There are 12 files that range in size from 300 MB to 900 MB and relate to email interactions.

Existing Environment. Product Data

POS1 contains a product list and related data. The data comes from the following three tables:

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Existing Environment. Azure
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Existing Environment. User Problems
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The data engineering team has successfully exported data from MAR1. The team experiences transient connectivity errors, which causes the data exports to fail.

Requirements. Planned Changes
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Lakehouse1: Will store both raw and cleansed data from the sources
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Requirements. Technical Requirements
The new lakehouses must follow a medallion architecture by using the following three layers: bronze, silver, and gold. There will be extensive data cleansing required to populate the MAR1 data in the silver layer, including deduplication, the handling of missing values, and the standardizing of capitalization.
Each layer must be fully populated before moving on to the next layer. If any step in populating the lakehouses fails, an email must be sent to the data engineers.
Data imports must run simultaneously, when possible.
The use of email data from the Amazon S3 bucket must meet the following requirements:
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Prevent saving a copy of the raw data in the lakehouses.
Items that relate to data ingestion must meet the following requirements:
The items must be source controlled alongside other workspace items.
Ingested data must land in the bronze layer of Lakehouse1 in the Delta format.
No changes other than changes to the file formats must be implemented before the data lands in the bronze layer.
Development effort must be minimized and a built-in connection must be used to import the source data.
In the event of a connectivity error, the ingestion processes must attempt the connection again.
Lakehouses, data pipelines, and notebooks must be stored in WorkspaceA. Semantic models, reports, and dataflows must be stored in WorkspaceB.
Once a week, old files that are no longer referenced by a Delta table log must be removed.

Requirements. Data Transformation

In the POS1 product data, ProductID values are unique. The product dimension in the gold layer must include only active products from product list. Active products are identified by an IsActive value of 1. Some product categories and subcategories are NOT assigned to any product. They are NOT analytically relevant and must be omitted from the product dimension in the gold layer.

Requirements. Data Security

Security in Fabric must meet the following requirements:

The data engineers must have read and write access to all the lakehouses, including the underlying files.

The data analysts must only have read access to the Delta tables in the gold layer.

The data analysts must NOT have access to the data in the bronze and silver layers.

The data engineers must be able to commit changes to source control in WorkspaceA.

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Existing Environment. Data Processing
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Existing Environment. Sales Data
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Existing Environment. Performance Issues

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Requirements. Data Requirements
Litware identifies the following data requirements:
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The data analysts must only have read access to the Delta tables in the gold layer.
The data analysts must NOT have access to the data in the bronze and silver layers.

The data engineers must be able to commit changes to source control in WorkspaceA.

1) You need to ensure that the data analysts can access the gold layer lakehouse. (Case Study 2)

What should you do?

A. Add the DataAnalyst group to the Viewer role for WorkspaceA.
B. Share the lakehouse with the DataAnalysts group and grant the Build reports on the default semantic model permission.
C. Share the lakehouse with the DataAnalysts group and grant the Read all SQL Endpoint data permission.
D. Share the lakehouse with the DataAnalysts group and grant the Read all Apache Spark permission.

1) You need to ensure that the data analysts can access the gold layer lakehouse. (Case Study 2)

What should you do?

A. Add the DataAnalyst group to the Viewer role for WorkspaceA.
B. Share the lakehouse with the DataAnalysts group and grant the Build reports on the default semantic model permission.
C. Share the lakehouse with the DataAnalysts group and grant the Read all SQL Endpoint data permission.
D. Share the lakehouse with the DataAnalysts group and grant the Read all Apache Spark permission.

Correct Answer: *C*

2) You have a Fabric workspace.
You have semi-structured data.
You need to read the data by using T-SQL, KQL, and Apache Spark. The data will only be written by using Spark.
What should you use to store the data?

A. a lakehouse
B. an eventhouse
C. a datamart
D. a warehouse

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You need to read the data by using T-SQL, KQL, and Apache Spark. The data will only be written by using Spark.
What should you use to store the data?

A. a lakehouse
B. an eventhouse
C. a datamart
D. a warehouse

Correct Answer: *A*

3) You have a Fabric workspace that contains a warehouse named Warehouse1.
You have an on-premises Microsoft SQL Server database named Database1 that is accessed by using an on-premises data gateway.
You need to copy data from Database1 to Warehouse1.
Which item should you use?

A. a Dataflow Gen1 dataflow
B. a data pipeline
C. a KQL queryset
D. a notebook

3) You have a Fabric workspace that contains a warehouse named Warehouse1.
You have an on-premises Microsoft SQL Server database named Database1 that is accessed by using an on-premises data gateway.
You need to copy data from Database1 to Warehouse1.
Which item should you use?

A. a Dataflow Gen1 dataflow
B. a data pipeline
C. a KQL queryset
D. a notebook

Correct Answer: *B*

4) You have a Fabric workspace that contains a warehouse named Warehouse1.
You have an on-premises Microsoft SQL Server database named Database1 that is accessed by using an on-premises data gateway.
You need to copy data from Database1 to Warehouse1.
Which item should you use?

A. an Apache Spark job definition
B. a data pipeline
C. a Dataflow Gen1 dataflow
D. an eventstream

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You have an on-premises Microsoft SQL Server database named Database1 that is accessed by using an on-premises data gateway.
You need to copy data from Database1 to Warehouse1.
Which item should you use?

A. an Apache Spark job definition
B. a data pipeline
C. a Dataflow Gen1 dataflow
D. an eventstream

Correct Answer: *B*

5) You have a Fabric F32 capacity that contains a workspace. The workspace contains a warehouse named DW1 that is modelled by using MD5 hash surrogate keys.
DW1 contains a single fact table that has grown from 200 million rows to 500 million rows during the past year.
You have Microsoft Power BI reports that are based on Direct Lake. The reports show year-over-year values.
Users report that the performance of some of the reports has degraded over time and some visuals show errors.
You need to resolve the performance issues. The solution must meet the following requirements:
Provide the best query performance.
Minimize operational costs.
Which should you do?

A. Change the MD5 hash to SHA256.
B. Increase the capacity.
C. Enable V-Order.
D. Modify the surrogate keys to use a different data type.
E. Create views.

5) You have a Fabric F32 capacity that contains a workspace. The workspace contains a warehouse named DW1 that is modelled by using MD5 hash surrogate keys.
DW1 contains a single fact table that has grown from 200 million rows to 500 million rows during the past year.
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Which should you do?

A. Change the MD5 hash to SHA256.
B. Increase the capacity.
C. Enable V-Order.
D. Modify the surrogate keys to use a different data type.
E. Create views.

Correct Answer: *D*

6) You have a Fabric workspace that contains a warehouse named DW1. DW1 contains the following tables and columns.

Table name	Column name	Description
SalesOrderDetail	ProductID	Contains the product ID of the ordered product
SalesOrderDetail	ModifiedDate	Contains the date of an order
SalesOrderDetail	OrderQty	Contains the order quantity
Product	ProductID	Contains the unique ID of a product
Product	Name	Contains a product name

You need to create an output that presents the summarized values of all the order quantities by year and product. The results must include a summary of the order quantities at the year level for all the products.
How should you complete the code? To answer, select the appropriate options in the answer area.
NOTE: Each correct selection is worth one point.

Answer Area

[dropdown] (SO.ModifiedDate) AS OrderDate

- SELECT CAST
- SELECT CONVERT
- SELECT YEAR

```
    ,P.Name AS ProductName
    ,SUM(SO.OrderQty) AS OrderQty
  FROM [dbo].[SalesOrderDetail] SO
  INNER JOIN [dbo].[Product] P
    ON P.ProductID = SO.ProductID
  GROUP BY
```

[dropdown]

- CUBE(YEAR(SO.ModifiedDate), P.Name)
- GROUPING SETS ((YEAR(SO.ModifiedDate), P.Name), (YEAR(SO.ModifiedDate)))
- ROLLUP(YEAR(SO.ModifiedDate), P.Name)
- YEAR(SO.ModifiedDate), P.Name

```
  ORDER BY OrderDate
```

Answer Area

[dropdown] (SO.ModifiedDate) AS OrderDate

- SELECT CAST
- SELECT CONVERT
- SELECT YEAR

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      ,P.Name AS ProductName
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- ROLLUP(YEAR(SO.ModifiedDate), P.Name)
- YEAR(SO.ModifiedDate), P.Name

```
    ORDER BY OrderDate
```

7) You have a Fabric workspace that contains a lakehouse named Lakehouse1. Data is ingested into Lakehouse1 as one flat table. The table contains the following columns.

Name	Description
TransactionID	Contains a unique ID for each transaction
Date	Contains the date of a transaction
ProductID	Contains a unique ID for each product
ProductColor	Contains a descriptive attribute that describes the color of each product
ProductName	Contains a unique name for each product
SalesAmount	Contains the sales amount of a transaction

You plan to load the data into a dimensional model and implement a star schema. From the original flat table, you create two tables named FactSales and DimProduct. You will track changes in DimProduct.

You need to prepare the data.

Which three columns should you include in the DimProduct table? Each correct answer presents part of the solution.

NOTE: Each correct selection is worth one point.

A. Date
B. ProductName
C. ProductColor
D. TransactionID
E. SalesAmount
F. ProductID

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Name	Description
TransactionID	Contains a unique ID for each transaction
Date	Contains the date of a transaction
ProductID	Contains a unique ID for each product
ProductColor	Contains a descriptive attribute that describes the color of each product
ProductName	Contains a unique name for each product
SalesAmount	Contains the sales amount of a transaction

You plan to load the data into a dimensional model and implement a star schema. From the original flat table, you create two tables named FactSales and DimProduct. You will track changes in DimProduct.
You need to prepare the data.
Which three columns should you include in the DimProduct table? Each correct answer presents part of the solution.
NOTE: Each correct selection is worth one point.

A. Date
B. ProductName
C. ProductColor
D. TransactionID
E. SalesAmount
F. ProductID

Correct Answer: *BCF*

8) You have a Fabric workspace named Workspace1 that contains a notebook named Notebook1.
In Workspace1, you create a new notebook named Notebook2.
You need to ensure that you can attach Notebook2 to the same Apache Spark session as Notebook1.
What should you do?

A. Enable high concurrency for notebooks.
B. Enable dynamic allocation for the Spark pool.
C. Change the runtime version.
D. Increase the number of executors.

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You need to ensure that you can attach Notebook2 to the same Apache Spark session as Notebook1.
What should you do?

A. Enable high concurrency for notebooks.
B. Enable dynamic allocation for the Spark pool.
C. Change the runtime version.
D. Increase the number of executors.

Correct Answer: *A*

9) You have a Fabric workspace named Workspace1 that contains a lakehouse named Lakehouse1. Lakehouse1 contains the following tables:
Orders -
Customer -
Employee -
The Employee table contains Personally Identifiable Information (PII).
A data engineer is building a workflow that requires writing data to the Customer table, however, the user does NOT have the elevated permissions required to view the contents of the Employee table.
You need to ensure that the data engineer can write data to the Customer table without reading data from the Employee table.
Which three actions should you perform? Each correct answer presents part of the solution.
NOTE: Each correct selection is worth one point.

A. Share Lakehouse1 with the data engineer.
B. Assign the data engineer the Contributor role for Workspace2.
C. Assign the data engineer the Viewer role for Workspace2.
D. Assign the data engineer the Contributor role for Workspace1.
E. Migrate the Employee table from Lakehouse1 to Lakehouse2.
F. Create a new workspace named Workspace2 that contains a new lakehouse named Lakehouse2.
G. Assign the data engineer the Viewer role for Workspace1.

9) You have a Fabric workspace named Workspace1 that contains a lakehouse named Lakehouse1. Lakehouse1 contains the following tables:
Orders -
Customer -
Employee -
The Employee table contains Personally Identifiable Information (PII).
A data engineer is building a workflow that requires writing data to the Customer table, however, the user does NOT have the elevated permissions required to view the contents of the Employee table.
You need to ensure that the data engineer can write data to the Customer table without reading data from the Employee table.
Which three actions should you perform? Each correct answer presents part of the solution.
NOTE: Each correct selection is worth one point.

A. Share Lakehouse1 with the data engineer.
B. Assign the data engineer the Contributor role for Workspace2.
C. Assign the data engineer the Viewer role for Workspace2.
D. Assign the data engineer the Contributor role for Workspace1.
E. Migrate the Employee table from Lakehouse1 to Lakehouse2.
F. Create a new workspace named Workspace2 that contains a new lakehouse named Lakehouse2.
G. Assign the data engineer the Viewer role for Workspace1.

Correct Answer: *DEF*

10) You have a Fabric warehouse named DW1. DW1 contains a table that stores sales data and is used by multiple sales representatives.
You plan to implement row-level security (RLS).
You need to ensure that the sales representatives can see only their respective data.
Which warehouse object do you require to implement RLS?

A. STORED PROCEDURE
B. CONSTRAINT
C. SCHEMA
D. FUNCTION

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Which warehouse object do you require to implement RLS?

A. STORED PROCEDURE
B. CONSTRAINT
C. SCHEMA
D. FUNCTION

Correct Answer: *D*

11) HOTSPOT -
You have a Fabric workspace named Workspace1_DEV that contains the following items:
10 reports

Four notebooks -

Three lakehouses -

Two data pipelines -

Two Dataflow Gen1 dataflows -

Three Dataflow Gen2 dataflows -
Five semantic models that each has a scheduled refresh policy
You create a deployment pipeline named Pipeline1 to move items from Workspace1_DEV to a new workspace named Workspace1_TEST.
You deploy all the items from Workspace1_DEV to Workspace1_TEST.
For each of the following statements, select Yes if the statement is true. Otherwise, select No.

NOTE: Each correct selection is worth one point.

Answer Area

Statements	Yes	No
Data from the semantic models will be deployed to the target stage.	○	○
The Dataflow Gen1 dataflows will be deployed to the target stage.	○	○
The scheduled refresh policies will be deployed to the target stage.	○	○

Statements	Yes	No
Data from the semantic models will be deployed to the target stage.	○	○
The Dataflow Gen1 dataflows will be deployed to the target stage.	○	○
The scheduled refresh policies will be deployed to the target stage.	○	○

12) You have a Fabric deployment pipeline that uses three workspaces named Dev, Test, and Prod.
You need to deploy an eventhouse as part of the deployment process.
What should you use to add the eventhouse to the deployment process?

A. GitHub Actions
B. a deployment pipeline
C. an Azure DevOps pipeline

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What should you use to add the eventhouse to the deployment process?

A. GitHub Actions
B. a deployment pipeline
C. an Azure DevOps pipeline

Correct Answer: *B*

13) You have a Fabric workspace named Workspace1 that contains a warehouse named Warehouse1.
You plan to deploy Warehouse1 to a new workspace named Workspace2.
As part of the deployment process, you need to verify whether Warehouse1 contains invalid references. The solution must minimize development effort.
What should you use?

A. a database project
B. a deployment pipeline
C. a Python script
D. a T-SQL script

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What should you use?

A. a database project
B. a deployment pipeline
C. a Python script
D. a T-SQL script

Correct Answer: *B*

14) You have a Fabric workspace that contains a Real-Time Intelligence solution and an eventhouse.
Users report that from OneLake file explorer, they cannot see the data from the eventhouse.
You enable OneLake availability for the eventhouse.
What will be copied to OneLake?

A. only data added to new databases that are added to the eventhouse
B. only the existing data in the eventhouse
C. no data
D. both new data and existing data in the eventhouse
E. only new data added to the eventhouse

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C. no data
D. both new data and existing data in the eventhouse
E. only new data added to the eventhouse

Correct Answer: *E*

15) You have a Fabric workspace named Workspace1.
You plan to integrate Workspace1 with Azure DevOps.
You will use a Fabric deployment pipeline named deployPipeline1 to deploy items from Workspace1 to higher environment workspaces as part of a medallion architecture. You will run deployPipeline1 by using an API call from an Azure DevOps pipeline.
You need to configure API authentication between Azure DevOps and Fabric.
Which type of authentication should you use?

A. service principal
B. Microsoft Entra username and password
C. managed private endpoint
D. workspace identity

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You need to configure API authentication between Azure DevOps and Fabric.
Which type of authentication should you use?

A. service principal
B. Microsoft Entra username and password
C. managed private endpoint
D. workspace identity

Correct Answer: *A*

16) You have a Google Cloud Storage (GCS) container named storage1 that contains the files shown in the following table.

Name	Size
ProductFile.parquet	8 MB
StoreFile.json	500 MB
TripsFile.csv	99 MB

You have a Fabric workspace named Workspace1 that has the cache for shortcuts enabled. Workspace1 contains a lakehouse named Lakehouse1. Lakehouse1 has the shortcuts shown in the following table.

Name	Source	Last accessed
Products	ProductFile	12 hours ago
Stores	StoreFile	4 hours ago
Trips	TripsFile	48 hours ago

You need to read data from all the shortcuts.
Which shortcuts will retrieve data from the cache?

A. Stores only
B. Products only
C. Stores and Products only
D. Products, Stores, and Trips
E. Trips only
F. Products and Trips only

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Which shortcuts will retrieve data from the cache?

A. Stores only
B. Products only
C. Stores and Products only
D. Products, Stores, and Trips
E. Trips only
F. Products and Trips only

Correct Answer: *C*

17) You have two Fabric workspaces named Workspace1 and Workspace2.

You have a Fabric deployment pipeline named deployPipeline1 that deploys items from Workspace1 to Workspace2. DeployPipeline1 contains all the items in Workspace1.

You recently modified the items in Workspaces1.

The workspaces currently contain the items shown in the following table.

Workspace	Items
Workspace1	Model1 Notebook1 Report1 Lakehouse1 Pipeline1
Workspace2	Model1 Notebook2 Report1 Lakehouse2

Items in Workspace1 that have the same name as items in Workspace2 are currently paired.

You need to ensure that the items in Workspace1 overwrite the corresponding items in Workspace2. The solution must minimize effort.

What should you do?

A. Delete all the items in Workspace2, and then run deployPipeline1.
B. Rename each item in Workspace2 to have the same name as the items in Workspace1.
C. Back up the items in Workspace2, and then run deployPipeline1.
D. Run deployPipeline1 without modifying the items in Workspace2.

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You have a Fabric deployment pipeline named deployPipeline1 that deploys items from Workspace1 to Workspace2. DeployPipeline1 contains all the items in Workspace1.

You recently modified the items in Workspaces1.

The workspaces currently contain the items shown in the following table.

Workspace	Items
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Items in Workspace1 that have the same name as items in Workspace2 are currently paired.

You need to ensure that the items in Workspace1 overwrite the corresponding items in Workspace2. The solution must minimize effort.

What should you do?

A. Delete all the items in Workspace2, and then run deployPipeline1.
B. Rename each item in Workspace2 to have the same name as the items in Workspace1.
C. Back up the items in Workspace2, and then run deployPipeline1.
D. Run deployPipeline1 without modifying the items in Workspace2.

Answer : **D**

When running a deployment pipeline in Fabric, if the items in Workspace1 are paired with the corresponding items in Workspace2 (based on the same name), the deployment pipeline will automatically overwrite the existing items in Workspace2 with the modified items from Workspace1. There's no need to delete, rename, or back up items manually unless you need to keep versions. By simply running deployPipeline1, the pipeline will handle overwriting the existing items in Workspace2 based on the pairing, ensuring the latest version of the items is deployed with minimal effort.

18) You have a Fabric capacity that contains a workspace named Workspace1. Workspace1 contains a lakehouse named Lakehouse1, a data pipeline, a notebook, and several Microsoft Power BI reports.

A user named User1 wants to use SQL to analyze the data in Lakehouse1.

You need to configure access for User1. The solution must meet the following requirements:

What should you do?

A. Share Lakehouse1 with User1 directly and select Read all SQL endpoint data.
B. Assign User1 the Viewer role for Workspace1. Share Lakehouse1 with User1 and select Read all SQL endpoint data.
C. Share Lakehouse1 with User1 directly and select Build reports on the default semantic model.
D. Assign User1 the Member role for Workspace1. Share Lakehouse1 with User1 and select Read all SQL endpoint data.

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You need to configure access for User1. The solution must meet the following requirements:

What should you do?

A. Share Lakehouse1 with User1 directly and select Read all SQL endpoint data.
B. Assign User1 the Viewer role for Workspace1. Share Lakehouse1 with User1 and select Read all SQL endpoint data.
C. Share Lakehouse1 with User1 directly and select Build reports on the default semantic model.
D. Assign User1 the Member role for Workspace1. Share Lakehouse1 with User1 and select Read all SQL endpoint data.

Answer : **B**

To meet the specified requirements for User1, the solution must ensure:

Read access to the table data in Lakehouse1: User1 needs permission to access the data within Lakehouse1. By sharing Lakehouse1 with User1 and selecting the Read all SQL endpoint data option, User1 will be able to query the data via SQL endpoints.

Prevent Apache Spark usage: By sharing the lakehouse directly and selecting the SQL endpoint data option, you specifically enable SQL-based access to the data, preventing User1 from using Apache Spark to query the data.

Prevent access to other items in Workspace1: Assigning User1 the Viewer role for Workspace1 ensures that User1 can only view the shared items (in this case, Lakehouse1), without accessing other resources such as notebooks, pipelines, or Power BI reports within Workspace1.

This approach provides the appropriate level of access while restricting User1 to only the required resources and preventing access to other workspace assets.

19) You have a Fabric workspace named Workspace1 that contains a warehouse named DW1 and a data pipeline named Pipeline1.

You plan to add a user named User3 to Workspace1.

You need to ensure that User3 can perform the following actions:

View all the items in Workspace1.

Update the tables in DW1.

The solution must follow the principle of least privilege.

You already assigned the appropriate object-level permissions to DW1.

Which workspace role should you assign to User3?

A. Admin

B. Member

C. Viewer

D. Contributor

19) You have a Fabric workspace named Workspace1 that contains a warehouse named DW1 and a data pipeline named Pipeline1.

You plan to add a user named User3 to Workspace1.

You need to ensure that User3 can perform the following actions:

View all the items in Workspace1.

Update the tables in DW1.

The solution must follow the principle of least privilege.

You already assigned the appropriate object-level permissions to DW1.

Which workspace role should you assign to User3?

A. Admin

B. Member

C. Viewer

D. Contributor

Answer : **D**

To ensure User3 can view all items in Workspace1 and update the tables in DW1, the most appropriate workspace role to assign is the Contributor role. This role allows User3 to:

View all items in Workspace1: The Contributor role provides the ability to view all objects within the workspace, such as data pipelines, warehouses, and other resources.

Update the tables in DW1: The Contributor role allows User3 to modify or update resources within the workspace, including the tables in DW1, assuming that appropriate object-level permissions are set for the warehouse.

This role adheres to the principle of least privilege, as it provides the necessary permissions without granting broader administrative rights.

20) Your company has a sales department that uses two Fabric workspaces named Workspace1 and Workspace2.

The company decides to implement a domain strategy to organize the workspaces.

You need to ensure that a user can perform the following tasks:

Create a new domain for the sales department.

Create two subdomains: one for the east region and one for the west region.

Assign Workspace1 to the east region subdomain.

Assign Workspace2 to the west region subdomain.

The solution must follow the principle of least privilege.

Which role should you assign to the user?

A. workspace Admin

B. domain admin

C. domain contributor

D. Fabric admin

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You need to ensure that a user can perform the following tasks:

Create a new domain for the sales department.

Create two subdomains: one for the east region and one for the west region.

Assign Workspace1 to the east region subdomain.

Assign Workspace2 to the west region subdomain.

The solution must follow the principle of least privilege.

Which role should you assign to the user?

- **A.** workspace Admin
- **B.** domain admin
- **C.** domain contributor
- **D.** Fabric admin

Answer : **B**

To implement a domain strategy and manage subdomains within Fabric, the domain admin role is the appropriate role for the user. A domain admin has the permissions necessary to:

Create a new domain (for the sales department).

Create subdomains (for the east and west regions).

Assign workspaces (such as Workspace1 and Workspace2) to the appropriate subdomains.

The domain admin role allows for managing the structure and organization of workspaces in the context of domains and subdomains while maintaining the principle of least privilege by limiting the user's access to managing the domain structure specifically.

21) You have an Azure Data Lake Storage Gen2 account named storage1 and an Amazon S3 bucket named storage2.

You have the Delta Parquet files shown in the following table.

Name	Stored in	Size	Description
ProductFile	storage1	50 MB	Contains a list of products and their details
TripsFile	storage2	2 GB	Contains one month's worth of taxi trip data
StoreFile	storage2	25 MB	Contains a list of stores and their addresses

You have a Fabric workspace named Workspace1 that has the cache for shortcuts enabled. Workspace1 contains a lakehouse named Lakehouse1.
Lakehouse1 has the following shortcuts:
The data from which shortcuts will be retrieved from the cache?

A. Trips and Stores only

B. Products and Store only

C. Stores only

D. Products only

E. Products. Stores, and Trips

21) You have an Azure Data Lake Storage Gen2 account named storage1 and an Amazon S3 bucket named storage2.

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You have a Fabric workspace named Workspace1 that has the cache for shortcuts enabled. Workspace1 contains a lakehouse named Lakehouse1. Lakehouse1 has the following shortcuts:
The data from which shortcuts will be retrieved from the cache?

A. Trips and Stores only

B. Products and Store only

C. Stores only

D. Products only

E. Products. Stores, and Trips

Answer : **B**

When the cache for shortcuts is enabled in Fabric, the data retrieval is governed by the caching behavior, which generally retains data for a specific period after it was last accessed. The data from the shortcuts will be retrieved from the cache if the data is stored in locations that support caching. Here's a breakdown based on the data's location:

Products: The ProductFile is stored in Azure Data Lake Storage Gen2 (storage1). Since Azure Data Lake is a supported storage system in Fabric and the file is relatively small (50 MB), this data is most likely cached and can be retrieved from the cache.

Stores: The StoreFile is stored in Amazon S3 (storage2), and even though it is stored in a different cloud provider, Fabric can cache data from Amazon S3 if caching is enabled. This data (25 MB) is likely cached and retrievable.

Trips: The TripsFile is stored in Amazon S3 (storage2) and is significantly larger (2 GB) compared to the other files. While Fabric can cache data from Amazon S3, the larger size of the file (2 GB) may exceed typical cache sizes or retention windows, causing this file to likely be retrieved directly from the source instead of the cache.

22) You have a Fabric workspace named Workspace1 that contains an Apache Spark job definition named Job1.

You have an Azure SQL database named Source1 that has public internet access disabled.

You need to ensure that Job1 can access the data in Source1.

What should you create?

A. an on-premises data gateway

B. a managed private endpoint

C. an integration runtime

D. a data management gateway

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You need to ensure that Job1 can access the data in Source1.

What should you create?

A. an on-premises data gateway

B. a managed private endpoint

C. an integration runtime

D. a data management gateway

Answer : **B**

To allow Job1 in Workspace1 to access an Azure SQL database (Source1) with public internet access disabled, you need to create a managed private endpoint. A managed private endpoint is a secure, private connection that enables services like Fabric (or other Azure services) to access resources such as databases, storage accounts, or other services within a virtual network (VNet) without requiring public internet access. This approach maintains the security and integrity of your data while enabling access to the Azure SQL database.

This is a case study. Case studies are not timed separately. You can use as much exam time as you would like to complete each case. However, there may be additional case studies and sections on this exam. You must manage your time to ensure that you are able to complete all questions included on this exam in the time provided.
To answer the questions included in a case study, you will need to reference information that is provided in the case study. Case studies might contain exhibits and other resources that provide more information about the scenario that is described in the case study. Each question is independent of the other questions in this case study.
At the end of this case study, a review screen will appear. This screen allows you to review your answers and to make changes before you move to the next section of the exam. After you begin a new section, you cannot return to this section.

To start the case study
To display the first question in this case study, click the Next button. Use the buttons in the left pane to explore the content of the case study before you answer the questions. Clicking these buttons displays information such as business requirements, existing environment, and problem statements. If the case study has an All Information tab, note that the information displayed is identical to the information displayed on the subsequent tabs. When you are ready to answer a question, click the Question button to return to the question.

Overview. Company Overview:
Contoso, Ltd. is an online retail company that wants to modernize its analytics platform by moving to Fabric. The company plans to begin using Fabric for marketing analytics.

Overview. IT Structure:
The company's IT department has a team of data analysts and a team of data engineers that use analytics systems.
The data engineers perform the ingestion, transformation, and loading of data. They prefer to use Python or SQL to transform the data.
The data analysts query data and create semantic models and reports. They are qualified to write queries in Power Query and T-SQL.

Existing Environment:
Fabric
Contoso has an F64 capacity named Cap1. All Fabric users are allowed to create items.
Contoso has two workspaces named WorkspaceA and WorkspaceB that currently use Pro license mode.

Existing Environment:
Source Systems
Contoso has a point of sale (POS) system named POS1 that uses an instance of SQL Server on Azure Virtual Machines in the same Microsoft Entra tenant as Fabric. The host virtual machine is on a private virtual network that has public access blocked. POS1 contains all the sales transactions that were processed on the company's website.
The company has a software as a service (SaaS) online marketing app named MAR1. MAR1 has seven entities. The entities contain data that relates to email open rates and interaction rates, as well as website interactions. The data can be exported from MAR1 by calling REST APIs. Each entity has a different endpoint.
Contoso has been using MAR1 for one year. Data from prior years is stored in Parquet files in an Amazon Simple Storage Service (Amazon S3) bucket. There are 12 files that range in size from 300 MB to 900 MB and relate to email interactions.

Existing Environment:
Product Data
POS1 contains a product list and related data. The data comes from the following three tables:
Products
ProductCategories
ProductSubcategories
In the data, products are related to product subcategories, and subcategories are related to product categories.

Existing Environment:
Azure
Contoso has a Microsoft Entra tenant that has the following mail-enabled security groups:
DataAnalysts: Contains the data analysts
DataEngineers: Contains the data engineers
Contoso has an Azure subscription.

The company has an existing Azure DevOps organization and creates a new project for repositories that relate to Fabric.

Existing Environment:
User Problems
The VP of marketing at Contoso requires analysis on the effectiveness of different types of email content. It typically takes a week to manually compile and analyze the data. Contoso wants to reduce the time to less than one day by using Fabric.
The data engineering team has successfully exported data from MAR1. The team experiences transient connectivity errors, which causes the data exports to fail.

Requirements:
Planned Changes
Contoso plans to create the following two lakehouses:
Lakehouse1: Will store both raw and cleansed data from the sources
Lakehouse2: Will serve data in a dimensional model to users for analytical queries
Additional items will be added to facilitate data ingestion and transformation.
Contoso plans to use Azure Repos for source control in Fabric.

Requirements:
Technical Requirements
The new lakehouses must follow a medallion architecture by using the following three layers: bronze, silver, and gold. There will be extensive data cleansing required to populate the MAR1 data in the silver layer, including deduplication, the handling of missing values, and the standardizing of capitalization.
Each layer must be fully populated before moving on to the next layer. If any step in populating the lakehouses fails, an email must be sent to the data engineers.

Data imports must run simultaneously, when possible.
The use of email data from the Amazon S3 bucket must meet the following requirements:
Minimize egress costs associated with cross-cloud data access.
Prevent saving a copy of the raw data in the lakehouses.
Items that relate to data ingestion must meet the following requirements:
The items must be source controlled alongside other workspace items.
Ingested data must land in the bronze layer of Lakehouse1 in the Delta format.
No changes other than changes to the file formats must be implemented before the data lands in the bronze layer.
Development effort must be minimized and a built-in connection must be used to import the source data.
In the event of a connectivity error, the ingestion processes must attempt the connection again.
Lakehouses, data pipelines, and notebooks must be stored in WorkspaceA. Semantic models, reports, and dataflows must be stored in WorkspaceB.
Once a week, old files that are no longer referenced by a Delta table log must be removed.

Requirements:
Data Transformation
In the POS1 product data, ProductID values are unique. The product dimension in the gold layer must include only active products from product list. Active products are identified by an IsActive value of 1.
Some product categories and subcategories are NOT assigned to any product. They are NOT analytically relevant and must be omitted from the product dimension in the gold layer.

Requirements:
Data Security
Security in Fabric must meet the following requirements:
The data engineers must have read and write access to all the lakehouses, including the underlying files.
The data analysts must only have read access to the Delta tables in the gold layer.
The data analysts must NOT have access to the data in the bronze and silver layers.
The data engineers must be able to commit changes to source control in WorkspaceA.

23) You need to schedule the population of the medallion layers to meet the technical requirements.

What should you do?

A. Schedule a data pipeline that calls other data pipelines.

B. Schedule a notebook.

C. Schedule an Apache Spark job.

D. Schedule multiple data pipelines.

23) You need to schedule the population of the medallion layers to meet the technical requirements.

What should you do?

A. Schedule a data pipeline that calls other data pipelines.

B. Schedule a notebook.

C. Schedule an Apache Spark job.

D. Schedule multiple data pipelines.

Answer: A

The technical requirements specify that: Why Use a Data Pipeline That Calls Other Data Pipelines?
- Sequential execution of child pipelines.
- Error handling to send email notifications upon failures.
- Parallel execution of tasks where possible (e.g., simultaneous imports into the bronze layer).

24) You have a Fabric notebook named Notebook1 that has been executing successfully for the last week.

During the last run, Notebook1executed nine jobs.

You need to view the jobs in a timeline chart.

What should you use?

A. Real-Time hub

B. Monitoring hub

C. the job history from the application run

D. Spark History Server

E. the run series from the details of the application run

24) You have a Fabric notebook named Notebook1 that has been executing successfully for the last week.

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What should you use?

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B. Monitoring hub

C. the job history from the application run

D. Spark History Server

E. the run series from the details of the application run

Answer: E

The run series from the details of the application run is the most detailed and relevant feature for visualizing job execution in a timeline format, making it the correct choice for this scenario. It provides an intuitive way to analyze job execution patterns and improve the efficiency of the notebook.

Explanation:The run series from the details of the application run is the most detailed and relevant feature for visualizing job execution in a timeline format, making it the correct choice for this scenario. It provides an intuitive way to analyze job execution patterns and improve the efficiency of the notebook.

25) You have a Fabric warehouse named DW1 that loads data by using a data pipeline named Pipeline1. Pipeline1 uses a Copy data activity with a dynamic SQL source. Pipeline1 is scheduled to run every 15minutes.

You discover that Pipeline1 keeps failing.

You need to identify which SQL query was executed when the pipeline failed.

What should you do?

A. From Monitoring hub, select the latest failed run of Pipeline1, and then view the output JSON.

B. From Monitoring hub, select the latest failed run of Pipeline1, and then view the input JSON.

C. From Real-time hub, select Fabric events, and then review the details of Microsoft.Fabric.ItemReadFailed.

D. From Real-time hub, select Fabric events, and then review the details of Microsoft. Fabric.ItemUpdateFailed.

25) You have a Fabric warehouse named DW1 that loads data by using a data pipeline named Pipeline1. Pipeline1 uses a Copy data activity with a dynamic SQL source. Pipeline1 is scheduled to run every 15minutes.

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You need to identify which SQL query was executed when the pipeline failed.

What should you do?

A. From Monitoring hub, select the latest failed run of Pipeline1, and then view the output JSON.

B. From Monitoring hub, select the latest failed run of Pipeline1, and then view the input JSON.

C. From Real-time hub, select Fabric events, and then review the details of Microsoft.Fabric.ItemReadFailed.

D. From Real-time hub, select Fabric events, and then review the details of Microsoft. Fabric.ItemUpdateFailed.

Answer: B

Explanation:The input JSON contains the configuration details and parameters passed to the Copy data activity during execution, including the dynamically generated SQL query.

Viewing the input JSON for the failed pipeline run provides direct insight into what query was executed at the time of failure.

26) You have a Fabric workspace that contains an eventstream named EventStream1. EventStream1 outputs events to a table in a lakehouse.

You need to remove files that are older than seven days and are no longer in use.

Which command should you run?

A. VACUUM

B. COMPUTE

C. OPTIMIZE

D. CLONE

26) You have a Fabric workspace that contains an eventstream named EventStream1. EventStream1 outputs events to a table in a lakehouse.

You need to remove files that are older than seven days and are no longer in use.

Which command should you run?

A. VACUUM

B. COMPUTE

C. OPTIMIZE

D. CLONE

Answer: A

Explanation: VACUUM is used to clean up storage by removing files no longer in use by a Delta table. It removes old and unreferenced files from Delta tables. For example, to remove files older than 7 days:

VACUUM delta.`/path_to_table` RETAIN 7 HOURS;

27) You have a Fabric workspace that contains a warehouse named Warehouse1. Data is loaded daily into Warehouse1 by using data pipelines and stored procedures.

You discover that the daily data load takes longer than expected.

You need to monitor Warehouse1 to identify the names of users that are actively running queries.

Which view should you use?

A. sys.dm_exec_connections

B. sys.dm_exec_requests

C. queryinsights.long_running_queries

D. queryinsights.frequently_run_queries

E. sys.dm_exec_sessions

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B. sys.dm_exec_requests

C. queryinsights.long_running_queries

D. queryinsights.frequently_run_queries

E. sys.dm_exec_sessions

Answer: E

Explanation:sys.dm_exec_sessions provides real-time information about all active sessions, including the user, session ID, and status of the session. You can filter on session status to see users actively running queries.

28) You have a Fabric workspace that contains a lakehouse named Lakehouse1.
In an external data source, you have data files that are 500GB each. A new file is added every day.
You need to ingest the data into Lakehouse1 without applying any transformations. The solution must meet the following requirements
Trigger the process when a new file is added.
Provide the highest throughput.
Which type of item should you use to ingest the data?

A. Data pipeline
B. Environment
C. KQL queryset
D. Dataflow Gen2

28) You have a Fabric workspace that contains a lakehouse named Lakehouse1.
In an external data source, you have data files that are 500GB each. A new file is added every day.
You need to ingest the data into Lakehouse1 without applying any transformations. The solution must meet the following requirements
Trigger the process when a new file is added.
Provide the highest throughput.
Which type of item should you use to ingest the data?

A. Data pipeline
B. Environment
C. KQL queryset
D. Dataflow Gen2

Correct Answer: A

To efficiently ingest large data files (500 GB each) into Lakehouse1 with high throughput and trigger the process when a new file is added, a Data pipeline is the most suitable solution. Data pipelines in Fabric are ideal for orchestrating data movement and can be configured to automatically trigger based on file arrivals or other events. This solution meets both requirements: ingesting the data without transformations (since you just need to copy the data) and triggering the process when new files are added.

29) You have a Fabric workspace that contains an eventhouse and a KQL database named Database1. Database1 has the following:
Policy1 sends data from Table1 to Table2.
The following is a sample of the data in Table2.

Timestamp (datetime)	DeviceId (guid)	StreamData (dynamic)
2024-05-18 12:45:17.16524	81416f30-60a2-4e75-9b19-2a84ea059735	[{ "index": 0, "eventid": "719afca0-be30-4559-bb5e-59feade642f6" }]
2024-05-18 12:45:21.76423	bb664e1e-02aa-4e17-8c8a-116cd4458d52	[{ "index": 0, "eventid": "782222b2-fbcb-43c0-82d6-ecd49a99dbf5 }]
2024-05-18 12:45:23.98642	717bfe7d-0e5d-498f-9f21-e60aaf258056	[{ "index": 0, "eventid": "d5730286-0da4-41f8-8e59-f75e209310a9 }]

Recently, the following actions were performed on Table1:
You plan to load additional records to Table2.
Which two records will load from Table1 to Table2? Each correct answer presents a complete solution.
NOTE: Each correct selection is worth one point.

A)

Timestamp (datetime)	DeviceId (guid)	StreamData (dynamic)
2024-05-18	81416f30-60a2-4e75-9b19-2a84ea059735	[{ "index": 40, "eventid": "729afca2-be30-4559-bb5e-59feade642f3", "temperature": 32 }]

B)

Timestamp (datetime)	DeviceId (guid)	StreamData (dynamic)
2024-05-21	81416f30	[{ "index": 0, "eventid": "719afca0-be30-4559-bb5e-5werade642f6", "temperature": 27 }]

C)

Timestamp (datetime)	DeviceId (guid)	StreamData (dynamic)
2024-05-23	81416f3060a24e759b192a84ea05973532dhdyte3	[{ "index": 0, "eventid": "719afca0-be30-4559-bb5e-59feade642f6" }]

D)

Timestamp (datetime)	DeviceId (guid)	StreamData (dynamic)
2024-05-24	81416f30-60a2-4e75-9b19-2a84ea059735	[{ "index": 0, "eventid": "719afca0-be30-4559-bb5e-59feade642f6" }]

A. Option A

B. Option B

C. Option c

D. Option D

A. Option A

B. Option B

C. Option c

D. Option D

Correct Answer: B, D

Changes to Table1 Structure:
Impact of Changes:
Record B:
Accepted because it fully matches Table2's structure and data types.
Record D:
Accepted because it fully matches Table2's structure and data types.

30) You have a Fabric workspace that contains a warehouse named DW1. DW1 is loaded by using a notebook named Notebook1.
You need to identify which version of Delta was used when Notebook1 was executed.
What should you use?

A.Real-Time hub

B.OneLake data hub

C.the Admin monitoring workspace

D.Fabric Monitor

E.the Microsoft Fabric Capacity Metrics app

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What should you use?

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C.the Admin monitoring workspace

D.Fabric Monitor

E.the Microsoft Fabric Capacity Metrics app

Answer(s): D

Case Study:

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Overview
Litware, Inc. is a publishing company that has an online bookstore and several retail bookstores worldwide. Litware also manages an online advertising business for the authors it represents.

Existing Environment:
Fabric Environment
Litware has a Fabric workspace named Workspace1. High concurrency is enabled for Workspace1.
The company has a data engineering team that uses Python for data processing.

Existing Environment:
Data Processing
The retail bookstores send sales data at the end of each business day, while the online bookstore constantly provides logs and sales data to a central enterprise resource planning (ERP) system.
Litware implements a medallion architecture by using the following three layers: bronze, silver, and gold. The sales data is ingested from the ERP system as Parquet files that land in the Files folder in a lakehouse. Notebooks are used to transform the files in a Delta table for the bronze and silver layers. The gold layer is in a warehouse that has V-Order disabled.
Litware has image files of book covers in Azure Blob Storage. The files are loaded into the Files folder.

Existing Environment:
Sales Data
Month-end sales data is processed on the first calendar day of each month. Data that is older than one month never changes.
In the source system, the sales data refreshes every six hours starting at midnight each day.
The sales data is captured in a Dataflow Gen1 dataflow. When the dataflow runs, new and historical data is captured. The dataflow captures the following fields of the source:
Sales Date
Author
Price
Units
SKU
A table named AuthorSales stores the sales data that relates to each author. The table contains a column named AuthorEmail. Authors authenticate to a guest Fabric tenant by using their email address.

Existing Environment:
Security Groups
Litware has the following security groups:
Sales
Fabric Admins
Streaming Admins

Existing Environment:
Performance Issues
Business users perform ad-hoc queries against the warehouse. The business users indicate that reports against the warehouse sometimes run for two hours and fail to load as expected. Upon further investigation, the data engineering team receives the following error message when the reports fail to load: "The SQL query failed while running."
The data engineering team wants to debug the issue and find queries that cause more than one failure.
When the authors have new book releases, there is often an increase in sales activity. This increase slows the data ingestion process.
The company's sales team reports that during the last month, the sales data has NOT been up-to-date when they arrive at work in the morning.

Requirements:
Planned Changes
Litware recently signed a contract to receive book reviews. The provider of the reviews exposes the data in Amazon Simple Storage Service (Amazon S3) buckets.
Litware plans to manage Search Engine Optimization (SEO) for the authors. The SEO data will be streamed from a REST API.
Requirements. Version Control
Litware plans to implement a version control solution in Fabric that will use GitHub integration and follow the principle of least privilege.
Requirements. Governance Requirements
To control data platform costs, the data platform must use only Fabric services and items. Additional Azure resources must NOT be provisioned.

Requirements. Data Requirements
Litware identifies the following data requirements:
Process the SEO data in near-real-time (NRT).
Make the book reviews available in the lakehouse without making a copy of the data.
When a new book cover image arrives in the Files folder, process the image as soon as possible.
You need to implement the solution for the book reviews.
Which should you do?

31) You need to implement the solution for the book reviews.
Which should you do? (case Study 3)

A.Create a Dataflow Gen2 dataflow.

B.Create a shortcut.

C.Enable external data sharing.

D.Create a data pipeline.

31) You need to implement the solution for the book reviews.
Which should you do? (case Study 3)

A.Create a Dataflow Gen2 dataflow.

B.Create a shortcut.

C.Enable external data sharing.

D.Create a data pipeline.

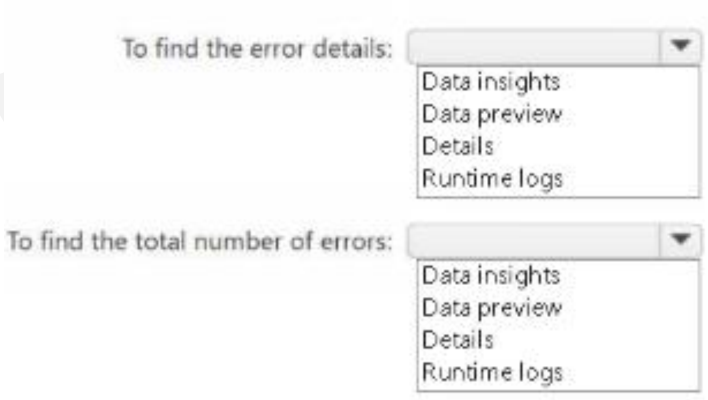
Answer(s): B

Explanation:
The requirement specifies that Litware plans to make the book reviews available in the lakehouse without making a copy of the data. In this case, creating a shortcut in Fabric is the most appropriate solution. A shortcut is a reference to the external data, and it allows Litware to access the book reviews stored in Amazon S3 without duplicating the data into the lakehouse.

32) You have a Fabric workspace that contains an eventstream named EventStream1.
You discover that an EventStream1 transformation fails.
You need to find the following error information:
The error details, including the occurrence time

The total number of errors -

What should you use? To answer, select the appropriate options in the answer area.
NOTE: Each correct selection is worth one point.



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You discover that an EventStream1 transformation fails.
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The total number of errors -

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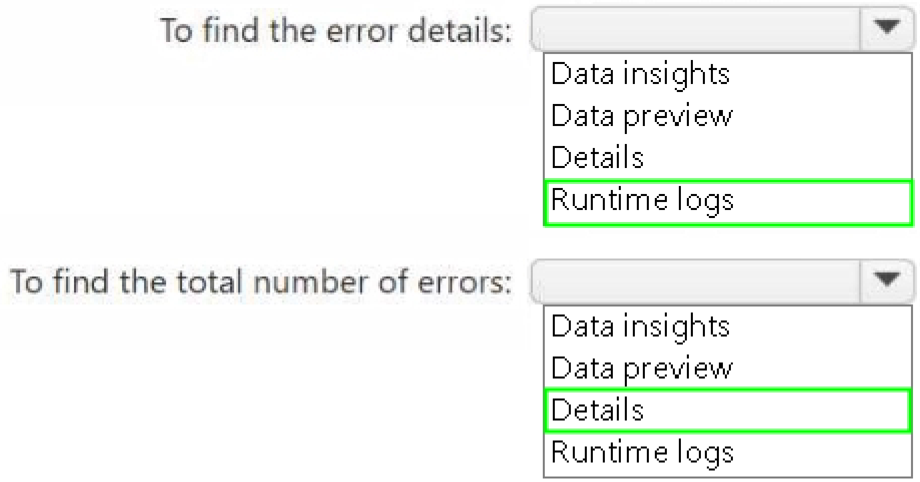

33) You have a Fabric workspace that contains a lakehouse named Lakehouse1. Lakehouse1 contains a Delta table named Table1. You analyze Table1 and discover that Table1 contains 2,000 Parquet files of 1MB each. You need to minimize how long it takes to query Table1. What should you do?

A- Disable V-Order and run the OPTIMIZE command.
B- Disable V-Order and run the VACUUM command.
C- Run the OPTIMIZE and VACUUM commands.

33) You have a Fabric workspace that contains a lakehouse named Lakehouse1. Lakehouse1 contains a Delta table named Table1. You analyze Table1 and discover that Table1 contains 2,000 Parquet files of 1MB each. You need to minimize how long it takes to query Table1. What should you do?

A- Disable V-Order and run the OPTIMIZE command.
B- Disable V-Order and run the VACUUM command.
C- Run the OPTIMIZE and VACUUM commands.

Answer: C

Commands and Their Roles:- Compacts small Parquet files into larger files to improve query performance.- It supports optional features like V-Order, which organizes data for efficient scanning.- Removes old, unreferenced data files and metadata from the Delta table.- Running VACUUM after OPTIMIZE ensures unnecessary files are cleaned up, reducing storage overhead and improving performance.

34) You have a KQL database that contains two tables named Stream and Reference.
Stream contains streaming data in the following format.

Column name	Data type
Timestamp	Datetime
GeoLocation	Dynamic
Temperature	Decimal
DeviceId	Int

Reference contains reference data in the following format.

Column name	Data type
DeviceId	Int
DeviceName	String

Both tables contain millions of rows. You have the following KQL queryset.

```
Stream
| extend lat = todecimal(GeoLocation.Latitude), long = todecimal(GeoLocation.Longitude)
| join kind=inner Reference on DeviceId
| project Timestamp, lat, long, Temperature, DeviceName
| filter Temperature >= 10
| render scatterchart with (kind = map)
```

You need to reduce how long it takes to run the KQL queryset.
Solution: You add the make_list() function to the output columns.
Does this meet the goal?

A: Yes
B: No

You need to reduce how long it takes to run the KQL queryset.
Solution: You add the make_list() function to the output columns.
Does this meet the goal?

A: Yes
B: No

Answer: B

Adding an aggregation like make_list() would require additional processing and memory, which could make the query slower.

35) You have a Fabric workspace that contains a warehouse named Warehouse1.
Warehouse1 contains the following tables and columns.

Table name	Column name	Data type
Employee	EmployeeID	Int
Employee	EmployeeName	Varchar(128)
Employee	EmployeePosition	Varchar(64)
Contract	EmployeeID	Int
Contract	ContractType	Varchar(64)
Contract	StartDate	Datetime2
Contract	EndDate	Datetime2

You need to denormalize the tables and include the ContractType and StartDate columns in the

Employee table. The solution must meet the following requirements: How should you complete the statement? To answer, select the appropriate options in the answer area.

NOTE: Each correct selection is worth one point.

Answer Area

```
WITH result AS(
SELECT e.EmployeeID
  , e.EmployeeName
  , e.EmployeePosition
  , c.ContractType
  ,    [dropdown] (date, c.StartDate) as StartDate
```

- CAST
- CONVERT
- REPLACE
- SUBSTRING

```
FROM Employee AS e
[dropdown] Contract AS c on c.EmployeeID = e.EmployeeID
```

- CROSS JOIN
- INNER JOIN
- LEFT OUTER JOIN
- RIGHT OUTER JOIN

```
)
SELECT COUNT(DISTINCT EmployeeID) AS TotalEmployees
 , ContractType
FROM result
GROUP BY ContractType
[dropdown] COUNT(DISTINCT EmployeeID) > 2
```

- CONTAINS
- HAVING
- UMIT
- WHERE

```
WITH result AS(

SELECT e.EmployeeID

  , e.EmployeeName

  , e.EmployeePosition

  , c.ContractType

  ,   [dropdown] (date, c.StartDate) as StartDate
        CAST
        CONVERT
        REPLACE
        SUBSTRING

FROM Employee AS e

[dropdown] Contract AS c on c.EmployeeID = e.EmployeeID
  CROSS JOIN
  INNER JOIN
  LEFT OUTER JOIN
  RIGHT OUTER JOIN
)

SELECT COUNT(DISTINCT EmployeeID) AS TotalEmployees

 , ContractType

FROM result

GROUP BY ContractType

[dropdown] COUNT(DISTINCT EmployeeID) > 2
  CONTAINS
  HAVING
  LIMIT
  WHERE
```

36) You have a Fabric workspace named Workspace1 that contains a data pipeline named Pipeline1 and a lakehouse named Lakehouse1. You have a deployment pipeline named deployPipeline1 that deploys Workspace1 to Workspace2. You restructure Workspace1 by adding a folder named Folder1 and moving Pipeline1 to Folder1. You use deployPipeline1 to deploy Workspace1 to Workspace2.

What occurs to Workspace2?

A. Folder1 is created, Pipeline1 moves to Folder1, and Lakehouse1 is deployed.
B. Only Pipeline1 and Lakehouse1 are deployed.
C. Folder1 is created, and Pipeline1 and Lakehouse1 move to Folder1.
D. Only Folder1 is created and Pipeline1 moves to Folder1.

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B. Only Pipeline1 and Lakehouse1 are deployed.
C. Folder1 is created, and Pipeline1 and Lakehouse1 move to Folder1.
D. Only Folder1 is created and Pipeline1 moves to Folder1.

Answer: A

Explanation: When you restructure Workspace1 by adding a new folder (Folder1) and moving Pipeline1 into it, deployPipeline1 will deploy the entire structure of Workspace1 to Workspace2, preserving the changes made in Workspace1.
This includes:
Folder1 will be created in Workspace2, mirroring the structure in Workspace1. Pipeline1 will be moved into Folder1 in Workspace2, maintaining the same folder structure.
Lakehouse1 will be deployed to Workspace2 as it exists in Workspace1.

37) You have a Fabric eventstream that loads data into a table named Bike_Location in a KQL database. The table contains the following columns: You need to apply transformation and filter logic to prepare the data for consumption. The solution must return data for a neighbourhood named Sands End when No_Bikes is at least 15. The results must be ordered by No_Bikes in ascending order. Solution: You use the following code segment:

```
bike_location
| filter Neighbourhood == "Sands End" and No_Bikes >= 15
| sort by No_Bikes asc
| project BikepointID, Street, Neighbourhood, No_Bikes, No_Empty_Docks, Timestamp
```

Does this meet the goal?

A. Yes
B. no

37) You have a Fabric eventstream that loads data into a table named Bike_Location in a KQL database. The table contains the following columns: You need to apply transformation and filter logic to prepare the data for consumption. The solution must return data for a neighbourhood named Sands End when No_Bikes is at least 15. The results must be ordered by No_Bikes in ascending order. Solution: You use the following code segment:

```
bike_location
| filter Neighbourhood == "Sands End" and No_Bikes >= 15
| sort by No_Bikes asc
| project BikepointID, Street, Neighbourhood, No_Bikes, No_Empty_Docks, Timestamp
```

Does this meet the goal?

A. Yes

B. no

Answer: A

Explanation:

Filter Condition: It correctly filters rows where Neighbourhood is "Sands End" and No_Bikes is greater than or equal to 15.

Sorting: The sorting is explicitly done by No_Bikes in ascending order using sort by No_Bikes asc.

Projection: It projects the required columns (BikepointID, Street, Neighbourhood, No_Bikes, No_Empty_Docks, Timestamp), which minimizes the data returned for consumption.

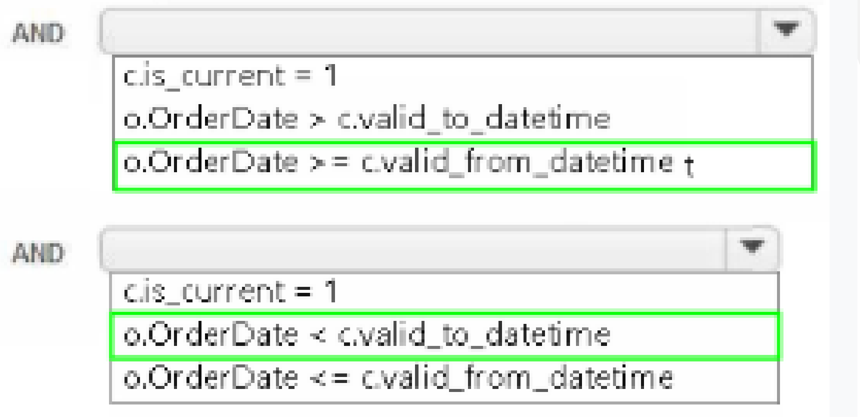
38) You have a Fabric workspace that contains two lakehouses named Lakehouse1 and Lakehouse2. Lakehouse1 contains staging data in a Delta table named Orderlines. Lakehouse2 contains a Type 2 slowly changing dimension (SCD) dimension table named Dim_Customer. You need to build a query that will combine data from Orderlines and Dim_Customer to create a new fact table named Fact_Orders. The new table must meet the following requirements: Enable the analysis of customer orders based on historical attributes. Enable the analysis of customer orders based on the current attributes. How should you complete the statement? To answer, select the appropriate options in the answer area. NOTE: Each correct selection is worth one point.

```
SELECT
  OrderLineID order_line_id
  ,OrderDate order_date
  ,c.customer_key
  ,c.customer_id
  ,Quantity order_quantity
  ,UnitPrice unit_price
  ,TaxRate tax_rate
FROM
  Lakehouse1.orderlines o
INNER JOIN
  Lakehouse2.dim_customer c
    ON o.customerid = c.customer_id

    AND [dropdown]
        c.is_current = 1
        o.OrderDate > c.valid_to_datetime
        o.OrderDate > = c.valid_from_datetime

    AND [dropdown]
        c.is_current = 1
        o.OrderDate < c.valid_to_datetime
        o.OrderDate < = c.valid_from_datetime
```

38) You have a Fabric workspace that contains two lakehouses named Lakehouse1 and Lakehouse2. Lakehouse1 contains staging data in a Delta table named Orderlines. Lakehouse2 contains a Type 2 slowly changing dimension (SCD) dimension table named Dim_Customer. You need to build a query that will combine data from Orderlines and Dim_Customer to create a new fact table named Fact_Orders. The new table must meet the following requirements: Enable the analysis of customer orders based on historical attributes. Enable the analysis of customer orders based on the current attributes. How should you complete the statement? To answer, select the appropriate options in the answer area. NOTE: Each correct selection is worth one point.



39) HOTSPOT (Drag and Drop is not supported)
Case Study:

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Overview
Litware, Inc. is a publishing company that has an online bookstore and several retail bookstores worldwide. Litware also manages an online advertising business for the authors it represents.

Existing Environment:
Fabric Environment
Litware has a Fabric workspace named Workspace1. High concurrency is enabled for Workspace1.
The company has a data engineering team that uses Python for data processing.

Existing Environment:
Data Processing
The retail bookstores send sales data at the end of each business day, while the online bookstore constantly provides logs and sales data to a central enterprise resource planning (ERP) system.
Litware implements a medallion architecture by using the following three layers: bronze, silver, and gold. The sales data is ingested from the ERP system as Parquet files that land in the Files folder in a lakehouse. Notebooks are used to transform the files in a Delta table for the bronze and silver layers. The gold layer is in a warehouse that has V-Order disabled.
Litware has image files of book covers in Azure Blob Storage. The files are loaded into the Files folder.

Existing Environment:
Sales Data
Month-end sales data is processed on the first calendar day of each month. Data that is older than one month never changes.
In the source system, the sales data refreshes every six hours starting at midnight each day.
The sales data is captured in a Dataflow Gen1 dataflow. When the dataflow runs, new and historical data is captured. The dataflow captures the following fields of the source:
Sales Date
Author
Price
Units
SKU
A table named AuthorSales stores the sales data that relates to each author. The table contains a column named AuthorEmail. Authors authenticate to a guest Fabric tenant by using their email address.

Existing Environment:
Security Groups
Litware has the following security groups:
Sales
Fabric Admins
Streaming Admins

Existing Environment:
Performance Issues
Business users perform ad-hoc queries against the warehouse. The business users indicate that reports against the warehouse sometimes run for two hours and fail to load as expected. Upon further investigation, the data engineering team receives the following error message when the reports fail to load: “The SQL query failed while running.”
The data engineering team wants to debug the issue and find queries that cause more than one failure.
When the authors have new book releases, there is often an increase in sales activity. This increase slows the data ingestion process.
The company’s sales team reports that during the last month, the sales data has NOT been up-to-date when they arrive at work in the morning.

Requirements:
Planned Changes
Litware recently signed a contract to receive book reviews. The provider of the reviews exposes the data in Amazon Simple Storage Service (Amazon S3) buckets.
Litware plans to manage Search Engine Optimization (SEO) for the authors. The SEO data will be streamed from a REST API.
Requirements. Version Control
Litware plans to implement a version control solution in Fabric that will use GitHub integration and follow the principle of least privilege.
Requirements. Governance Requirements
To control data platform costs, the data platform must use only Fabric services and items. Additional Azure resources must NOT be provisioned.

Requirements. Data Requirements

Litware identifies the following data requirements:

Process the SEO data in near-real-time (NRT).

Make the book reviews available in the lakehouse without making a copy of the data.

When a new book cover image arrives in the Files folder, process the image as soon as possible.

You need to troubleshoot the ad-hoc query issue.

How should you complete the statement? To answer, select the appropriate options in the answer area.

NOTE: Each correct selection is worth one point.

Answer Area

```
SELECT last_run_start_time, last_run_command
FROM [dropdown]
```

- queryinsights.exec_requests_history
- queryinsights.exec_sessions_history
- queryinsights.frequently_run_queries
- queryinsights.long_running_queries

```
WHERE last_run_total_elapsed_time_ms > 7200000
AND [dropdown]
```

- max_run_total_elapsed_time_ms > 7200000
- median_total_elapsed_time_ms > 7200000
- number_of_canceled_runs > 1
- number_of_failed_runs > 1
- number_of_runs > 1

Explanation:

```
SELECT last_run_start_time, last_run_command
FROM
    queryinsights.exec_requests_history
    queryinsights.exec_sessions_history
    queryinsights.frequently_run_queries
    queryinsights.long_running_queries
WHERE last_run_total_elapsed_time_ms > 7200000
AND
    max_run_total_elapsed_time_ms > 7200000
    median_total_elapsed_time_ms > 7200000
    number_of_canceled_runs > 1
    number_of_failed_runs > 1
    number_of_runs > 1
```

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The company's sales team reports that during the last month, the sales data has NOT been up-to-date when they arrive at work in the morning.

Requirements:
Planned Changes
Litware recently signed a contract to receive book reviews. The provider of the reviews exposes the data in Amazon Simple Storage Service (Amazon S3) buckets.
Litware plans to manage Search Engine Optimization (SEO) for the authors. The SEO data will be streamed from a REST API.
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Requirements. Governance Requirements
To control data platform costs, the data platform must use only Fabric services and items. Additional Azure resources must NOT be provisioned.
Requirements. Data Requirements
Litware identifies the following data requirements:
Process the SEO data in near-real-time (NRT).
Make the book reviews available in the lakehouse without making a copy of the data.

When a new book cover image arrives in the Files folder, process the image as soon as possible.
You need to resolve the sales data issue. The solution must minimize the amount of data transferred.
What should you do?

A. Spilt the dataflow into two dataflows.

B. Configure scheduled refresh for the dataflow.

C. Configure incremental refresh for the dataflow. Set Store rows from the past to 1 Month.

D. Configure incremental refresh for the dataflow. Set Refresh rows from the past to 1 Year.

E. Configure incremental refresh for the dataflow. Set Refresh rows from the past to 1 Month.

When a new book cover image arrives in the Files folder, process the image as soon as possible.
You need to resolve the sales data issue. The solution must minimize the amount of data transferred.
What should you do?

- A Spilt the dataflow into two dataflows.
- B Configure scheduled refresh for the dataflow.
- C Configure incremental refresh for the dataflow. Set Store rows from the past to 1 Month.
- D Configure incremental refresh for the dataflow. Set Refresh rows from the past to 1 Year.
- E Configure incremental refresh for the dataflow. Set Refresh rows from the past to 1 Month.

Answer(s): E

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Overview. Company Overview:

Contoso, Ltd. is an online retail company that wants to modernize its analytics platform by moving to Fabric. The company plans to begin using Fabric for marketing analytics.

Overview. IT Structure:

The company's IT department has a team of data analysts and a team of data engineers that use analytics systems.

The data engineers perform the ingestion, transformation, and loading of data. They prefer to use Python or SQL to transform the data.

The data analysts query data and create semantic models and reports. They are qualified to write queries in Power Query and T-SQL.

Existing Environment:

Fabric

Contoso has an F64 capacity named Cap1. All Fabric users are allowed to create items.

Contoso has two workspaces named WorkspaceA and WorkspaceB that currently use Pro license mode.

Existing Environment:

Source Systems

Contoso has a point of sale (POS) system named POS1 that uses an instance of SQL Server on Azure Virtual Machines in the same Microsoft Entra tenant as Fabric. The host virtual machine is on a private virtual network that has public access blocked. POS1 contains all the sales transactions that were processed on the company's website.

The company has a software as a service (SaaS) online marketing app named MAR1. MAR1 has seven entities. The entities contain data that relates to email open rates and interaction rates, as well as website interactions. The data can be exported from MAR1 by calling REST APIs. Each entity has a different endpoint.

Contoso has been using MAR1 for one year. Data from prior years is stored in Parquet files in an Amazon Simple Storage Service (Amazon S3) bucket. There are 12 files that range in size from 300 MB to 900 MB and relate to email interactions.

Existing Environment:

Product Data

POS1 contains a product list and related data. The data comes from the following three tables:

Products

ProductCategories

ProductSubcategories

In the data, products are related to product subcategories, and subcategories are related to product categories.

Existing Environment:

Azure

Contoso has a Microsoft Entra tenant that has the following mail-enabled security groups:

DataAnalysts: Contains the data analysts

DataEngineers: Contains the data engineers

The company has an existing Azure DevOps organization and creates a new project for repositories that relate to Fabric.

Existing Environment:
User Problems
The VP of marketing at Contoso requires analysis on the effectiveness of different types of email content. It typically takes a week to manually compile and analyze the data. Contoso wants to reduce the time to less than one day by using Fabric.
The data engineering team has successfully exported data from MAR1. The team experiences transient connectivity errors, which causes the data exports to fail.

Requirements:
Planned Changes
Contoso plans to create the following two lakehouses:
Lakehouse1: Will store both raw and cleansed data from the sources
Lakehouse2: Will serve data in a dimensional model to users for analytical queries
Additional items will be added to facilitate data ingestion and transformation.
Contoso plans to use Azure Repos for source control in Fabric.

Requirements:
Technical Requirements
The new lakehouses must follow a medallion architecture by using the following three layers: bronze, silver, and gold. There will be extensive data cleansing required to populate the MAR1 data in the silver layer, including deduplication, the handling of missing values, and the standardizing of capitalization.
Each layer must be fully populated before moving on to the next layer. If any step in populating the lakehouses fails, an email must be sent to the data engineers.
Data imports must run simultaneously, when possible.
The use of email data from the Amazon S3 bucket must meet the following requirements:
Minimize egress costs associated with cross-cloud data access.

Prevent saving a copy of the raw data in the lakehouses.
Items that relate to data ingestion must meet the following requirements:
The items must be source controlled alongside other workspace items.
Ingested data must land in the bronze layer of Lakehouse1 in the Delta format.
No changes other than changes to the file formats must be implemented before the data lands in the bronze layer.
Development effort must be minimized and a built-in connection must be used to import the source data.
In the event of a connectivity error, the ingestion processes must attempt the connection again.
Lakehouses, data pipelines, and notebooks must be stored in WorkspaceA. Semantic models, reports, and dataflows must be stored in WorkspaceB.
Once a week, old files that are no longer referenced by a Delta table log must be removed.

Requirements:
Data Transformation
In the POS1 product data, ProductID values are unique. The product dimension in the gold layer must include only active products from product list. Active products are identified by an IsActive value of 1.
Some product categories and subcategories are NOT assigned to any product. They are NOT analytically relevant and must be omitted from the product dimension in the gold layer.

Requirements:
Data Security
Security in Fabric must meet the following requirements:

The data engineers must have read and write access to all the lakehouses, including the underlying files.

The data analysts must only have read access to the Delta tables in the gold layer.

The data analysts must NOT have access to the data in the bronze and silver layers.

The data engineers must be able to commit changes to source control in WorkspaceA.

You need to recommend a method to populate the POS1 data to the lakehouse medallion layers.

What should you recommend for each layer? To answer, select the appropriate options in the answer area.

NOTE: Each correct selection is worth one point.

Answer Area

Bronze layer:

- A Dataflow Gen2 dataflow
- A notebook
- A pipeline Copy activity
- A pipeline stored procedure

Silver layer:

- A Dataflow Gen2 dataflow
- A notebook
- A pipeline Copy activity
- A pipeline stored procedure

Explanation:

Bronze layer:
- A Dataflow Gen2 dataflow
- A notebook
- A pipeline Copy activity
- A pipeline stored procedure

Silver layer:
- A Dataflow Gen2 dataflow
- A notebook
- A pipeline Copy activity
- A pipeline stored procedure

42)

DRAG DROP (Drag and Drop is not supported)

You have a Fabric eventhouse that contains a KQL database. The database contains a table named TaxiData. The following is a sample of the data in TaxiData.

You need to build two KQL queries. The solution must meet the following requirements:

One of the queries must partition RunningTotalAmount by VendorID.

The other query must create a column named FirstPickupDateTime that shows the first value of each hour from tpep_pickup_datetime partitioned by payment_type.

How should you complete each query? To answer, drag the appropriate values the correct targets. Each value may be used once, more than once, or not at all. You may need to drag the split bar between panes or scroll to view content.

NOTE: Each correct selection is worth one point.

VendorID	tpep_pickup_datetime	tpep_dropoff_datetime	passenger_count	trip_distance	PULocationID	DOLocationID	payment_type	total_amount
2	2022-06-06T11:08:32Z	2022-06-06T11:22:17Z	1	0.17	231	50	2	7.12
2	2022-06-06T11:12:05Z	2022-06-06T11:20:43Z	1	1.02	161	163	1	10.56
1	2022-06-06T11:15:00Z	2022-06-06T11:25:32Z	1	1.07	142	230	2	17.12
2	2022-06-06T11:29:54Z	2022-06-06T11:49:34Z	2	2.07	162	236	2	12.01
1	2022-06-06T11:50:50Z	2022-06-06T12:07:24Z	2	2.65	140	142	1	7.89

Values

- Row_cumsum
- Row_rank_dense
- Row_rank_min
- Row_window_session

Answer Area

Statement1:

```
TaxiData
| sort by VendorID asc
| extend RunningTotalAmount = [          ] (total_amount, VendorID != prev(VendorID))
```

Statement2:

```
TaxiData
| sort by tpep_pickup_datetime asc, payment_type asc
| extend FirstPickupDateTime = [          ] (tpep_pickup_datetime, 1h, 0m, payment_type != prev(payment_type))
```

Explanation:

Values

- Row_cumsum
- Row_rank_dense
- Row_rank_min
- Row_window_session

Answer Area

Statement1:

```
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| sort by VendorID asc
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| extend FirstPickupDateTime = [          ] (tpep_pickup_datetime, 1h, 0m, payment_type != prev(payment_type))
```

43)

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The data analysts must NOT have access to the data in the bronze and silver layers.
The data engineers must be able to commit changes to source control in WorkspaceA.
You need to ensure that usage of the data in the Amazon S3 bucket meets the technical requirements.
What should you do?

A. Create a workspace identity and enable high concurrency for the notebooks.

B. Create a shortcut and ensure that caching is disabled for the workspace.

C. Create a workspace identity and use the identity in a data pipeline.

D. Create a shortcut and ensure that caching is enabled for the workspace.

The data engineers must have read and write access to all the lakehouses, including the underlying files.
The data analysts must only have read access to the Delta tables in the gold layer.
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What should you do?

A. Create a workspace identity and enable high concurrency for the notebooks.

B. Create a shortcut and ensure that caching is disabled for the workspace.

C. Create a workspace identity and use the identity in a data pipeline.

D. Create a shortcut and ensure that caching is enabled for the workspace.

Answer(s): B

44)

You are processing streaming data from an external data provider.

You have the following code segment.

For each of the following statements, select Yes if the statement is true. Otherwise, select No.

NOTE: Each correct selection is worth one point.

```
datatable (Location:string, Company:string, UnitsSold:long)
[
  "New York", "Contoso", 300,
  "New York", "Litware", 1000,
  "New York", "Relecloud", 300,
  "New York", "Fabrikam", 200,
  "Seattle", "Contoso", 300,
  "Seattle", "Litware", 100,
  "Seattle", "Fabrikam", 100,
  "San Francisco", "Relecloud", 500,
  "San Francisco", "Litware", 500,
  "Washington DC", "Litware", 300,
  "Washington DC", "Contoso", 400
]
| sort by Location desc, UnitsSold desc
| extend Rank=row_rank_dense(UnitsSold, prev(Location) != Location)
```

Answer Area

Statements	Yes	No
Litware from New York will be displayed at the top of the result set.	○	○
Fabrikam in Seattle will have value = 2 in the Rank column.	○	○
Litware in San Francisco will have the same value in the Rank column as Litware in New York.	○	○

44)

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You have the following code segment.
For each of the following statements, select Yes if the statement is true. Otherwise, select No.
NOTE: Each correct selection is worth one point.

```
datatable (Location:string, Company:string, UnitsSold:long)
[
  "New York", "Contoso", 300,
  "New York", "Litware", 1000,
  "New York", "Relecloud", 300,
  "New York", "Fabrikam", 200,
  "Seattle", "Contoso", 300,
  "Seattle", "Litware", 100,
  "Seattle", "Fabrikam", 100,
  "San Francisco", "Relecloud", 500,
  "San Francisco", "Litware", 500,
  "Washington DC", "Litware", 300,
  "Washington DC", "Contoso", 400
]
| sort by Location desc, UnitsSold desc
| extend Rank=row_rank_dense(UnitsSold, prev(Location) != Location)
```

Answer Area

Statements	Yes	No
Litware from New York will be displayed at the top of the result set.	☑	○
Fabrikam in Seattle will have value = 2 in the Rank column.	○	☑
Litware in San Francisco will have the same value in the Rank column as Litware in New York.	○	☑

45)

This is a case study. Case studies are not timed separately. You can use as much exam time as you would like to complete each case. However, there may be additional case studies and sections on this exam. You must manage your time to ensure that you are able to complete all questions included on this exam in the time provided.
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To start the case study
To display the first question in this case study, click the Next button. Use the buttons in the left pane to explore the content of the case study before you answer the questions. Clicking these buttons displays information such as business requirements, existing environment, and problem statements. If the case study has an All Information tab, note that the information displayed is identical to the information displayed on the subsequent tabs. When you are ready to answer a question, click the Question button to return to the question.

Overview. Company Overview:
Contoso, Ltd. is an online retail company that wants to modernize its analytics platform by moving to Fabric. The company plans to begin using Fabric for marketing analytics.

Overview. IT Structure:
The company's IT department has a team of data analysts and a team of data engineers that use analytics systems.
The data engineers perform the ingestion, transformation, and loading of data. They prefer to use Python or SQL to transform the data.
The data analysts query data and create semantic models and reports. They are qualified to write queries in Power Query and T-SQL.

Existing Environment:
Fabric
Contoso has an F64 capacity named Cap1. All Fabric users are allowed to create items.
Contoso has two workspaces named WorkspaceA and WorkspaceB that currently use Pro license mode.

Existing Environment:
Source Systems
Contoso has a point of sale (POS) system named POS1 that uses an instance of SQL Server on Azure Virtual Machines in the same Microsoft Entra tenant as Fabric. The host virtual machine is on a private virtual network that has public access blocked. POS1 contains all the sales transactions that were processed on the company's website.
The company has a software as a service (SaaS) online marketing app named MAR1. MAR1 has seven entities. The entities contain data that relates to email open rates and interaction rates, as well as website interactions. The data can be exported from MAR1 by calling REST APIs. Each entity has a different endpoint.
Contoso has been using MAR1 for one year. Data from prior years is stored in Parquet files in an Amazon Simple Storage Service (Amazon S3) bucket. There are 12 files that range in size from 300 MB to 900 MB and relate to email interactions.

Existing Environment:
Product Data
POS1 contains a product list and related data. The data comes from the following three tables:
Products
ProductCategories
ProductSubcategories
In the data, products are related to product subcategories, and subcategories are related to product categories.

Existing Environment:
Azure
Contoso has a Microsoft Entra tenant that has the following mail-enabled security groups:
DataAnalysts: Contains the data analysts
DataEngineers: Contains the data engineers
Contoso has an Azure subscription.
The company has an existing Azure DevOps organization and creates a new project for repositories that relate to Fabric.

Existing Environment:
User Problems
The VP of marketing at Contoso requires analysis on the effectiveness of different types of email content. It typically takes a week to manually compile and analyze the data. Contoso wants to reduce the time to less than one day by using Fabric.
The data engineering team has successfully exported data from MAR1. The team experiences transient connectivity errors, which causes the data exports to fail.

Requirements:
Planned Changes
Contoso plans to create the following two lakehouses:
Lakehouse1: Will store both raw and cleansed data from the sources
Lakehouse2: Will serve data in a dimensional model to users for analytical queries
Additional items will be added to facilitate data ingestion and transformation.
Contoso plans to use Azure Repos for source control in Fabric.

Requirements:
Technical Requirements
The new lakehouses must follow a medallion architecture by using the following three layers: bronze, silver, and gold. There will be extensive data cleansing required to populate the MAR1 data in the silver layer, including deduplication, the handling of missing values, and the standardizing of capitalization.
Each layer must be fully populated before moving on to the next layer. If any step in populating the lakehouses fails, an email must be sent to the data engineers.
Data imports must run simultaneously, when possible.
The use of email data from the Amazon S3 bucket must meet the following requirements:
Minimize egress costs associated with cross-cloud data access.
Prevent saving a copy of the raw data in the lakehouses.
Items that relate to data ingestion must meet the following requirements:
The items must be source controlled alongside other workspace items.
Ingested data must land in the bronze layer of Lakehouse1 in the Delta format.
No changes other than changes to the file formats must be implemented before the data lands in the bronze layer.
Development effort must be minimized and a built-in connection must be used to import the source data.
In the event of a connectivity error, the ingestion processes must attempt the connection again.
Lakehouses, data pipelines, and notebooks must be stored in WorkspaceA. Semantic models, reports, and dataflows must be stored in WorkspaceB.
Once a week, old files that are no longer referenced by a Delta table log must be removed.

Requirements:
Data Transformation
In the POS1 product data, ProductID values are unique. The product dimension in the gold layer must include only active products from product list. Active products are identified by an IsActive value of 1.
Some product categories and subcategories are NOT assigned to any product. They are NOT analytically relevant and must be omitted from the product dimension in the gold layer.

Requirements:

Data Security

Security in Fabric must meet the following requirements:

The data engineers must have read and write access to all the lakehouses, including the underlying files.

The data analysts must only have read access to the Delta tables in the gold layer.

The data analysts must NOT have access to the data in the bronze and silver layers.

The data engineers must be able to commit changes to source control in WorkspaceA.

You need to create the product dimension.

How should you complete the Apache Spark SQL code? To answer, select the appropriate options in the answer area.

NOTE: Each correct selection is worth one point.

Answer Area

```
SELECT ProductID, ProductNumber, ProductName, ModelName, SubCategoryName, CategoryName
FROM ContosoLake.Products p
[dropdown] ContosoLake.ProductSubCategories s ON p.SubCategoryID = s.SubCategoryID
[dropdown] ContosoLake.ProductCategories c ON c.CategoryID = s.CategoryID
WHERE [dropdown]
```

First dropdown options:

- FULL JOIN
- INNER JOIN
- LEFT ANTI JOIN
- LEFT OUTER JOIN
- OUTER JOIN

Second dropdown options:

- FULL JOIN
- INNER JOIN
- LEFT ANTI JOIN
- LEFT OUTER JOIN
- OUTER JOIN

WHERE dropdown options:

- CategoryID = 1;
- CategoryName is not null;
- IsActive = 1;
- IsActive is not null;
- ProductNumber is not null;
- SubCategoryID = 1;
- SubCategoryName is not null;

Explanation:

```
SELECT ProductID, ProductNumber, ProductName, ModelName, SubCategoryName, CategoryName

FROM ContosoLake.Products p

[dropdown] ContosoLake.ProductSubCategories s ON p.SubCategoryID = s.SubCategoryID
```

- FULL JOIN
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[dropdown] ContosoLake.ProductCategories c ON c.CategoryID = s.CategoryID
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- FULL JOIN
- INNER JOIN
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- LEFT OUTER JOIN
- OUTER JOIN

```
WHERE [dropdown]
```

- CategoryID = 1;
- CategoryName is not null;
- IsActive = 1;
- IsActive is not null;
- ProductNumber is not null;
- SubCategoryID = 1;
- SubCategoryName is not null;

46)

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The data engineers must be able to commit changes to source control in WorkspaceA.
You need to populate the MAR1 data in the bronze layer.
Which two types of activities should you include in the pipeline? Each correct answer presents part of the solution.
NOTE: Each correct selection is worth one point.

A ForEach

B Copy data

C WebHook

D Stored procedure

Requirements:
Data Security
Security in Fabric must meet the following requirements:
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You need to populate the MAR1 data in the bronze layer.
Which two types of activities should you include in the pipeline? Each correct answer presents part of the solution.
NOTE: Each correct selection is worth one point.

A. ForEach

B. Copy data

C. WebHook

D. Stored procedure

Answer(s): A,B

47)

You have an Azure Event Hubs data source that contains weather data.

You ingest the data from the data source by using an eventstream named Eventstream1. Eventstream1 uses a lakehouse as the destination.

You need to batch ingest only rows from the data source where the City attribute has a value of Kansas. The filter must be added before the destination. The solution must minimize development effort.

What should you use for the data processor and filtering? To answer, select the appropriate options in the answer area.

NOTE: Each correct selection is worth one point.

Answer Area

Data processor:
- A data pipeline
- A Dataflow Gen2 dataflow
- An eventstream with a custom endpoint
- An eventstream with an external data source

Filtering:
- A Filter activity in a data pipeline
- A filter in a Dataflow Gen2 dataflow
- A KQL statement
- An eventstream processor

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- A data pipeline
- A Dataflow Gen2 dataflow
- An eventstream with a custom endpoint
- An eventstream with an external data source

Filtering:

- A Filter activity in a data pipeline
- A filter in a Dataflow Gen2 dataflow
- A KQL statement
- An eventstream processor

48)

You have a Fabric workspace that contains an eventstream named Eventstream1. Eventstream1 processes data from a thermal sensor by using event stream processing, and then stores the data in a lakehouse.
You need to modify Eventstream1 to include the standard deviation of the temperature.
Which transform operator should you include in the Eventstream1 logic?

A Expand

B Group by

C Union

D Aggregate

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Which transform operator should you include in the Eventstream1 logic?

A Expand

B Group by

C Union

D Aggregate

Answer(s): D

49)

You have an Azure event hub. Each event contains the following fields:
BikepointID
Street
Neighbourhood
Latitude
Longitude
No_Bikes
No_Empty_Docks
You need to ingest the events. The solution must only retain events that have a Neighbourhood value of Chelsea, and then store the retained events in a Fabric lakehouse.
What should you use?

A a KQL queryset

B an eventstream

C a streaming dataset

D Apache Spark Structured Streaming

49)

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BikepointID

Street

Neighbourhood

Latitude

Longitude

No_Bikes

No_Empty_Docks

You need to ingest the events. The solution must only retain events that have a Neighbourhood value of Chelsea, and then store the retained events in a Fabric lakehouse.

What should you use?

- A a KQL queryset
- B an eventstream
- C a streaming dataset
- D Apache Spark Structured Streaming

Answer(s): B

50) You have a Fabric workspace that contains a lakehouse and a notebook named Notebook1. Notebook1 reads data into a DataFrame from a table named Table1 and applies transformation logic. The data from the DataFrame is then written to a new Delta table named Table2 by using a merge operation.
You need to consolidate the underlying Parquet files in Table1.
Which command should you run?

A. VACUUM
B. CACHE
C. OPTIMIZE
D. BROADCAST

50) You have a Fabric workspace that contains a lakehouse and a notebook named Notebook1. Notebook1 reads data into a DataFrame from a table named Table1 and applies transformation logic. The data from the DataFrame is then written to a new Delta table named Table2 by using a merge operation.
You need to consolidate the underlying Parquet files in Table1.
Which command should you run?

A. VACUUM
B. CACHE
C. OPTIMIZE
D. BROADCAST

Correct Answer: C

51) You have a Fabric workspace that contains a semantic model named Model1.
You need to dynamically execute and monitor the refresh progress of Model1.
What should you use?

A. dynamic management views in Azure Data Studio
B. Monitoring hub
C. dynamic management views in Microsoft SQL Server Management Studio
D. a semantic link in a notebook

51) You have a Fabric workspace that contains a semantic model named Model1.
You need to dynamically execute and monitor the refresh progress of Model1.
What should you use?

A. dynamic management views in Azure Data Studio
B. Monitoring hub
C. dynamic management views in Microsoft SQL Server Management Studio
D. a semantic link in a notebook

Correct Answer: D

52) You are building a data loading pattern for Fabric notebook workloads.
You have the following code segment:

```
def loading_pattern_sample(df_source):
    try:
        deltaTable = DeltaTable.forName(spark, target_table)
    except Exception:
        try:
            df_source.write.format('delta').mode('overwrite').saveAsTable(f"{target_table}")
        except Exception as e:
            print(f':Load for table {target_table} failed with error: {str(e)}')
            raise
        return

    try:
        change_detection_columns = [col for col in df_source.columns if col not in candidate_key]

        match_condition = ' AND '.join([f'target.{col} = source.{col}' for col in candidate_key])
        update_condition = ' OR '.join([f'target.{col} != source.{col}' for col in change_detection_columns])

        update_expr = {col: f'source.{col}' for col in df_source.columns}

        merge_operation = deltaTable.alias('target').merge(
            source=df_source.alias('source'),
            condition=match_condition
        ).whenMatchedUpdate(
            condition=update_condition,
            set=update_expr
        ).whenNotMatchedInsertAll()

        merge_operation.execute()
    except Exception as e:
            print(f'Insert operation for table {target_table} failed with error: {str(e)}')
    return
```

For each of the following statements, select Yes if the statement is true. Otherwise, select No.
NOTE: Each correct selection is worth one point.

Answer Area

Statements	Yes	No
The target table will always be overwritten.	○	○
The merge operation will always run.	○	○
The loading pattern supports both full and incremental loading requirements.	○	○

For each of the following statements, select Yes if the statement is true. Otherwise, select No.
NOTE: Each correct selection is worth one point.

Statements	Yes	No
The target table will always be overwritten.	○	○
The merge operation will always run.	○	○
The loading pattern supports both full and incremental loading requirements.	○	○

53) You are building a Fabric notebook named MasterNotebook1 in a workspace. MasterNotebook1 contains the following code.

```
DAG = {
    "activities": [
        {   "name": "execute_notebook_1",
            "path": "notebook_01",
            "timeoutPerCellInSeconds": 600,
            "args": {
                        "input_value": "999"
                },
            "retry": 1,
            "retryIntervalInSeconds": 30
        },
        {   "name": "execute_notebook_2",
            "path": "notebook_02",
            "timeoutPerCellInSeconds": 400,
            "args": {
                        "input_value": "888"
                    },
            "retry": 1,
            "retryIntervalInSeconds": 30
        },
        {   "name": "execute_notebook_3",
            "path": "notebook_03",
            "timeoutPerCellInSeconds": 600,
            "args": {
                        "input_value": "777"
                    },
            "retry": 1,
            "retryIntervalInSeconds": 30
        },
        {   "name": "execute_notebook_3",
            "path": "notebook_03",
            "timeoutPerCellInSeconds": 600,
            "args": {
                        "input_value": "777"
                    },
            "retry": 1,
            "retryIntervalInSeconds": 30
        }
        ],
    "timeoutInSeconds": 43200,
    "concurrency": 0
}

mssparkutils.notebook.runMultiple(DAG,{"displayDAGViaGraphviz": True})
```

You need to ensure that the notebooks are executed in the following sequence:

1. Notebook_03
2. Notebook_01
3. Notebook_02

Which two actions should you perform? Each correct answer presents part of the solution.
NOTE: Each correct selection is worth one point.

A. Move the declaration of Notebook_02 to the bottom of the Directed Acyclic Graph (DAG) definition.

B. Add dependencies to the execution of Notebook_03.

C. Split the Directed Acyclic Graph (DAG) definition into three separate definitions.

D. Add dependencies to the execution of Notebook_02.

E. Change the concurrency to 3.

F. Move the declaration of Notebook_03 to the top of the Directed Acyclic Graph (DAG) definition.

Which two actions should you perform? Each correct answer presents part of the solution.
NOTE: Each correct selection is worth one point.

A. Move the declaration of Notebook_02 to the bottom of the Directed Acyclic Graph (DAG) definition.

B. Add dependencies to the execution of Notebook_03.

C. Split the Directed Acyclic Graph (DAG) definition into three separate definitions.

D. Add dependencies to the execution of Notebook_02.

E. Change the concurrency to 3.

F. Move the declaration of Notebook_03 to the top of the Directed Acyclic Graph (DAG) definition.

Answer: *DF*

54) You need to recommend a Fabric streaming solution that will use the sources shown in the following table.

Name	Message size	Description
Source1	10 MB	Contains semi-structured data that has a bigint column in the messages
Source2	25 MB	Contains structured data that has 19 columns
Source3	5 MB	Contains unstructured data that has images in the messages

The solution must minimize development effort.

What should you include in the recommendation for each source? To answer, select the appropriate options in the answer area.

NOTE: Each correct selection is worth one point.

Answer Area

Source1:
- Apache Spark Structured Streaming
- An eventstream
- A data pipeline
- A streaming dataflow

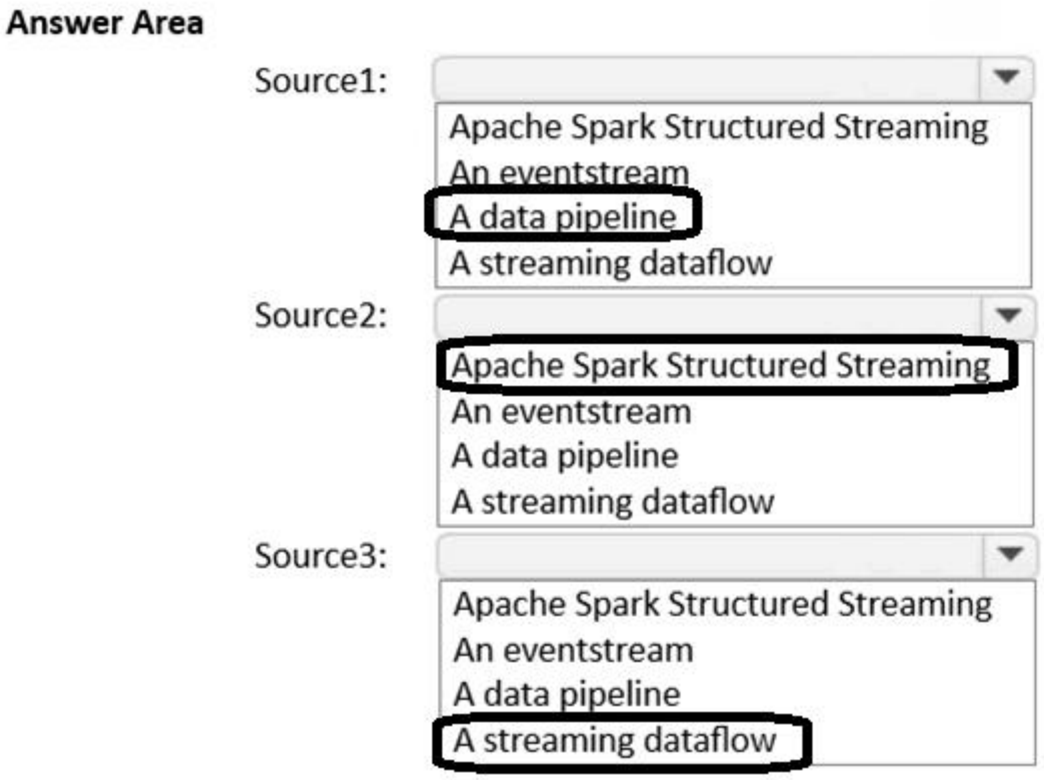
Source2:
- Apache Spark Structured Streaming
- An eventstream
- A data pipeline
- A streaming dataflow

Source3:
- Apache Spark Structured Streaming
- An eventstream
- A data pipeline
- A streaming dataflow

The solution must minimize development effort.

What should you include in the recommendation for each source? To answer, select the appropriate options in the answer area.

NOTE: Each correct selection is worth one point.



55) Note: This question is part of a series of questions that present the same scenario. Each question in the series contains a unique solution that might meet the stated goals. Some question sets might have more than one correct solution, while others might not have a correct solution.
After you answer a question in this section, you will NOT be able to return to it. As a result, these questions will not appear in the review screen.
You have an Azure key vault named KeyVault1 that contains secrets.
You have a Fabric workspace named Workspace1. Workspace contains a notebook named Notebook1 that performs the following tasks:

- Loads stage data to the target tables in a lakehouse
- Triggers the refresh of a semantic model

You plan to add functionality to Notebook1 that will use the Fabric API to monitor the semantic model refreshes.
You need to retrieve the registered application ID and secret from KeyVault1 to generate the authentication token.
Solution: You use the following code segment:
Use notebookutils.credentials.putSecret and specify the key vault URL and key vault secret.
Does this meet the goal?

55) Note: This question is part of a series of questions that present the same scenario. Each question in the series contains a unique solution that might meet the stated goals. Some question sets might have more than one correct solution, while others might not have a correct solution.
After you answer a question in this section, you will NOT be able to return to it. As a result, these questions will not appear in the review screen.
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Solution: You use the following code segment:
Use notebookutils.credentials.putSecret and specify the key vault URL and key vault secret.
Does this meet the goal?

A. Yes
B. No

55) Note: This question is part of a series of questions that present the same scenario. Each question in the series contains a unique solution that might meet the stated goals. Some question sets might have more than one correct solution, while others might not have a correct solution.
After you answer a question in this section, you will NOT be able to return to it. As a result, these questions will not appear in the review screen.
You have an Azure key vault named KeyVault1 that contains secrets.
You have a Fabric workspace named Workspace1. Workspace contains a notebook named Notebook1 that performs the following tasks:

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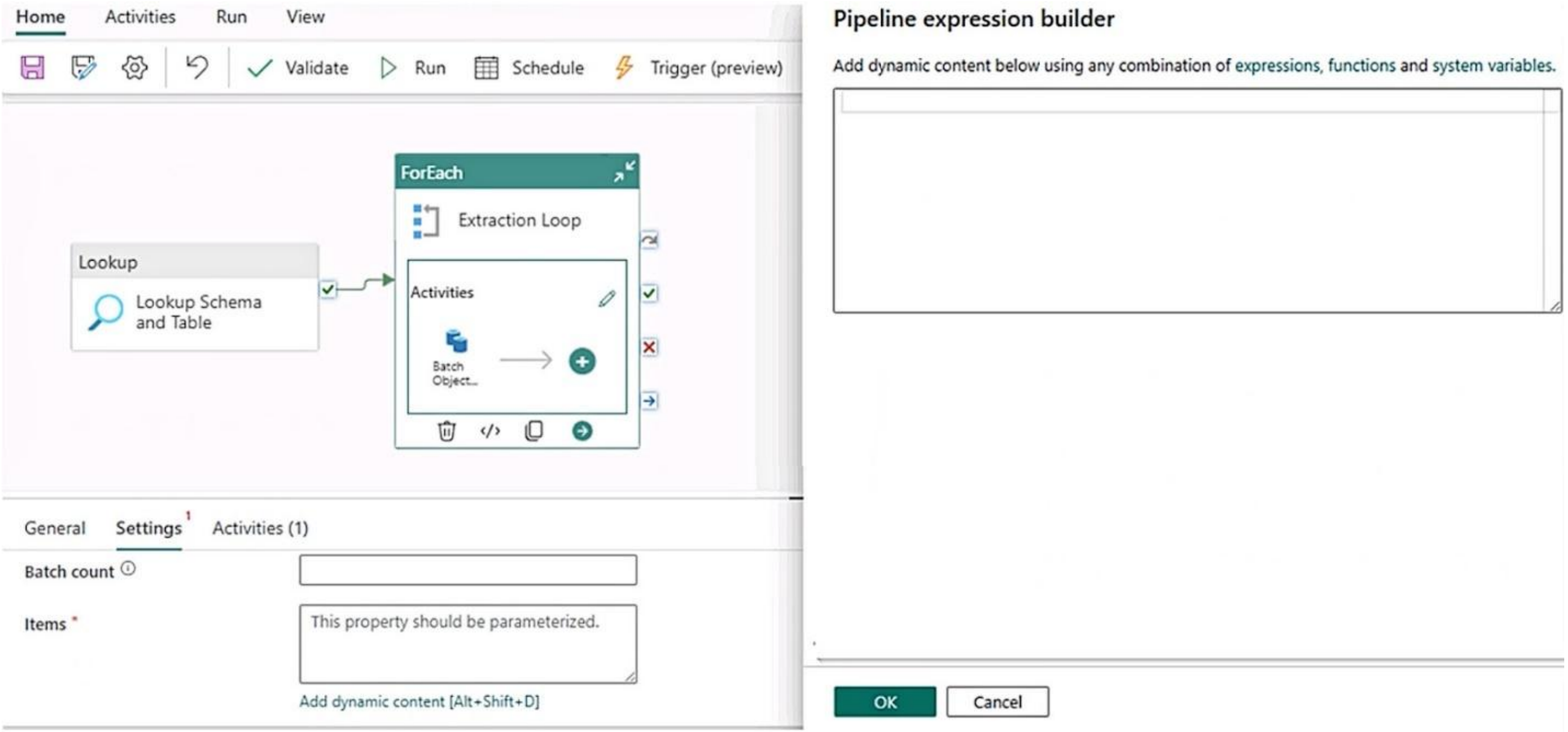
You need to retrieve the registered application ID and secret from KeyVault1 to generate the authentication token.
Solution: You use the following code segment:
Use notebookutils.credentials.putSecret and specify the key vault URL and key vault secret.
Does this meet the goal?

A. Yes
B. No

Answer: *B*

56) You are building a data orchestration pattern by using a Fabric data pipeline named Dynamic Data Copy as shown in the exhibit. (Click the Exhibit tab.)



Dynamic Data Copy does NOT use parametrization.

You need to configure the ForEach activity to receive the list of tables to be copied.

How should you complete the pipeline expression? To answer, select the appropriate options in the answer area.

NOTE: Each correct selection is worth one point.

Answer Area

@activity(' [dropdown 1] ').[dropdown 2]

Dropdown 1	Dropdown 2
Batch Object Copy	output
Dynamic Data Copy	output.count
Extraction Loop	output.pipelineReturnValue
Lookup Schema and Table	output.value

Dynamic Data Copy does NOT use parametrization.

You need to configure the ForEach activity to receive the list of tables to be copied.

How should you complete the pipeline expression? To answer, select the appropriate options in the answer area.

NOTE: Each correct selection is worth one point.

Answer Area

@activity(' [dropdown] ').[dropdown]

First dropdown	Second dropdown
Batch Object Copy	output
Dynamic Data Copy	output.count
Extraction Loop	output.pipelineReturnValue
Lookup Schema and Table	output.value

57) You need to ensure that usage of the data in the Amazon S3 bucket meets the technical requirements. (Case Study 4)
What should you do?

A. Create a shortcut and ensure that caching is enabled for the workspace.
B. Create a workspace identity and use the identity in a data pipeline.
C. Create a shortcut and ensure that caching is disabled for the workspace.
D. Create a workspace identity and enable high concurrency for the notebooks.

57) You need to ensure that usage of the data in the Amazon S3 bucket meets the technical requirements. (Case Study 4)
What should you do?

A. Create a shortcut and ensure that caching is enabled for the workspace.
B. Create a workspace identity and use the identity in a data pipeline.
C. Create a shortcut and ensure that caching is disabled for the workspace.
D. Create a workspace identity and enable high concurrency for the notebooks.

Correct Answer: C

To ensure that the usage of the data in the Amazon S3 bucket meets the technical requirements, we must address two key points:
Minimize egress costs associated with cross-cloud data access: Using a shortcut ensures that Fabric does not replicate the data from the S3 bucket into the lakehouse but rather provides direct access to the data in its original location. This minimizes cross-cloud data transfer and avoids additional egress costs.
Prevent saving a copy of the raw data in the lakehouses: Disabling caching ensures that the raw data is not copied or persisted in the Fabric workspace. The data is accessed on- demand directly from the Amazon S3 bucket.

58) You are developing a data pipeline named Pipeline1.
You need to add a Copy data activity that will copy data from a Snowflake data source to a Fabric warehouse.
What should you configure?

A. Degree of copy parallelism
B. Fault tolerance
C. Enable staging
D. Enable logging

58) You are developing a data pipeline named Pipeline1.
You need to add a Copy data activity that will copy data from a Snowflake data source to a Fabric warehouse.
What should you configure?

A. Degree of copy parallelism
B. Fault tolerance
C. Enable staging
D. Enable logging

Correct Answer: C

59) You have a Fabric workspace named Workspace1 that contains a lakehouse named Lakehouse1. Workspace1 contains the following items:

• A Dataflow Gen2 dataflow that copies data from an on-premises Microsoft SQL Server database to Lakehouse1
• A notebook that transforms files and loads the data to Lakehouse1
• A data pipeline that loads a CSV file to Lakehouse1

You need to develop an orchestration solution in Fabric that will load each item one after the other. The solution must be scheduled to run every 15 minutes.

Which type of item should you use?

A. notebook
B. warehouse
C. Dataflow Gen2 dataflow
D. data pipeline

59) You have a Fabric workspace named Workspace1 that contains a lakehouse named Lakehouse1. Workspace1 contains the following items:

• A Dataflow Gen2 dataflow that copies data from an on-premises Microsoft SQL Server database to Lakehouse1
• A notebook that transforms files and loads the data to Lakehouse1
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Which type of item should you use?

A. notebook
B. warehouse
C. Dataflow Gen2 dataflow
D. data pipeline

Correct Answer: D

60) (Case Study 4)

You need to populate the MAR1 data in the bronze layer.
Which two types of activities should you include in the pipeline? Each correct answer presents part of the solution.
NOTE: Each correct selection is worth one point.

A. ForEach
B. Copy data
C. WebHook
D. Stored procedure

60) (Case Study 4)

You need to populate the MAR1 data in the bronze layer.
Which two types of activities should you include in the pipeline? Each correct answer presents part of the solution.
NOTE: Each correct selection is worth one point.

A. ForEach
B. Copy data
C. WebHook
D. Stored procedure

Correct Answer: AB

Explanation:
MAR1 has seven entities, each accessible via a different API endpoint. A ForEach activity is required to iterate over these endpoints to fetch data from each one. It enables dynamic execution of API calls for each entity.
The Copy data activity is the primary mechanism to extract data from REST APIs and load it into the bronze layer in Delta format. It supports native connectors for REST APIs and Delta, minimizing development effort.

61) You are implementing a medallion architecture in a Fabric lakehouse.

You plan to create a dimension table that will contain the following columns:

- ID
- CustomerCode
- CustomerName
- CustomerAddress
- CustomerLocation
- ValidFrom
- ValidTo

You need to ensure that the table supports the analysis of historical sales data by customer location at the time of each sale.

Which type of slowly changing dimension (SCD) should you use?

A. Type 2
B. Type 0
C. Type 1
D. Type 3

61) You are implementing a medallion architecture in a Fabric lakehouse.

You plan to create a dimension table that will contain the following columns:

• ID
• CustomerCode
• CustomerName
• CustomerAddress
• CustomerLocation
• ValidFrom
• ValidTo

You need to ensure that the table supports the analysis of historical sales data by customer location at the time of each sale.

Which type of slowly changing dimension (SCD) should you use?

A. Type 2
B. Type 0
C. Type 1
D. Type 3

Correct Answer: A

62) HOTSPOT -

You have three users named User1, User2, and User3.

You have the Fabric workspaces shown in the following table.

Name	Workspace admin
Workspace1	User1
Workspace2	User2

You have a security group named Group1 that contains User1 and User3.

The Fabric admin creates the domains shown in the following table.

Name	Domain admin
Domain1	User1
Domain2	User2

User1 creates a new workspace named Workspace3.

You add Group1 to the default domain of Domain1.

For each of the following statements, select Yes if the statement is true. Otherwise, select No.

NOTE: Each correct selection is worth one point.

Answer Area

Statements	Yes	No
User3 has Viewer role access to Workspace3.	○	○
User3 has Domain contributor access to Domain1.	○	○
User2 has Contributor role access to Workspace3.	○	○

Statements	Yes	No
User3 has Viewer role access to Workspace3.	☒	○
User3 has Domain contributor access to Domain1.	○	☒
User2 has Contributor role access to Workspace3.	○	☒

63) You have a Fabric workspace named Workspace1 that contains the items shown in the following table.

Name	Type
Notebook1	Notebook
Notebook2	Notebook
Lakehouse1	Lakehouse
Pipeline1	Data pipeline
Model1	Semantic model

For Model1, the Keep your Direct Lake data up to date option is disabled.
You need to configure the execution of the items to meet the following requirements:
Notebook1 must execute every weekday at 8:00 AM.
Notebook2 must execute when a file is saved to an Azure Blob Storage container.
Model1 must refresh when Notebook1 has executed successfully.
How should you orchestrate each item? To answer, select the appropriate options in the answer area.
NOTE: Each correct selection is worth one point.

Answer Area

Notebook1:

- Add Notebook1 to an Apache Spark job definition.
- Add Notebook1 to Pipeline1.
- From Real-Time hub, configure the execution of Notebook1.

Notebook2:

- Add Notebook2 to an Apache Spark job definition.
- Add Notebook2 to Pipeline1.
- From Real-Time hub, configure the execution of Notebook2.

Pipeline1:

- Add Pipeline1 to an Apache Spark job definition.
- Configure the execution of Pipeline1 by using a schedule.
- From Real-Time hub, configure the execution of Pipeline1.

Model1:

- Add Model1 to Pipeline1.
- From Real-Time hub, configure Model1 to refresh.
- Set Keep your Direct Lake data up to date to On.

Notebook1:

- Add Notebook1 to an Apache Spark job definition.
- Add Notebook1 to Pipeline1.
- From Real-Time hub, configure the execution of Notebook1.

Notebook2:

- Add Notebook2 to an Apache Spark job definition.
- Add Notebook2 to Pipeline1.
- From Real-Time hub, configure the execution of Notebook2.

Pipeline1:

- Add Pipeline1 to an Apache Spark job definition.
- Configure the execution of Pipeline1 by using a schedule.
- From Real-Time hub, configure the execution of Pipeline1.

Model1:

- Add Model1 to Pipeline1.
- From Real-Time hub, configure Model1 to refresh.
- Set Keep your Direct Lake data up to date to On.

64) You have an Azure event hub. Each event contains the following fields:

BikepointID -
Street -
Neighbourhood -
Latitude -
Longitude -
No_Bikes -
No_Empty_Docks -
You need to ingest the events. The solution must only retain events that have a Neighbourhood value of Chelsea, and then store the retained events in a Fabric lakehouse.
What should you use?

A. a KQL queryset
B. an eventstream
C. a streaming dataset
D. Apache Spark Structured Streaming

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BikepointID -
Street -
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You need to ingest the events. The solution must only retain events that have a Neighbourhood value of Chelsea, and then store the retained events in a Fabric lakehouse.
What should you use?

A. a KQL queryset
B. an eventstream
C. a streaming dataset
D. Apache Spark Structured Streaming

Answer: *B*

65) You have two Fabric notebooks named Load_Salesperson and Load_Orders that read data from Parquet files in a lakehouse. Load_Salesperson writes to a Delta table named dim_salesperson. Load_Orders writes to a Delta table named fact_orders and is dependent on the successful execution of Load_Salesperson.

You need to implement a pattern to dynamically execute Load_Salesperson and Load_Orders in the appropriate order by using a notebook.

How should you complete the code? To answer, drag the appropriate values the correct targets. Each value may be used once, more than once, or not at all. You may need to drag the split bar between panes or scroll to view content.

NOTE: Each correct selection is worth one point.

Values

- activities
- broadcast
- dependencies
- execute
- notebooks
- runMultiple

Answer Area

```
DAG = {
    "          ": [
        {
            "name": "Load_Salesperson",
            "path": "Load_Salesperson",
            "timeoutPerCellInSeconds": 300,
        },
        {
            "name": "Load_Orders",
            "path": "Load_Orders",
            "timeoutPerCellInSeconds": 600,
            "          ": ["Load_Salesperson"]
        }
    ],
    "timeoutInSeconds": 43200
}
mssparkutils.notebook.          (DAG)
```

Answer Area

```
DAG = {
    " activities ": [
        {
            "name": "Load_Salesperson",
            "path": "Load_Salesperson",
            "timeoutPerCellInSeconds": 300,
        },
        {
            "name": "Load_Orders",
            "path": "Load_Orders",
            "timeoutPerCellInSeconds": 600,
            " dependencies ": ["Load_Salesperson"]
        }
    ],
    "timeoutInSeconds": 43200
}
mssparkutils.notebook. runMultiple (DAG)
```

66) You plan to process the following three datasets by using Fabric:

Dataset1: This dataset will be added to Fabric and will have a unique primary key between the source and the destination. The unique primary key will be an integer and will start from 1 and have an increment of 1.
Dataset2: This dataset contains semi-structured data that uses bulk data transfer. The dataset must be handled in one process between the source and the destination. The data transformation process will include the use of custom visuals to understand and work with the dataset in development mode.
Dataset3: This dataset is in a lakehouse. The data will be bulk loaded. The data transformation process will include row-based windowing functions during the loading process.

You need to identify which type of item to use for the datasets. The solution must minimize development effort and use built-in functionality, when possible.

What should you identify for each dataset? To answer, select the appropriate options in the answer area.

NOTE: Each correct selection is worth one point.

Answer Area

Dataset1:

- Dataflow Gen2 dataflow
- A notebook
- A T-SQL statement

Dataset2:

- Dataflow Gen2 dataflow
- A notebook
- A T-SQL statement

Dataset3:

- Dataflow Gen2 dataflow
- A KQL queryset
- A T-SQL statement

Answer Area

Dataset1:
- Dataflow Gen2 dataflow
- A notebook
- A T-SQL statement

Dataset2:
- Dataflow Gen2 dataflow
- A notebook
- A T-SQL statement

Dataset3:
- Dataflow Gen2 dataflow
- A KQL queryset
- A T-SQL statement

67) You have a Fabric workspace named Workspace1 that contains a warehouse named Warehouse2.

A team of data analysts has Viewer role access to Workspace1.

You create a table by running the following statement.

```
CREATE TABLE [warehouse2].[dbo].[CreditCard]
(
  CreditCard varchar(20) NOT NULL
  ,CreditCardType varchar(10) NOT NULL)
GO
```

You need to ensure that the team can view only the first two characters and the last four characters of the CreditCard attribute.

How should you complete the statement? To answer, select the appropriate options in the answer area.

NOTE: Each correct selection is worth one point.

Answer Area

[▼] TABLE dbo.CreditCard

- ALTER
- CREATE
- DEFAULT
- DROP
- EMAIL
- PARTIAL
- REPLACE
- UPDATE

[▼] COLUMN [CreditCard]

- ALTER
- CREATE
- DEFAULT
- DROP
- EMAIL
- PARTIAL
- REPLACE
- UPDATE

ADD MASKED

WITH (FUNCTION = ' [▼] (2,"XXXXXXXXXX",4)')

- ALTER
- CREATE
- DEFAULT
- DROP
- EMAIL
- PARTIAL
- REPLACE
- UPDATE

Answer Area

[ALTER / CREATE / DEFAULT / DROP / EMAIL / PARTIAL / REPLACE / UPDATE] TABLE dbo.CreditCard

- ALTER
- CREATE
- DEFAULT
- DROP
- EMAIL
- PARTIAL
- REPLACE
- UPDATE

[ALTER / CREATE / DEFAULT / DROP / EMAIL / PARTIAL / REPLACE / UPDATE] COLUMN [CreditCard]

- ALTER
- CREATE
- DEFAULT
- DROP
- EMAIL
- PARTIAL
- REPLACE
- UPDATE

ADD MASKED

WITH (FUNCTION = ' [ALTER / CREATE / DEFAULT / DROP / EMAIL / PARTIAL / REPLACE / UPDATE] (2,"XXXXXXXXXX",4)')

- ALTER
- CREATE
- DEFAULT
- DROP
- EMAIL
- PARTIAL
- REPLACE
- UPDATE

68) Case Study. This is a case study. Case studies are not timed separately. You can use as much exam time as you would like to complete each case. However, there may be additional case studies and sections on this exam. You must manage your time to ensure that you are able to complete all questions included on this exam in the time provided.

To answer the questions included in a case study, you will need to reference information that is provided in the case study. Case studies might contain exhibits and other resources that provide more information about the scenario that is described in the case study. Each question is independent of the other questions in this case study.

At the end of this case study, a review screen will appear. This screen allows you to review your answers and to make changes before you move to the next section of the exam. After you begin a new section, you cannot return to this section.

To start the case study

-

To display the first question in this case study, click the Next button. Use the buttons in the left pane to explore the content of the case study before you answer the questions. Clicking these buttons displays information such as business requirements, existing environment, and problem statements. If the case study has an All Information tab, note that the information displayed is identical to the information displayed on the subsequent tabs. When you are ready to answer a question, click the Question button to return to the question.

Overview. Company Overview

-

Contoso, Ltd. is an online retail company that wants to modernize its analytics platform by moving to Fabric. The company plans to begin using Fabric for marketing analytics.

Overview. IT Structure

-

The company's IT department has a team of data analysts and a team of data engineers that use analytics systems.

The data engineers perform the ingestion, transformation, and loading of data. They prefer to use Python or SQL to transform the data.

The data analysts query data and create semantic models and reports. They are qualified to write queries in Power Query and T-SQL.

Existing Environment. Fabric

-

Contoso has an F64 capacity named Cap1. All Fabric users are allowed to create items.
Contoso has two workspaces named WorkspaceA and WorkspaceB that currently use Pro license mode.
Existing Environment. Source Systems
Contoso has a point of sale (POS) system named POS1 that uses an instance of SQL Server on Azure Virtual Machines in the same Microsoft Entra tenant as Fabric. The host virtual machine is on a private virtual network that has public access blocked. POS1 contains all the sales transactions that were processed on the company's website.

The company has a software as a service (SaaS) online marketing app named MAR1. MAR1 has seven entities. The entities contain data that relates to email open rates and interaction rates, as well as website interactions. The data can be exported from MAR1 by calling REST APIs. Each entity has a different endpoint.

Contoso has been using MAR1 for one year. Data from prior years is stored in Parquet files in an Amazon Simple Storage Service (Amazon S3) bucket. There are 12 files that range in size from 300 MB to 900 MB and relate to email interactions.
Existing Environment. Product Data

POS1 contains a product list and related data. The data comes from the following three tables:
• Products
• ProductCategories
• ProductSubcategories

In the data, products are related to product subcategories, and subcategories are related to product categories.

Existing Environment. Azure
Contoso has a Microsoft Entra tenant that has the following mail-enabled security groups:

• DataAnalysts: Contains the data analysts
• DataEngineers: Contains the data engineers

Contoso has an Azure subscription.

The company has an existing Azure DevOps organization and creates a new project for repositories that relate to Fabric.

Existing Environment. User Problems

The VP of marketing at Contoso requires analysis on the effectiveness of different types of email content. It typically takes a week to manually compile and analyze the data. Contoso wants to reduce the time to less than one day by using Fabric.

The data engineering team has successfully exported data from MAR1. The team experiences transient connectivity errors, which causes the data exports to fail.

Requirements. Planned Changes

Contoso plans to create the following two lakehouses:

- Lakehouse1: Will store both raw and cleansed data from the sources
- Lakehouse2: Will serve data in a dimensional model to users for analytical queries

Additional items will be added to facilitate data ingestion and transformation.

Contoso plans to use Azure Repos for source control in Fabric.

Requirements. Technical Requirements

The new lakehouses must follow a medallion architecture by using the following three layers: bronze, silver, and gold. There will be extensive data cleansing required to populate the MAR1 data in the silver layer, including deduplication, the handling of missing values, and the standardizing of capitalization.

Each layer must be fully populated before moving on to the next layer. If any step in populating the lakehouses fails, an email must be sent to the data engineers.

Data imports must run simultaneously, when possible.

The use of email data from the Amazon S3 bucket must meet the following requirements:

• Minimize egress costs associated with cross-cloud data access.
• Prevent saving a copy of the raw data in the lakehouses.

Items that relate to data ingestion must meet the following requirements:

• The items must be source controlled alongside other workspace items.
• Ingested data must land in the bronze layer of Lakehouse1 in the Delta format.
• No changes other than changes to the file formats must be implemented before the data lands in the bronze layer.
• Development effort must be minimized and a built-in connection must be used to import the source data.
• In the event of a connectivity error, the ingestion processes must attempt the connection again.

Lakehouses, data pipelines, and notebooks must be stored in WorkspaceA. Semantic models, reports, and dataflows must be stored in WorkspaceB.

Once a week, old files that are no longer referenced by a Delta table log must be removed.

Requirements. Data Transformation

In the POS1 product data, ProductID values are unique. The product dimension in the gold layer must include only active products from product list. Active products are identified by an IsActive value of 1.

Some product categories and subcategories are NOT assigned to any product. They are NOT analytically relevant and must be omitted from the product dimension in the gold layer.

Requirements. Data Security

-

Security in Fabric must meet the following requirements:

• The data engineers must have read and write access to all the lakehouses, including the underlying files.
• The data analysts must only have read access to the Delta tables in the gold layer.
• The data analysts must NOT have access to the data in the bronze and silver layers.
• The data engineers must be able to commit changes to source control in WorkspaceA.

You need to ensure that the data engineers are notified if any step in populating the lakehouses fails. The solution must meet the technical requirements and minimize development effort.

What should you use? To answer, select the appropriate options in the answer area.
NOTE: Each correct selection is worth one point.

Answer Area

To identify the failure:

- A Fail activity
- An If condition activity
- An On failure dependency condition
- An On completion dependency condition

To send the notification:

- A Teams activity
- An Invoke pipeline activity
- An Office365Outlook activity

Answer Area

To identify the failure:

- A Fail activity
- An If condition activity
- An On failure dependency condition
- An On completion dependency condition

To send the notification:

- A Teams activity
- An Invoke pipeline activity
- An Office365Outlook activity

69) You have a Fabric workspace that contains a lakehouse named Lakehouse1.
In an external data source, you have data files that are 500 GB each. A new file is added every day.
You need to ingest the data into Lakehouse1 without applying any transformations. The solution must meet the following requirements
Trigger the process when a new file is added.
Provide the highest throughput.
Which type of item should you use to ingest the data?

A. Eventstream

B. Dataflow Gen2

C. Streaming dataset

D. Data pipeline

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Which type of item should you use to ingest the data?

A. Eventstream

B. Dataflow Gen2

C. Streaming dataset

D. Data pipeline

Answer: *D*

70) You have a Fabric workspace that contains a lakehouse named Lakehouse1.

In an external data source, you have data files that are 500 GB each. A new file is added every day.

You need to ingest the data into Lakehouse1 without applying any transformations. The solution must meet the following requirements:

• Trigger the process when a new file is added.
• Provide the highest throughput.

Which type of item should you use to ingest the data?

A. KQL queryset

B. Streaming dataset

C. Notebook

D. Dataflow Gen2

70) You have a Fabric workspace that contains a lakehouse named Lakehouse1.

In an external data source, you have data files that are 500 GB each. A new file is added every day.

You need to ingest the data into Lakehouse1 without applying any transformations. The solution must meet the following requirements:

• Trigger the process when a new file is added.
• Provide the highest throughput.

Which type of item should you use to ingest the data?

A. KQL queryset

B. Streaming dataset

C. Notebook

D. Dataflow Gen2

Answer: *D*

71) You need to recommend a method to populate the POS1 data to the lakehouse medallion layers. (Case Study 3)
What should you recommend for each layer? To answer, select the appropriate options in the answer area.

NOTE: Each correct selection is worth one point.

Answer Area

Bronze layer:
- A Dataflow Gen2 dataflow
- A notebook
- A pipeline Copy activity
- A pipeline stored procedure

Silver layer:
- A Dataflow Gen2 dataflow
- A notebook
- A pipeline Copy activity
- A pipeline stored procedure

71) You need to recommend a method to populate the POS1 data to the lakehouse medallion layers. (Case Study 3)
What should you recommend for each layer? To answer, select the appropriate options in the answer area.

NOTE: Each correct selection is worth one point.

Bronze layer:

- A Dataflow Gen2 dataflow
- A notebook
- A pipeline Copy activity
- A pipeline stored procedure

Silver layer:

- A Dataflow Gen2 dataflow
- A notebook
- A pipeline Copy activity
- A pipeline stored procedure

72) You need to troubleshoot the ad-hoc query issue.
How should you complete the statement? To answer, select the appropriate options in the answer area. (Case Study Q.1)
NOTE: Each correct selection is worth one point.

Answer Area

```
SELECT last_run_start_time, last_run_command
FROM
```

- queryinsights.exec_requests_history
- queryinsights.exec_sessions_history
- queryinsights.frequently_run_queries
- queryinsights.long_running_queries

```
WHERE last_run_total_elapsed_time_ms > 7200000
AND
```

- max_run_total_elapsed_time_ms > 7200000
- median_total_elapsed_time_ms > 7200000
- number_of_canceled_runs > 1
- number_of_failed_runs > 1
- number_of_runs > 1

72) You need to troubleshoot the ad-hoc query issue.
How should you complete the statement? To answer, select the appropriate options in the answer area. (Case Study Q.1)
NOTE: Each correct selection is worth one point.

```
SELECT last_run_start_time, last_run_command
FROM [dropdown]
WHERE last_run_total_elapsed_time_ms > 7200000
AND [dropdown]
```

FROM options:
- queryinsights.exec_requests_history
- queryinsights.exec_sessions_history
- queryinsights.frequently_run_queries
- queryinsights.long_running_queries

AND options:
- max_run_total_elapsed_time_ms > 7200000
- median_total_elapsed_time_ms > 7200000
- number_of_canceled_runs > 1
- number_of_failed_runs > 1
- number_of_runs > 1

73) You have a KQL database that contains a table named Readings.

You need to build a KQL query to compare the MeterReading value of each row to the previous row base on the Timestamp value.

A sample of the expected output is shown in the following table.

City	Area	MeterReading	Timestamp	PrevMeterReading	PrevTimestamp
Kansas	Area1	1500	2024-07-30 10:00:00		
Kansas	Area2	1520	2024-07-30 11:00:00	1500	2024-07-30 10:00:00
Kansas	Area1	1550	2024-07-30 12:00:00	1520	2024-07-30 11:00:00
Kansas	Area2	1580	2024-07-30 13:00:00	1550	2024-07-30 12:00:00

How should you complete the query? To answer, drag the appropriate values the correct targets. Each value may be used once, more than once, or not at all. You may need to drag the split bar between panes or scroll to view content.

NOTE: Each correct selection is worth one point.

Values

- evaluate
- extend
- lookup
- project
- sort
- summarize
- take

Answer Area

```
Readings
| filter City == "Kansas"
[          ] by Timestamp
[          ] PrevMeterReading = prev(MeterReading),
      PrevTimestamp = prev(Timestamp)
[          ] City, Area, MeterReading, Timestamp, PrevMeterReading, PrevTimeStamp
```

How should you complete the query? To answer, drag the appropriate values the correct targets. Each value may be used once, more than once, or not at all. You may need to drag the split bar between panes or scroll to view content.

NOTE: Each correct selection is worth one point.

Values

- evaluate
- extend
- lookup
- project
- sort
- summarize
- take

Answer Area

```
Readings
| filter City == "Kansas"
[sort] by Timestamp
[extend] PrevMeterReading = prev(MeterReading),
      PrevTimestamp = prev(Timestamp)
[project] City, Area, MeterReading, Timestamp, PrevMeterReading, PrevTimeStamp
```

74) You have a Fabric workspace that contains an eventhouse named Eventhouse1.

In Eventhouse1, you plan to create a table named DeviceStreamData in a KQL database. The table will contain data based on the following sample.

Timestamp	DeviceId	StreamData
2024-05-18 12:45:17.16524	81416f30-60a2-4e75-9b19-2a84ea059735	[{ "index": 0, "eventid": "719afca0-be30-4559-bb5e-59feade642f6", "isActive": false, "latitude": 5.390012, "longitude": -40.100235, "tags": ["tempor"] }]
2024-05-18 12:45:21.76423	bb664e1e-02aa-4e17-8c8a-116cd4458d52	[{ "index": 0, "eventid": "782222b2-fbcb-43c0-82d6-ecd49a99dbf5", "isActive": true, "latitude": -56.153786, "longitude": 130.870907, "tags": ["adipisicing"] }]
2024-05-18 12:45:23.98642	717bfe7d-0e5d-498f-9f21-e60aaf258056	[{ "index": 0, "eventid": "d5730286-0da4-41f8-8e59-f75e209310a9", "isActive": true, "latitude": -21.39289, "longitude": 123.959442, "tags": ["ad"] }]
2024-05-18 12:45:25.39523	1a390e71-4faf-4df5-a479-2238d84001f7	[{ "index": 0, "eventid": "9572e141-8692-4d16-89e8-002f9b7e22b6", "isActive": true, "latitude": -84.926214, "longitude": -11.499007, "tags": ["ex"] }]
2024-05-18 12:45:27.43343	2f0ba7d0-6dff-4081-bf7f-d0d39d4c3260	[{ "index": 0, "eventid": "08a42b87-ce84-4bb2-99f0-4fb5c75ff63f", "isActive": true, "latitude": -49.909339, "longitude": -177.505775, "tags": ["laboris"] }]

You need to use a KQL query to develop the solution for Eventhouse1.

Which three code segments should you run in sequence? To answer, move the appropriate code segments from the list of code segments to the answer area and arrange them in the correct order.

Code Segments

```
StreamData:long)
```

```
.create table EventStreamData (
```

```
StreamData:dynamic )
```

```
.create function EventStreamData (
```

```
TimeStamp:datetime, DeviceId:string
```

Answer Area

1.

2.

3.

You need to use a KQL query to develop the solution for Eventhouse1.

Which three code segments should you run in sequence? To answer, move the appropriate code segments from the list of code segments to the answer area and arrange them in the correct order.

Code Segments

```
StreamData:long)
```

```
.create table EventStreamData (
```

```
StreamData:dynamic )
```

```
.create function EventStreamData (
```

```
TimeStamp:datetime, DeviceId:string
```

Answer Area

1. `.create table EventStreamData (`
2. `TimeStamp:datetime, DeviceId:string`
3. `StreamData:dynamic )`

75) You need to create the product dimension.
How should you complete the Apache Spark SQL code? To answer, select the appropriate options in the answer area. (Case Study 4)
NOTE: Each correct selection is worth one point.

Answer Area

```
SELECT ProductID, ProductNumber, ProductName, ModelName, SubCategoryName, CategoryName

FROM ContosoLake.Products p

[▼] ContosoLake.ProductSubCategories s ON p.SubCategoryID = s.SubCategoryID
```

- FULL JOIN
- INNER JOIN
- LEFT ANTI JOIN
- LEFT OUTER JOIN
- OUTER JOIN

```
[▼] ContosoLake.ProductCategories c ON c.CategoryID = s.CategoryID
```

- FULL JOIN
- INNER JOIN
- LEFT ANTI JOIN
- LEFT OUTER JOIN
- OUTER JOIN

```
WHERE [▼]
```

- CategoryID = 1;
- CategoryName is not null;
- IsActive = 1;
- IsActive is not null;
- ProductNumber is not null;
- SubCategoryID = 1;
- SubCategoryName is not null;

75) You need to create the product dimension.
How should you complete the Apache Spark SQL code? To answer, select the appropriate options in the answer area. (Case Study 4)
NOTE: Each correct selection is worth one point.

```
SELECT ProductID, ProductNumber, ProductName, ModelName, SubCategoryName, CategoryName

FROM ContosoLake.Products p

[dropdown 1] ContosoLake.ProductSubCategories s ON p.SubCategoryID = s.SubCategoryID

[dropdown 2] ContosoLake.ProductCategories c ON c.CategoryID = s.CategoryID

WHERE [dropdown 3]
```

Dropdown 1:
- FULL JOIN
- INNER JOIN
- LEFT ANTI JOIN
- LEFT OUTER JOIN
- OUTER JOIN

Dropdown 2:
- FULL JOIN
- INNER JOIN
- LEFT ANTI JOIN
- LEFT OUTER JOIN
- OUTER JOIN

Dropdown 3:
- CategoryID = 1;
- CategoryName is not null;
- IsActive = 1;
- IsActive is not null;
- ProductNumber is not null;
- SubCategoryID = 1;
- SubCategoryName is not null;

76) You have an Azure SQL database named DB1.
In a Fabric workspace, you deploy an eventstream named EventStreamDBI to stream record changes from DB1 into a lakehouse.
You discover that events are NOT being propagated to EventStreamDBI.
You need to ensure that the events are propagated to EventStreamDBI. What should you do?

A. Create a read-only replica of DB1.
B. Create an Azure Stream Analytics job.
C. Enable Extended Events for DB1.
D. Enable change data capture (CDC) for DB1.

76) You have an Azure SQL database named DB1.
In a Fabric workspace, you deploy an eventstream named EventStreamDBI to stream record changes from DB1 into a lakehouse.
You discover that events are NOT being propagated to EventStreamDBI.
You need to ensure that the events are propagated to EventStreamDBI. What should you do?

A.Create a read-only replica of DB1.
B.Create an Azure Stream Analytics job.
C.Enable Extended Events for DB1.
D.Enable change data capture (CDC) for DB1.

Answer: D

77) You have a Fabric workspace that contains a warehouse named Warehouse1.
While monitoring Warehouse1, you discover that query performance has degraded during the last 60 minutes.

You need to isolate all the queries that were run during the last 60 minutes. The results must include the username of the users that submitted the queries and the query statements.

What should you use?

A. the Microsoft Fabric Capacity Metrics app
B. views from the queryinsights schema
C. Query activity
D. the sys.dm_exec_requests dynamic management view

77) You have a Fabric workspace that contains a warehouse named Warehouse1.
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What should you use?

A. the Microsoft Fabric Capacity Metrics app
B. views from the queryinsights schema
C. Query activity
D. the sys.dm_exec_requests dynamic management view

Answer: B

78) You need to ensure that the authors can see only their respective sales data. (Case Study 1)

How should you complete the statement? To answer, drag the appropriate values the correct targets. Each value may be used once, more than once, or not at all. You may need to drag the split bar between panes or scroll to view content.

NOTE: Each correct selection is worth one point.

Values

- AuthorSales
- AuthorEmail
- AuthorSales.AuthorEmail
- BLOCK
- FILTER
- INLINE
- SCHEMABINDING
- USER_NAME()

Answer Area

```
CREATE FUNCTION dbo.tvf_rlspredicate(@Author AS varchar(50))
    RETURNS TABLE
WITH [          ]
AS
    RETURN SELECT 1 AS tvf_rlspredicate_result
WHERE @Author = [          ]
GO

CREATE SECURITY POLICY RLSFilter
ADD FILTER PREDICATE Security.tvf_rlspredicate(AuthorEmail)
ON [          ]
WITH (STATE = ON)
```

Answer Area

```
CREATE FUNCTION dbo.tvf_rlspredicate(@Author AS varchar(50))
    RETURNS TABLE
WITH  [SCHEMABINDING]
AS
    RETURN SELECT 1 AS tvf_rlspredicate_result
WHERE @Author = [USER_NAME()]
GO

CREATE SECURITY POLICY RLSFilter
ADD FILTER PREDICATE Security.tvf_rlspredicate(AuthorEmail)
ON [AuthorSales]
WITH (STATE = ON)
```

79) You need to resolve the sales data issue. The solution must minimize the amount of data transferred.
What should you do? (Case Study 3)

A. Spilt the dataflow into two dataflows.
B. Configure scheduled refresh for the dataflow.
C. Configure incremental refresh for the dataflow. Set Store rows from the past to 1 Month.
D. Configure incremental refresh for the dataflow. Set Refresh rows from the past to 1 Year.
E. Configure incremental refresh for the dataflow. Set Refresh rows from the past to 1 Month.

79) You need to resolve the sales data issue. The solution must minimize the amount of data transferred.
What should you do? (Case Study 3)

A. Spilt the dataflow into two dataflows.
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C. Configure incremental refresh for the dataflow. Set Store rows from the past to 1 Month.
D. Configure incremental refresh for the dataflow. Set Refresh rows from the past to 1 Year.
E. Configure incremental refresh for the dataflow. Set Refresh rows from the past to 1 Month.

Answer: *E*

80) You have a Fabric workspace that contains an eventstream named EventStream1. EventStream1 outputs events to a table named Table1 in a lakehouse. The streaming data is sourced from motorway sensors and represents the speed of cars.

You need to add a transformation to EventStream1 to average the car speeds. The speeds must be grouped by non-overlapping and contiguous time intervals of one minute. Each event must belong to exactly one window.

Which windowing function should you use?

A. sliding

B. hopping

C. tumbling

D. session

80) You have a Fabric workspace that contains an eventstream named EventStream1. EventStream1 outputs events to a table named Table1 in a lakehouse. The streaming data is sourced from motorway sensors and represents the speed of cars.

You need to add a transformation to EventStream1 to average the car speeds. The speeds must be grouped by non-overlapping and contiguous time intervals of one minute. Each event must belong to exactly one window.

Which windowing function should you use?

A. sliding

B. hopping

C. tumbling

D. session

Answer: *C*

81) You have an Azure Event Hubs data source that contains weather data.
You ingest the data from the data source by using an eventstream named Eventstream1. Eventstream1 uses a lakehouse as the destination.
You need to batch ingest only rows from the data source where the City attribute has a value of Kansas. The filter must be added before the destination. The solution must minimize development effort.
What should you use for the data processor and filtering? To answer, select the appropriate options in the answer area.

NOTE: Each correct selection is worth one point.

Answer Area

Data processor:

- A data pipeline
- A Dataflow Gen2 dataflow
- An eventstream with a custom endpoint
- An eventstream with an external data source

Filtering:

- A Filter activity in a data pipeline
- A filter in a Dataflow Gen2 dataflow
- A KQL statement
- An eventstream processor

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You ingest the data from the data source by using an eventstream named Eventstream1. Eventstream1 uses a lakehouse as the destination.
You need to batch ingest only rows from the data source where the City attribute has a value of Kansas. The filter must be added before the destination. The solution must minimize development effort.
What should you use for the data processor and filtering? To answer, select the appropriate options in the answer area.

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Data processor:

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- An eventstream with a custom endpoint
- An eventstream with an external data source

Filtering:

- A Filter activity in a data pipeline
- A filter in a Dataflow Gen2 dataflow
- A KQL statement
- An eventstream processor

82) You are building a data loading pattern by using a Fabric data pipeline. The source is an Azure SQL database that contains 25 tables. The destination is a lakehouse.

In a warehouse, you create a control table named Control.Object as shown in the exhibit. (Click the Exhibit tab.)

	schema_name	table_name
1	Warehouse	ColdRoomTemperatures
2	Warehouse	Colors
3	Warehouse	PackageTypes
4	Warehouse	StockGroups
5	Warehouse	StockItems
6	dbo	BuildVersion
7	dbo	ErrorLog
8	Application	SystemParameters
9	Purchasing	PurchaseOrderLines
10	Purchasing	PurchaseOrders

You need to build a data pipeline that will support the dynamic ingestion of the tables listed in the control table by using a single execution.

Which three actions should you perform in sequence? To answer, move the appropriate actions from the list of actions to the answer area and arrange them in the correct order.

Actions	Answer Area
Add a ForEach activity to iterate over the list of tables and copy the source data to the lakehouse Delta tables.	1.
Add a Get metadata activity to query Control.Object and generate a list of schemas and tables to copy.	2.
Add an Until activity to iterate over the list of tables and copy the source data to the lakehouse Delta tables.	3.
Add a Lookup activity to query Control.Object and generate a list of the schemas and tables to copy.	
Add a Copy data activity as an inner activity to the iterator activity.	

You need to build a data pipeline that will support the dynamic ingestion of the tables listed in the control table by using a single execution.

Which three actions should you perform in sequence? To answer, move the appropriate actions from the list of actions to the answer area and arrange them in the correct order.

Actions

- Add a ForEach activity to iterate over the list of tables and copy the source data to the lakehouse Delta tables.
- Add a Get metadata activity to query Control.Object and generate a list of schemas and tables to copy.
- Add an Until activity to iterate over the list of tables and copy the source data to the lakehouse Delta tables.
- Add a Lookup activity to query Control.Object and generate a list of the schemas and tables to copy.
- Add a Copy data activity as an inner activity to the iterator activity.

Answer Area

1. Add a Lookup activity to query Control.Object and generate a list of the schemas and tables to copy.
2. Add a ForEach activity to iterate over the list of tables and copy the source data to the lakehouse Delta tables.
3. Add a Copy data activity as an inner activity to the iterator activity.

83) You have a Fabric eventhouse that contains a KQL database. The database contains a table named TaxiData. The following is a sample of the data in TaxiData.

VendorID	tpep_pickup_datetime	tpep_dropoff_datetime	passenger_count	trip_distance	PULocationID	DOLocationID	payment_type	total_amount
2	2022-06-06T11:08:32Z	2022-06-06T11:22:17Z	1	0.17	231	50	2	7.12
2	2022-06-06T11:12:05Z	2022-06-06T11:20:43Z	1	1.02	161	163	1	10.56
1	2022-06-06T11:15:00Z	2022-06-06T11:25:32Z	1	1.07	142	230	2	17.12
2	2022-06-06T11:29:54Z	2022-06-06T11:49:34Z	2	2.07	162	236	2	12.01
1	2022-06-06T11:50:50Z	2022-06-06T12:07:24Z	2	2.65	140	142	1	7.89

You need to build two KQL queries. The solution must meet the following requirements:
One of the queries must partition RunningTotalAmount by VendorID.

The other query must create a column named FirstPickupDateTime that shows the first value of each hour from tpep_pickup_datetime partitioned by payment_type.

How should you complete each query? To answer, drag the appropriate values the correct targets. Each value may be used once, more than once, or not at all. You may need to drag the split bar between panes or scroll to view content.

NOTE: Each correct selection is worth one point.

Values

- Row_cumsum
- Row_rank_dense
- Row_rank_min
- Row_window_session

Answer Area

Statement1:

```
TaxiData
| sort by VendorID asc
| extend RunningTotalAmount = [          ] (total_amount, VendorID != prev(VendorID))
```

Statement2:

```
TaxiData
| sort by tpep_pickup_datetime asc, payment_type asc
| extend FirstPickupDateTime = [          ] (tpep_pickup_datetime, 1h, 0m, payment_type != prev(payment_type))
```

Values

- Row_cumsum
- Row_rank_dense
- Row_rank_min
- Row_window_session

Answer Area

Statement1:

```
TaxiData
| sort by VendorID asc
| extend RunningTotalAmount = [          ] (total_amount, VendorID != prev(VendorID))
```

Statement2:

```
TaxiData
| sort by tpep_pickup_datetime asc, payment_type asc
| extend FirstPickupDateTime = [          ] (tpep_pickup_datetime, 1h, 0m, payment_type != prev(payment_type))
```

84) You have a Fabric workspace named Workspace1 that contains the following items:

• A Microsoft Power BI report named Report1
• A Power BI dashboard named Dashboard1
• A semantic model named Model1
• A lakehouse name Lakehouse1

Your company requires that specific governance processes be implemented for the items.

Which items can you endorse in Fabric?

A. Lakehouse1, Model1, and Dashboard1 only
B. Lakehouse1, Model1, Report1 and Dashboard1
C. Report1 and Dashboard1 only
D. Model1, Report1, and Dashboard1 only
E. Lakehouse1, Model1, and Report1 only

84) You have a Fabric workspace named Workspace1 that contains the following items:

- A Microsoft Power BI report named Report1
- A Power BI dashboard named Dashboard1
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B. Lakehouse1, Model1, Report1 and Dashboard1
C. Report1 and Dashboard1 only
D. Model1, Report1, and Dashboard1 only
E. Lakehouse1, Model1, and Report1 only

Answer: *E*

85) You need to ensure that processes for the bronze and silver layers run in isolation. How should you configure the Apache Spark settings? (Case Study 1)

A.Disable high concurrency.

B.Create a custom pool.

C.Modify the number of executors.

D.Set the default environment.

85) You need to ensure that processes for the bronze and silver layers run in isolation. How should you configure the Apache Spark settings? (Case Study 1)

A.Disable high concurrency.

B.Create a custom pool.

C.Modify the number of executors.

D.Set the default environment.

Answer(s): B

Explanation:
Isolated Compute
The Isolated Compute option provides more security to Spark compute resources from untrusted services by dedicating the physical compute resource to a single customer. Isolated compute option is best suited for workloads that require a high degree of isolation from other customer's workloads for reasons that include meeting compliance and regulatory requirements. The Isolate Compute option is only available with the XXXLarge (80 vCPU / 504 GB) node size and only available in the following regions. *The isolated compute option can be enabled or disabled after pool creation although the instance might need to be restarted*.
Scenario:

Existing Environment. Data Processing
Litware implements a medallion architecture by using the following three layers: bronze, silver, and gold. The sales data is ingested from the ERP system as Parquet files that land in the Files folder in a lakehouse. Notebooks are used to transform the files in a Delta table for the bronze and silver layers. The gold layer is in a warehouse that has V-Order disabled.

86) You need to ensure that WorkspaceA can be configured for source control.
Which two actions should you perform? Each correct answer presents part of the solution.
Note: Each correct selection is worth one point. (Case Study 2)

A.From Tenant setting, set Users can synchronize workspace items with their Git repositories to Enabled.

B.From Tenant setting, set Users can sync workspace items with GitHub repositories to Enabled.

C.Configure WorkspaceA to use a Premium Per User (PPU) license.

D.Assign WorkspaceA to Cap1.

86) You need to ensure that WorkspaceA can be configured for source control.
Which two actions should you perform? Each correct answer presents part of the solution.
Note: Each correct selection is worth one point. (Case Study 2)

A.From Tenant setting, set Users can synchronize workspace items with their Git repositories to Enabled.

B.From Tenant setting, set Users can sync workspace items with GitHub repositories to Enabled.

C.Configure WorkspaceA to use a Premium Per User (PPU) license.

D.Assign WorkspaceA to Cap1.

Answer(s): A,D

Explanation:
A F64 capacity is a Microsoft Fabric Capacity (Pay-as-you-go or Reservation).
Note: Microsoft Fabric, Get started with Git integration
Prerequisites
To integrate Git with your Microsoft Fabric workspace, you need to set up the following prerequisites for both Fabric and Git.
Fabric prerequisites

To access the Git integration feature, you need a Fabric capacity. A Fabric capacity is required to use all supported Fabric items [D]. If you don't have one yet, sign up for a free trial. Customers that already have a Power BI Premium capacity, can use that capacity, but keep in mind that certain Power BI SKUs only support Power BI items.
In addition, the following tenant switches must be enabled from the Admin portal:
* Users can create Fabric items
*-> Users can synchronize workspace items with their Git repositories [A, not B] For GitHub users only: Users can synchronize workspace items with GitHub repositories Scenario:

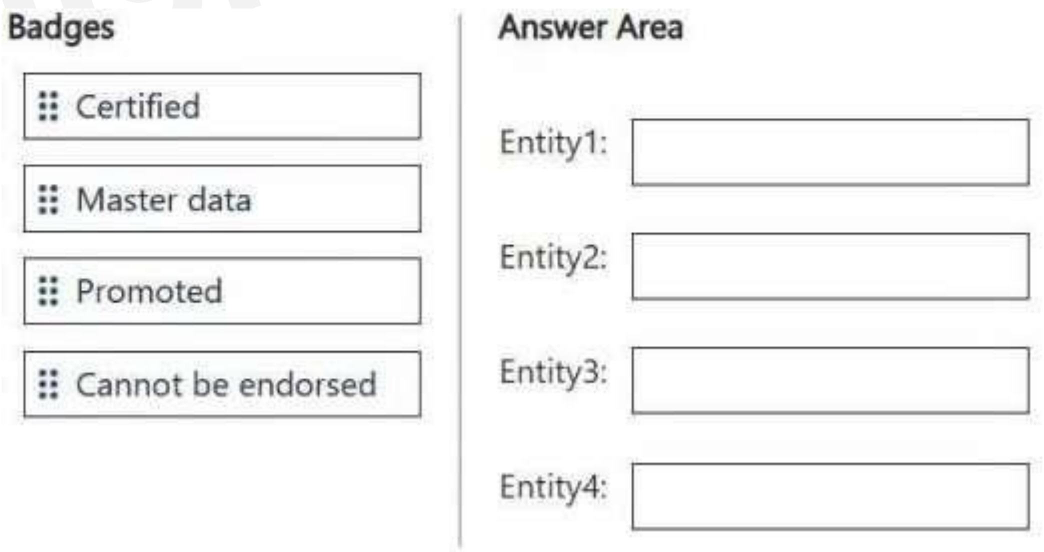
Existing Environment. Fabric
Contoso has an F64 capacity named Cap1 [D]. All Fabric users are allowed to create items.
Contoso has two workspaces named WorkspaceA and WorkspaceB that currently use Pro license mode.

Requirements. Planned Changes
Contoso plans to use Azure Repos for source control in Fabric.

Requirements. Technical Requirements
Items that relate to data ingestion must meet the following requirements:
*-> The items must be source controlled alongside other workspace items.

Requirements. Data Security
Security in Fabric must meet the following requirements:
*-> The data engineers must be able to commit changes to source control in WorkspaceA.

87) You are implementing the following data entities in a Fabric environment:
1. Entity1: Available in a lakehouse and contains data that will be used as a core organization entity
2. Entity2: Available in a semantic model and contains data that meets organizational standards
3. Entity3: Available in a Microsoft Power BI report and contains data that is ready for sharing and reuse
4. Entity4: Available in a Power BI dashboard and contains approved data for executive-level decision making
Your company requires that specific governance processes be implemented for the data.
You need to apply endorsement badges to the entities based on each entity's use case.
Which badge should you apply to each entity? To answer, drag the appropriate badges the correct entities.
Each badge may be used once, more than once, or not at all. You may need to drag the split bar between panes or scroll to view content.
Note: Each correct selection is worth one point.
Select and Place:



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Which badge should you apply to each entity? To answer, drag the appropriate badges the correct entities.
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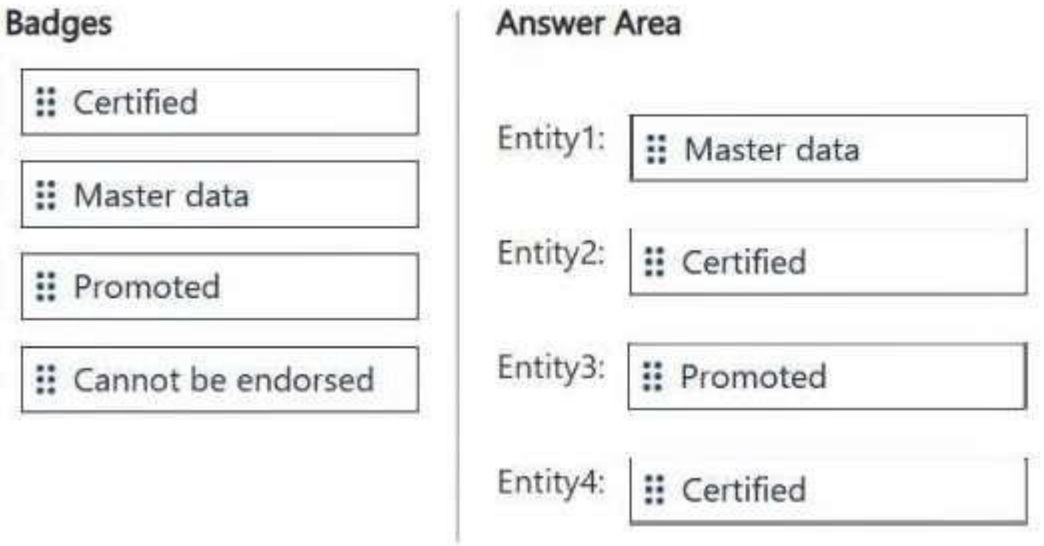
Note: Each correct selection is worth one point.
Select and Place:



Entity1: Master data
Entity1 is available in a lakehouse and contains data that will be used as a core organizational entity. This type of data, essential for the organization's operation, qualifies as Master data, which is foundational and authoritative within the business.
Entity2: Certified
Entity2 is available in a semantic model and contains data that meets organizational standards. Since it conforms to defined standards and has been validated, it should be Certified, signifying that it has passed the required data quality checks and is trusted for reporting and analysis.
Entity3: Promoted
Entity3 is available in a Microsoft Power BI report and contains data that is ready for sharing and reuse. Since the data is actively promoted for sharing and reuse within the organization, it receives the Promoted badge. This signifies that the data has been curated for wider consumption and distribution.
Entity4: Certified
Entity4 is available in a Power BI dashboard and contains approved data for executive-level decision making. This data is trusted and vetted for high-level decision-making and is thus Certified, indicating that it meets rigorous standards and is fit for executive-level use.

88) Your company has a team of developers. The team creates Python libraries of reusable code that is used to transform data.
You create a Fabric workspace name Workspace1 that will be used to develop extract, transform, and load (ETL) solutions by using notebooks.
You need to ensure that the libraries are available by default to new notebooks in Workspace1.
Which three actions should you perform in sequence? To answer, move the appropriate actions from the list of actions to the answer area and arrange them in the correct order.
Select and Place:

Actions	Answer Area
Change the runtime version.	
Install the libraries.	
Create a pool.	
Create an environment.	
Set the default environment.	

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Select and Place:

Actions

- Change the runtime version.
- Install the libraries.
- Create a pool.
- Create an environment.
- Set the default environment.

Answer Area

- Create an environment.
- Install the libraries.
- Set the default environment.

To ensure that your Python libraries are available by default to new notebooks in Workspace1, you need to follow these steps:
1. Create an environment: First, you need to create an environment where your libraries can be installed. This environment will be used by notebooks to access the required libraries.
2. Install the libraries: After creating the environment, you install the required Python libraries into that environment. These libraries can then be used by notebooks within Workspace1.
3. Set the default environment: Finally, to make sure that the libraries are available to new notebooks by default, you set this environment as the default environment for notebooks.
When you create a new notebook, it will automatically use the environment with the installed libraries.

89) You have five Fabric workspaces.
You are monitoring the execution of items by using Monitoring hub.
You need to identify in which workspace a specific item runs.
Which column should you view in Monitoring hub?

A.Start time

B.Capacity

C.Activity name

D.Submitter

E.Item type

F.Job type

G.Location

89) You have five Fabric workspaces.
You are monitoring the execution of items by using Monitoring hub.
You need to identify in which workspace a specific item runs.
Which column should you view in Monitoring hub?

A.Start time

B.Capacity

C.Activity name

D.Submitter

E.Item type

F.Job type

G.Location

Answer(s): G
Explanation:

To identify in which workspace a specific item runs in Monitoring hub, you should view the Location column.
This column indicates the workspace where the item is executed. Since you have multiple workspaces and need to track the execution of items across them, the Location column will show you the exact workspace associated with each item or job execution.

90) You have a Fabric workspace that contains a warehouse named Warehouse1.
In Warehouse1, you create a table named DimCustomer by running the following statement.

```
CREATE TABLE dbo.DimCustomer (
  CustomerKey VARCHAR(255) NOT NULL,
  Name VARCHAR(255) NOT NULL,
  Email VARCHAR(255) NOT NULL
  );
```

You need to set the Customerkey column as a primary key of the DimCustomer table.
Which three code segments should you run in sequence? To answer, move the appropriate code segments from the list of code segments to the answer area and arrange them in the correct order.
Select and Place:

Code Segments

- `DROP CONSTRAINT PK_DimCustomer`
- `ADD CONSTRAINT PK_DimCustomer PRIMARY KEY NONCLUSTERED (CustomerKey)`
- `NOT ENFORCED`
- `ALTER TABLE dbo.DimCustomer`
- `ADD CONSTRAINT PK_DimCustomer PRIMARY KEY CLUSTERED (CustomerKey)`
- `ENFORCED`

Answer Area

90) You have a Fabric workspace that contains a warehouse named Warehouse1.
In Warehouse1, you create a table named DimCustomer by running the following statement.

```
CREATE TABLE dbo.DimCustomer (
  CustomerKey VARCHAR(255) NOT NULL,
  Name VARCHAR(255) NOT NULL,
  Email VARCHAR(255) NOT NULL
);
```

You need to set the Customerkey column as a primary key of the DimCustomer table.
Which three code segments should you run in sequence? To answer, move the appropriate code segments from the list of code segments to the answer area and arrange them in the correct order.
Select and Place:

Code Segments

- `DROP CONSTRAINT PK_DimCustomer`
- `ADD CONSTRAINT PK_DimCustomer PRIMARY KEY NONCLUSTERED (CustomerKey)`
- `NOT ENFORCED`
- `ALTER TABLE dbo.DimCustomer`
- `ADD CONSTRAINT PK_DimCustomer PRIMARY KEY CLUSTERED (CustomerKey)`
- `ENFORCED`

Answer Area

- `ALTER TABLE dbo.DimCustomer`
- `ADD CONSTRAINT PK_DimCustomer PRIMARY KEY CLUSTERED (CustomerKey)`
- `ENFORCED`

To set the CustomerKey column as the primary key of the DimCustomer table, the steps need to follow the logical sequence of ensuring that the table structure can accommodate the primary key constraint.
1. ALTER TABLE dbo.DimCustomer
This command prepares the table for modification. It ensures the DimCustomer table is being updated to include the primary key.
2. ADD CONSTRAINT PK_DimCustomer PRIMARY KEY CLUSTERED (CustomerKey) This command creates a clustered primary key constraint on the CustomerKey column. A primary key uniquely identifies each record and enforces uniqueness.
3. ENFORCED
This ensures that the primary key constraint is enforced, meaning it will actively validate and enforce uniqueness and nullability on the CustomerKey column.

91) You have a Fabric workspace that contains a warehouse named Warehouse1. Warehouse1 contains a table named DimCustomers. DimCustomers contains the following columns:
CustomerName
CustomerID
BirthDate
EmailAddress
You need to configure security to meet the following requirements:
BirthDate in DimCustomer must be masked and display 1900-01-01.
EmailAddress in DimCustomer must be masked and display only the first leading character and the last five characters.
How should you complete the statement? To answer, select the appropriate options in the answer area.
Note: Each correct selection is worth one point.
Hot Area:

Answer Area

```
ALTER TABLE DimCustomer

ALTER COLUMN BirthDate

ADD MASKED WITH (FUNCTION =
```

'default()'
'partial(1900-01-01)'
'random(1900-01-01, 1900-01-01)'

```
ALTER TABLE DimCustomer

ALTER COLUMN EmailAddress

ADD MASKED WITH (FUNCTION =
```

'default()'
'email()'
'partial(1,"@",5)'
'random(1,"@",5)'

Answer Area

```
ALTER TABLE DimCustomer

ALTER COLUMN BirthDate

ADD MASKED WITH (FUNCTION =
```

'default()'
'partial(1900-01-01)'
'random(1900-01-01, 1900-01-01)'

```
ALTER TABLE DimCustomer

ALTER COLUMN EmailAddress

ADD MASKED WITH (FUNCTION =
```

'default()'
'email()'
'partial(1,"@",5)'
'random(1,"@",5)'

Box 1: random(1900-01-01, 1900-01-01)
BirthDate in DimCustomer must be masked and display 1900-01-01.
Random
A random masking function for use on any numeric type to mask the original value with a random value within a specified range. Example definition syntax: Account_Number bigint MASKED WITH (FUNCTION = 'random ([start range], [end range])')
Example of alter syntax: ALTER COLUMN [Month] ADD MASKED WITH (FUNCTION = 'random(1, 12)')
Box 2: 'partial(1,"@",5)'
EmailAddress in DimCustomer must be masked and display only the first leading character and the last five characters.
Custom String Masking method that exposes the first and last letters and adds a custom padding string in the middle. prefix,[padding],suffix
If the original value is too short to complete the entire mask, part of the prefix or suffix isn't exposed.
xample definition syntax: FirstName varchar(100) MASKED WITH (FUNCTION = 'partial(prefix, [padding],suffix)')
NULL
Example of alter syntax: ALTER COLUMN [Phone Number] ADD MASKED WITH (FUNCTION = 'partial (1,"XXXXXXX",0)')
This turns a phone number like 555.123.1234 into 5XXXXXXX.
Incorrect:
* 'default()'
Full masking according to the data types of the designated fields.
default() displays XXXX.

92) You have a Fabric workspace named Workspace1.
Your company acquires GitHub licenses.
You need to configure source control for Workpace1 to use GitHub. The solution must follow the principle of least privilege.
Which permissions do you require to ensure that you can commit code to GitHub?

A.Actions (Read and write) and Contents (Read and write)

B.Actions (Read and write) only

C.Contents (Read and write) only

D.Contents (Read) and Commit statuses (Read and write)

92) You have a Fabric workspace named Workspace1.
Your company acquires GitHub licenses.
You need to configure source control for Workpace1 to use GitHub. The solution must follow the principle of least privilege.
Which permissions do you require to ensure that you can commit code to GitHub?

A.Actions (Read and write) and Contents (Read and write)

B.Actions (Read and write) only

C.Contents (Read and write) only

D.Contents (Read) and Commit statuses (Read and write)

Answer(s): C

Explanation:
Microsoft Fabric Git integration
Git roles
The following table describes the Git permissions needed to perform various common operations:
If you're using a fine-grained access token, the following permissions are needed for common operations:
* Commit workspace changes to Git
Contents= Access: Read and write branch policy should allow direct commit
Note:
* Commit workspace changes to Git
All of the following roles:
Contributor in the workspace (WRITE permission on all items) Owner of the item (if the tenant switch blocks updates for nonowners) BUILD on external dependencies (where applicable)

93) You have a Fabric workspace named Workspace1.
You plan to configure Git integration for Workspace1 by using an Azure DevOps Git repository.
An Azure DevOps admin creates the required artifacts to support the integration of Workspace1.
Which details do you require to perform the integration?

A.the organization, project, Git repository, and branch

B.the personal access token (PAT) for Git authentication and the Git repository URL

C.the project, Git repository, branch, and Git folder

D.the Git repository URL and the Git folder

93) You have a Fabric workspace named Workspace1.
You plan to configure Git integration for Workspace1 by using an Azure DevOps Git repository.
An Azure DevOps admin creates the required artifacts to support the integration of Workspace1.
Which details do you require to perform the integration?

A.the organization, project, Git repository, and branch

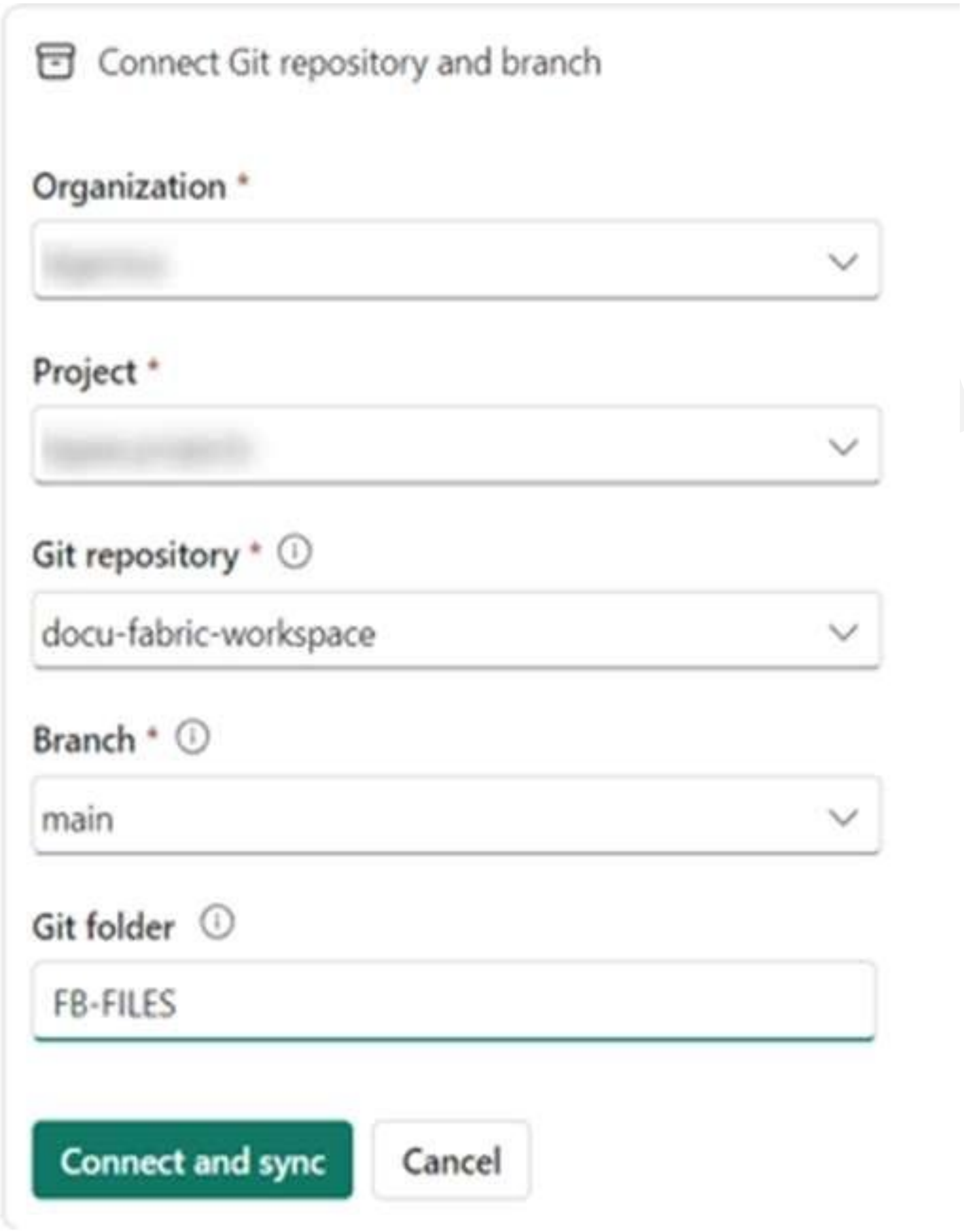
B.the personal access token (PAT) for Git authentication and the Git repository URL

C.the project, Git repository, branch, and Git folder

D.the Git repository URL and the Git folder

Answer(s): A

Explanation:
Microsoft Fabric Git integration
Connect to your new Azure DevOps git repository, then click on the Connect and sync button:


Connect Git repository and branch
Organization *
Project *
Git repository *
docu-fabric-workspace
Branch *
main
Git folder
FB-FILES
Connect and sync
Cancel

94) You have a Fabric workspace that contains a lakehouse and a semantic model named Model1.
You use a notebook named Notebook1 to ingest and transform data from an external data source.
You need to execute Notebook1 as part of a data pipeline named Pipeline1. The process must meet the following requirements:
Run daily at 07:00 AM UTC.
Attempt to retry Notebook1 twice if the notebook fails.
After Notebook1 executes successfully, refresh Model1.
Which three actions should you perform? Each correct answer presents part of the solution.
Note: Each correct selection is worth one point.

A.Place the Semantic model refresh activity after the Notebook activity and link the activities by using the On success condition.

B.From the Schedule settings of Pipeline1, set the time zone to UTC.

C.Set the Retry setting of the Notebook activity to 2.

D.From the Schedule settings of Notebook1, set the time zone to UTC.

E.Set the Retry setting of the Semantic model refresh activity to 2.

F.Place the Semantic model refresh activity after the Notebook activity and link the activities by using an On completion condition.

94) You have a Fabric workspace that contains a lakehouse and a semantic model named Model1.
You use a notebook named Notebook1 to ingest and transform data from an external data source.
You need to execute Notebook1 as part of a data pipeline named Pipeline1. The process must meet the following requirements:
Run daily at 07:00 AM UTC.
Attempt to retry Notebook1 twice if the notebook fails.
After Notebook1 executes successfully, refresh Model1.
Which three actions should you perform? Each correct answer presents part of the solution.
Note: Each correct selection is worth one point.

A.Place the Semantic model refresh activity after the Notebook activity and link the activities by using the On success condition.

B.From the Schedule settings of Pipeline1, set the time zone to UTC.

C.Set the Retry setting of the Notebook activity to 2.

D.From the Schedule settings of Notebook1, set the time zone to UTC.

E.Set the Retry setting of the Semantic model refresh activity to 2.

F.Place the Semantic model refresh activity after the Notebook activity and link the activities by using an On completion condition.

Answer(s): A,B,C

95) You have a Fabric workspace that contains a lakehouse named Lakehouse1.
You plan to create a data pipeline named Pipeline1 to ingest data into Lakehouse1. You will use a parameter named param1 to pass an external value into Pipeline1. The param1 parameter has a data type of int.
You need to ensure that the pipeline expression returns param1 as an int value.
How should you specify the parameter value?

A."@pipeline().parameters.param1"

B."@{pipeline().parameters.param1}"

C."@{pipeline().parameters.[param1]}"

D."@@{pipeline().parameters.param1}"

95) You have a Fabric workspace that contains a lakehouse named Lakehouse1.
You plan to create a data pipeline named Pipeline1 to ingest data into Lakehouse1. You will use a parameter named param1 to pass an external value into Pipeline1. The param1 parameter has a data type of int.
You need to ensure that the pipeline expression returns param1 as an int value.
How should you specify the parameter value?

A."@pipeline().parameters.param1"

B."@{pipeline().parameters.param1}"

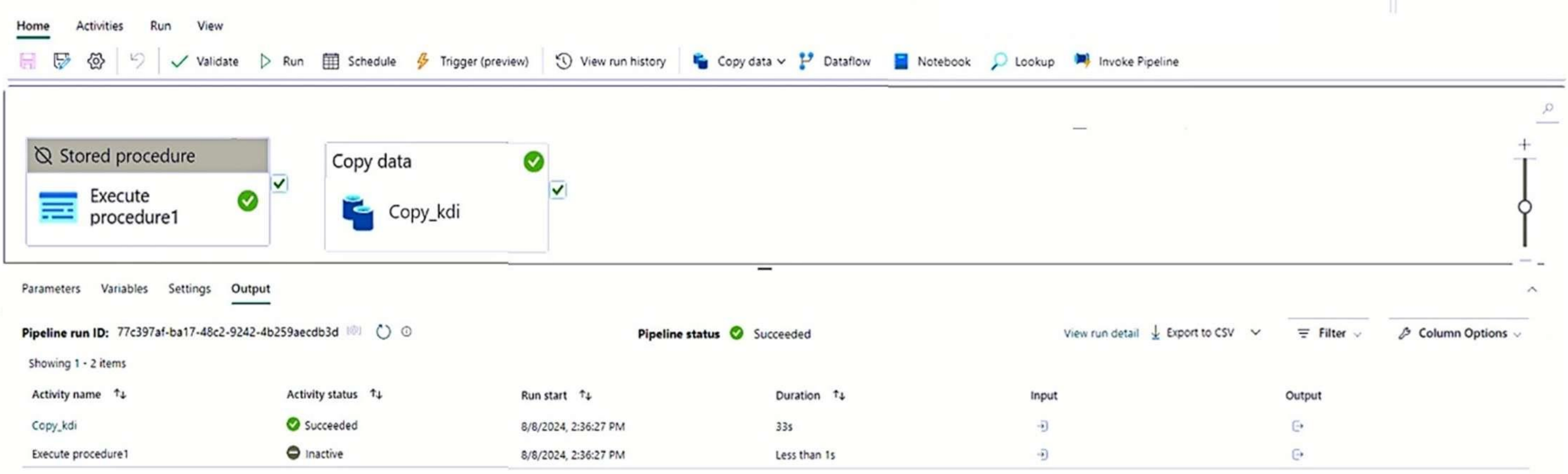
C."@{pipeline().parameters.[param1]}"

D."@@{pipeline().parameters.param1}"

Answer(s): B

Explanation:
Parameters for Data Factory in Microsoft Fabric
Expressions can appear inside strings, using a feature called string interpolation where expressions are wrapped in @{ ... }. For example, the following string includes parameter values and literal string values: "First Name: @{pipeline().parameters.firstName} Last Name: @{pipeline().parameters.lastName}" Using string interpolation, the result is always a string. For example, if you defined myNumber as 42:
"@pipeline().parameters.myNumber"
Returns 42 as a number.

96) You have a Fabric workspace that contains a data pipeline named Pipeline1 as shown in the exhibit. (Click the Exhibit tab.)



What will occur the next time Pipeline1 runs?

A.Copy_kdi will run first, and then Execute procedure1 will run.

B.Execute procedure1 will run first, and then Copy_kdi will run.

C.Execute procedure1 will run and Copy_kdi will be skipped.

D.Copy_kdi will run and Execute procedure1 will be skipped.

E.Both activities will run simultaneously.

F.Both activities will be skipped.

What will occur the next time Pipeline1 runs?

A.Copy_kdi will run first, and then Execute procedure1 will run.

B.Execute procedure1 will run first, and then Copy_kdi will run.

C.Execute procedure1 will run and Copy_kdi will be skipped.

D.Copy_kdi will run and Execute procedure1 will be skipped.

E.Both activities will run simultaneously.

F.Both activities will be skipped.

Answer(s): F

97) You have a Fabric workspace that contains a warehouse named Warehouse1. Warehouse1 contains a table named Customer. Customer contains the following data.

CustomerID	FirstName	LastName	Phone	CreditCard
1	John	Doe	555-123-4567	1234567812345670
2	Jane	Smith	555-987-6543	8765432187654320
3	Michael	Johnson	555-555-5555	1234987654321230
4	Emily	Davis	555-222-3333	4321123456789870
5	David	Brown	555-444-5555	5678123498761230

You have an internal Microsoft Entra user named User1 that has an email address of user1@contoso.com.
You need to provide User1 with access to the Customer table. The solution must prevent User1 from accessing the CreditCard column.
How should you complete the statement? To answer, select the appropriate options in the answer area.
Note: Each correct selection is worth one point.
Hot Area:

Answer Area

GRANT [dropdown] ON

- ALTER
- EXECUTE
- READ
- SELECT
- VIEW

Customers(CustomerID, FirstName, LastName, Phone)

TO [dropdown]

- User1
- [User1]
- [user1@contoso.com]

Answer Area

GRANT [dropdown] ON

- ALTER
- EXECUTE
- READ
- SELECT
- VIEW

`Customers(CustomerID, FirstName, LastName, Phone)`

TO [dropdown]

- User1
- [User1]
- [user1@contoso.com]

Box 1: SELECT
Implement column-level security in Fabric data warehousing
Column-level security (CLS) in Microsoft Fabric allows you to control access to columns in a table based on specific grants on these tables.
Define column-level access for tables
1. Identify user or roles and the data tables you want to secure with column-level security.
2. Implement column-level security with the GRANT T-SQL statement and a column list. For simplicity of management, assigning permissions to roles is preferred to using individuals.
-- Grant select to subset of columns of a table
GRANT SELECT ON YourSchema.YourTable
(Column1, Column2, Column3, Column4, Column5)
TO [SomeGroup];
Replace YourSchema with the name of your schema and YourTable with the name of your target table.
3. Replace SomeGroup with the name of your User/Group.
4. Replace the comma-delimited columns list with the columns you want to give the role access to.
5. Repeat these steps to grant specific column access for other tables if needed.
Box 2: [User1]

98) You have a Fabric workspace named Workspace1 that contains a warehouse named Warehouse1.
You plan to deploy Warehouse 1 to a new workspace named Workspace2.
As part of the deployment process, you need to verify whether Warehouse1 contains invalid references. The solution must minimize development effort and provide detailed information about the invalid references.
What should you use?

A.a dbt project

B.a deployment pipeline

C.a Python script

D.a database project

98) You have a Fabric workspace named Workspace1 that contains a warehouse named Warehouse1.
You plan to deploy Warehouse 1 to a new workspace named Workspace2.
As part of the deployment process, you need to verify whether Warehouse1 contains invalid references. The solution must minimize development effort and provide detailed information about the invalid references.
What should you use?

A.a dbt project

B.a deployment pipeline

C.a Python script

D.a database project

Answer(s): B

Explanation:
Correct:
* a deployment pipeline
Incorrect:
* a database project
* a dbt project
* a Python script
* a T-SQL script
Note:
A deployment pipeline in Fabric allows you to deploy assets like warehouses, datasets, and reports between different workspaces (such as from Workspace1 to Workspace2). One of the key features of a deployment pipeline is the ability to check for invalid references before deployment. This can help identify issues with assets, such as broken links or dependencies, ensuring the deployment is successful without introducing errors. This is the most efficient way to verify references and manage the deployment with minimal development effort.

Reference:
https://learn.microsoft.com/en-us/fabric/cicd/deployment-pipelines/understand-the-deployment-process

99) You need to create a workflow for the new book cover images.
Which two components should you include in the workflow? Each correct answer presents part of the solution. (Case Study 3)
Note: Each correct selection is worth one point.

A.an activator item

B.a data pipeline

C.a blob storage action

D.a time-based schedule

E.a streaming dataflow

F.a notebook that uses Apache Spark Structured Streaming

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C.a blob storage action

D.a time-based schedule

E.a streaming dataflow

F.a notebook that uses Apache Spark Structured Streaming

Answer(s): C,E

100) You need to create a workflow for the new book cover images.
Which two components should you include in the workflow? Each correct answer presents part of the solution. (Case Study 3)
Note: Each correct selection is worth one point.

A.a time-based schedule

B.a streaming dataflow

C.a blob storage action

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A.a time-based schedule

B.a streaming dataflow

C.a blob storage action

D.a data pipeline

E.a notebook that uses Apache Spark Structured Streaming

F.a reflex item

Answer(s): B,C

101) What should you recommend that the data engineering team use to ingest the SEO data? (Case Study 3)

A.a streaming dataflow

B.a streaming dataset

C.a notebook that uses Apache Spark Structured Streaming

D.an eventstream

101) What should you recommend that the data engineering team use to ingest the SEO data? (Case Study 3)

A.a streaming dataflow

B.a streaming dataset

C.a notebook that uses Apache Spark Structured Streaming

D.an eventstream

Answer(s): D

Explanation:
Microsoft Fabric Real-Time Intelligence, Eventstream REST API The Microsoft Fabric REST APIs enable you to automate Fabric procedures and processes, helping your organization complete tasks more efficiently and accurately. By automating these workflows, you can reduce errors, improve productivity, and achieve cost savings across your operations.
In Fabric, an item represents a set of capabilities within a specific experience. For example, Eventstream is an item under the Real-time Intelligence experience. Each item in Fabric is defined by an item definition--an object that outlines the structure, format, and key components that make up the item.
Scenario:

Requirements. Planned Changes
Litware plans to manage Search Engine Optimization (SEO) for the authors. The SEO data will be streamed from a REST API.

Requirements. Data Requirements
Litware identifies the following data requirements:
*-> Process the SEO data in near-real-time (NRT).

102) You need to recommend a solution to resolve the MAR1 connectivity issues.
The solution must minimize development effort. (Case Study 4)
What should you recommend?

A.Add a ForEach activity to the data pipeline.

B.Configure retries for the Copy data activity.

C.Call a notebook from the data pipeline.

D.Configure Fault tolerance for the Copy data activity.

102) You need to recommend a solution to resolve the MAR1 connectivity issues.
The solution must minimize development effort. (Case Study 4)
What should you recommend?

A.Add a ForEach activity to the data pipeline.

B.Configure retries for the Copy data activity.

C.Call a notebook from the data pipeline.

D.Configure Fault tolerance for the Copy data activity.

Answer(s): B

Explanation:
Retry Feature in Azure Data Factory
Enhancing Data Pipeline Reliability with Retry Feature in Azure Data Factory Understanding the Retry Feature
The Retry feature in ADF's Copy activity allows users to automatically retry failed operations in the event of transient failures. These failures could be caused by factors such as network disruptions, Java Native Interface errors, or temporary service unavailability. By configuring the retry settings, users can specify the number of retry attempts and the interval between retries, providing flexibility to tailor the retry behavior according to specific requirements.
Scenario:
Requirements. Technical Requirements
Items that relate to data ingestion must meet the following requirements:
*-> In the event of a connectivity error, the ingestion processes must attempt the connection again.
* Etc.
Note: Existing environment:
The company has a software as a service (SaaS) online marketing app named MAR1. MAR1 has seven entities. The entities contain data that relates to email open rates and interaction rates, as well as website interactions. The data can be exported from MAR1 by calling REST APIs. Each entity has a different endpoint.
Contoso has been using MAR1 for one year.
Requirements. Technical Requirements
The new lakehouses must follow a medallion architecture by using the following three layers: bronze, silver, and gold. There will be extensive data cleansing required to populate the MAR1 data in the silver layer, including deduplication, the handling of missing values, and the standardizing of capitalization.

103) You need to recommend a solution for handling old files. The solution must meet the technical requirements. (Case Study 4)
What should you include in the recommendation?

A.a data pipeline that includes a Copy data activity

B.a data pipeline that includes a Delete data activity

C.a notebook that runs the VACUUM command

D.a notebook that runs the OPTIMIZE command

103) You need to recommend a solution for handling old files. The solution must meet the technical requirements. (Case Study 4)
What should you include in the recommendation?

A.a data pipeline that includes a Copy data activity

B.a data pipeline that includes a Delete data activity

C.a notebook that runs the VACUUM command

D.a notebook that runs the OPTIMIZE command

Answer(s): C
Explanation:

Scenario:
Technical requirements include:
Once a week, old files that are no longer referenced by a Delta table log must be removed.
Note: Remove unused data files with vacuum
Predictive optimization automatically runs VACUUM on Unity Catalog managed tables. Databricks recommends enabling predictive optimizations for all Unity Catalog managed tables to simplify data maintenance and reduce storage costs.

You can remove data files no longer referenced by a Delta table that are older than the retention threshold by running the VACUUM command on the table. Running VACUUM regularly is important for cost and compliance because of the following considerations:
Deleting unused data files reduces cloud storage costs.
Data files removed by VACUUM might contain records that have been modified or deleted. Permanently removing these files from cloud storage ensures these records are no longer accessible.

104) Your company has three newly created data engineering teams named Team1, Team2, and Team3 that plan to use Fabric. The teams have the following personas:

• Team1 consists of members who currently use Microsoft Power BI. The team wants to transform data by using by a low-code approach.
• Team2 consists of members that have a background in Python programming. The team wants to use PySpark code to transform data.
• Team3 consists of members who currently use Azure Data Factory. The team wants to move data between source and sink environments by using the least amount of effort.

You need to recommend tools for the teams based on their current personas.

What should you recommend for each team? To answer, select the appropriate options in the answer area.

NOTE: Each correct selection is worth one point.

Answer Area

Team1:

- Data pipelines
- Notebooks
- Dataflow Gen2 dataflow

Team2:

- Data pipelines
- Notebooks
- Dataflow Gen2 dataflow

Team3:

- Data pipelines
- Notebooks
- Dataflow Gen2 dataflow

Answer Area

Team1:

- Data pipelines
- Notebooks
- Dataflow Gen2 dataflow

Team2:

- Data pipelines
- Notebooks
- Dataflow Gen2 dataflow

Team3:

- Data pipelines
- Notebooks
- Dataflow Gen2 dataflow

105) You have a Fabric workspace that contains a warehouse named DW1. DW1 contains the following tables and columns.

Table name	Column name
SalesOrderDetail	ProductID
SalesOrderDetail	ModifiedDate
SalesOrderDetail	OrderQty
Product	ProductID
Product	Name

You need to summarize order quantities by year and product. The solution must include the yearly sum of order quantities for all the products in each row.

How should you complete the T-SQL statement? To answer, select the appropriate options in the answer area.

NOTE: Each correct selection is worth one point.

Answer Area

```
SELECT [dropdown] (SO.ModifiedDate) AS OrderDate
```

CAST
CONVERT
YEAR

```
  ,P.Name AS ProductName
  ,SUM(SO.OrderQty) AS OrderQty
FROM [dbo].[SalesOrderDetail] SO
INNER JOIN [dbo].[Product] P
  ON P.ProductID = SO.ProductID
GROUP BY
```

CUBE(YEAR(SO.ModifiedDate), P.Name)
GROUPING SETS ((YEAR(SO.ModifiedDate), P.Name), (YEAR(SO.ModifiedDate)))
ROLLUP(YEAR(SO.ModifiedDate), P.Name)
YEAR(SO. Modified Date), P.Name

```
ORDER BY OrderDate
```

Answer Area

```
SELECT [ CAST | CONVERT | YEAR ] (SO.ModifiedDate) AS OrderDate
  ,P.Name AS ProductName
  ,SUM(SO.OrderQty) AS OrderQty
FROM [dbo].[SalesOrderDetail] SO
INNER JOIN [dbo].[Product] P
  ON P.ProductID = SO.ProductID
GROUP BY
  [ CUBE(YEAR(SO.ModifiedDate), P.Name)
  | GROUPING SETS ((YEAR(SO.ModifiedDate), P.Name), (YEAR(SO.ModifiedDate)))
  | ROLLUP(YEAR(SO.ModifiedDate), P.Name)
  | YEAR(SO. Modified Date), P.Name ]
ORDER BY OrderDate
```

106) You have a Fabric warehouse named DW1 that contains four staging tables named ProductCategory, ProductSubcategory, Product, and SalesOrder. ProductCategory, ProductSubcategory, and Product are used often in analytical queries.

You need to implement a star schema for DW1. The solution must minimize development effort.

Which design approach should you use? To answer, select the appropriate options in the answer area.

NOTE: Each correct selection is worth one point.

Answer Area

ProductCategory, ProductSubcategory and Product must be:	Added to the model as individual tables Denormalized by being added to the SalesOrder table Denormalized into a single product dimension table
The joining key must be:	The product name and the date The unique system generated identifier The product category name

106) You have a Fabric warehouse named DW1 that contains four staging tables named ProductCategory, ProductSubcategory, Product, and SalesOrder. ProductCategory, ProductSubcategory, and Product are used often in analytical queries.

You need to implement a star schema for DW1. The solution must minimize development effort.

Which design approach should you use? To answer, select the appropriate options in the answer area.

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Answer Area

ProductCategory, ProductSubcategory and Product must be:

- Added to the model as individual tables
- Denormalized by being added to the SalesOrder table
- Denormalized into a single product dimension table

The joining key must be:

- The product name and the date
- The unique system generated identifier
- The product category name

107) You have a Fabric warehouse named DW1 that contains a Type 2 slowly changing dimension (SCD) dimension table named DimCustomer. DimCustomer contains 100 columns and 20 million rows. The columns are of various data types, including int, varchar, date, and varbinary.
You need to identify incoming changes to the table and update the records when there is a change. The solution must minimize resource consumption.

What should you use to identify changes to attributes?

A. a hash function to compare the attributes in the source table.

B. a direct attributes comparison across the attributes in the DimCustomer table.

C. a direct attributes comparison for the attributes in the source table.

D. a hash function to compare the attributes in the DimCustomer table.

107) You have a Fabric warehouse named DW1 that contains a Type 2 slowly changing dimension (SCD) dimension table named DimCustomer. DimCustomer contains 100 columns and 20 million rows. The columns are of various data types, including int, varchar, date, and varbinary.
You need to identify incoming changes to the table and update the records when there is a change. The solution must minimize resource consumption.

What should you use to identify changes to attributes?

A. a hash function to compare the attributes in the source table.

B. a direct attributes comparison across the attributes in the DimCustomer table.

C. a direct attributes comparison for the attributes in the source table.

D. a hash function to compare the attributes in the DimCustomer table.

Answer: *A*

108) You have a table in a Fabric lakehouse that contains the following data.

SalesOrderNumber	OrderDate	CustomerName	Email
SO49172	2021-01-01	Brian Howard	brian23@adventure-works.com
SO49173	2021-01-01	Linda Alvarez	linda19@adventure-works.com
SO49174	2021-01-01	Gina Hernandez	gina4@adventure-works.com
SO49178	2021-01-01	Beth Ruiz	beth4@adventure-works.com
SO49179	2021-01-01	Evan Ward	evan13@adventure-works.com

You have a notebook that contains the following code segment.

```
df = df.withColumn("CustomerName", when((col("CustomerName").isNull() | (col("CustomerName")=="")),lit("Unknown")).otherwise(col("CustomerName")))
df = df.withColumn("Username",split(col("Email"), "@").getItem(1))
df = df.dropDuplicates(["OrderDate"]).select(col("OrderDate"), year("OrderDate").alias("Year"), ("CustomerName"), ("Username"))
display(df.head(10))
```

Answer Area

Statements	Yes	No
Line 01 will replace all the null and empty values in the CustomerName column with the Unknown value.	○	○
Line 02 will extract the value before the @ character and generate a new column named Username.	○	○
Line 03 will extract the year value from the OrderDate column and keep only the first occurrence for each year.	○	○

Answer Area

Statements	Yes	No
Line 01 will replace all the null and empty values in the CustomerName column with the Unknown value.	○	○
Line 02 will extract the value before the @ character and generate a new column named Username.	○	○
Line 03 will extract the year value from the OrderDate column and keep only the first occurrence for each year.	○	○

109) You have a Fabric workspace that contains a write-intensive warehouse named DW1. DW1 stores staging tables that are used to load a dimensional model. The tables are often read once, dropped, and then recreated to process new data.

You need to minimize the load time of DW1.

What should you do?

A. Enable V-Order.

B. Create statistics.

C. Drop statistics.

D. Disable V-Order.

109) You have a Fabric workspace that contains a write-intensive warehouse named DW1. DW1 stores staging tables that are used to load a dimensional model. The tables are often read once, dropped, and then recreated to process new data.

You need to minimize the load time of DW1.

What should you do?

A. Enable V-Order.

B. Create statistics.

C. Drop statistics.

D. Disable V-Order.

Answer: *D*

110) You have a Fabric workspace that contains a lakehouse named Lakehouse1. Lakehouse1 contains a table named Status_Target that has the following columns:

• Key
• Status
• LastModified

The data source contains a table named Status_Source that has the same columns as Status_Target. Status_Source is used to populate Status_Target.
In a notebook name Notebook1, you load Status_Source to a DataFrame named sourceDF and Status_Target to a DataFrame named targetDF.
You need to implement an incremental loading pattern by using Notebook1. The solution must meet the following requirements:
• For all the matching records that have the same value of key, update the value of LastModified in Status_Target to the value of LastModified in Status_Source.
• Insert all the records that exist in Status_Source that do NOT exist in Status_Target.
• Set the value of Status in Status_Target to inactive for all the records that were last modified more than seven days ago and that do NOT exist in Status_Source.
How should you complete the statement? To answer, select the appropriate options in the answer area.

NOTE: Each correct selection is worth one point.

Answer Area

```
...
(targetDF
  .merge(sourceDF, "sourceDF.Key" = "targetDF.Key")
   [ ▼ ]
   .whenMatchedInsert(
   .whenMatchedUpdate(
   .whenNotMatchedBySourceInsert(
   .whenNotMatchedBySourceUpdate(
   .whenNotMatchedInsert(
   .whenNotMatchedUpdate(
   set = {"targetDF.LastModified": "sourceDF.LastModified"}
 )
   [ ▼ ]
   .whenMatchedInsert(
   .whenMatchedUpdate(
   .whenNotMatchedBySourceInsert(
   .whenNotMatchedBySourceUpdate(
   .whenNotMatchedInsert(
   .whenNotMatchedUpdate(
    values = {
      "targetDF.Key": "sourceDF.Key",
      "targetDF.LastModified": "sourceDF.LastModified",
      "targetDF.Status": "sourceDF.Status"
    }
  )
    [ ▼ ]
    .whenMatchedInsert(
    .whenMatchedUpdate(
    .whenNotMatchedBySourceInsert(
    .whenNotMatchedBySourceUpdate(
    .whenNotMatchedInsert(
    .whenNotMatchedUpdate(
    condition="targetDF.LastModified > (current_date() - INTERVAL '7' DAY)",
    set = {"targetDF.Status": "'inactive'"}
  )
  .execute()
)
...
```

Answer Area

```
...
(targetDF
  .merge(sourceDF, "sourceDF.Key" = "targetDF.Key")
```

- .whenMatchedInsert(
- .whenMatchedUpdate(
- .whenNotMatchedBySourceInsert(
- .whenNotMatchedBySourceUpdate(
- .whenNotMatchedInsert(
- .whenNotMatchedUpdate(

```
   set = {"targetDF.LastModified": "sourceDF.LastModified"}
 )
```

- .whenMatchedInsert(
- .whenMatchedUpdate(
- .whenNotMatchedBySourceInsert(
- .whenNotMatchedBySourceUpdate(
- .whenNotMatchedInsert(
- .whenNotMatchedUpdate(

```
   values = {
     "targetDF.Key": "sourceDF.Key",
     "targetDF.LastModified": "sourceDF.LastModified",
     "targetDF.Status": "sourceDF.Status"
   }
  )
```

- .whenMatchedInsert(
- .whenMatchedUpdate(
- .whenNotMatchedBySourceInsert(
- .whenNotMatchedBySourceUpdate(
- .whenNotMatchedInsert(
- .whenNotMatchedUpdate(

```
   condition="targetDF.LastModified > (current_date() - INTERVAL '7' DAY)",
   set = {"targetDF.Status": "'inactive'"}
  )
  .execute()
)
...
```

111) Note: This question is part of a series of questions that present the same scenario. Each question in the series contains a unique solution that might meet the stated goals. Some question sets might have more than one correct solution, while others might not have a correct solution.
After you answer a question in this section, you will NOT be able to return to it. As a result, these questions will not appear in the review screen.
You have a Fabric eventstream that loads data into a table named Bike_Location in a KQL database. The table contains the following columns:
BikepointID -

Street -

Neighbourhood -

No_Bikes -

No_Empty_Docks -

Timestamp -
You need to apply transformation and filter logic to prepare the data for consumption. The solution must return data for a neighbourhood named Sands End when No_Bikes is at least 15. The results must be ordered by No_Bikes in ascending order.
Solution: You use the following code segment:

```
bike_location
| filter Neighbourhood == "Sands End" and No_Bikes >= 15
| sort by No_Bikes asc
| project BikepointID, Street, Neighbourhood, No_Bikes, No_Empty_Docks, Timestamp
```

Does this meet the goal?

A. Yes

B. No

```
bike_location
| filter Neighbourhood == "Sands End" and No_Bikes >= 15
| sort by No_Bikes asc
| project BikepointID, Street, Neighbourhood, No_Bikes, No_Empty_Docks, Timestamp
```

Does this meet the goal?

A. Yes

B. No

Answer: *A*

112) You have a Fabric workspace that contains an eventstream named Eventstream1. Eventstream1 processes data from a thermal sensor by using event stream processing, and then stores the data in a lakehouse.
You need to modify Eventstream1 to include the standard deviation of the temperature.
Which transform operator should you include in the Eventstream1 logic?

A. Expand

B. Group by

C. Union

D. Aggregate

112) You have a Fabric workspace that contains an eventstream named Eventstream1. Eventstream1 processes data from a thermal sensor by using event stream processing, and then stores the data in a lakehouse.
You need to modify Eventstream1 to include the standard deviation of the temperature.
Which transform operator should you include in the Eventstream1 logic?

A. Expand

B. Group by

C. Union

D. Aggregate

Answer: *B*

113) You have a Fabric tenant that contains a warehouse named WH1.

You run the following T-SQL query against WH1.

```
SELECT e.[WWI Employee ID],
       e.Employee,
       e.[Preferred Name],
       gdr.[WWI Employee ID] AS [Direct Report ID],
       gdr.Employee AS [Direct Report]
FROM Dimension.Employee AS e
OUTER APPLY Dimension.GetDirectReports(e.[Employee Key]) AS gdr;
```

For each of the following statements, select Yes if the statement is true. Otherwise, select No.

NOTE: Each correct selection is worth one point.

Answer Area

Statements	Yes	No
`Dimension.GetDirectReports` is a scalar T-SQL function.	○	○
The `Dimension.GetDirectReports` function will run only once when the query runs.	○	○
The output rows will include at least one row for each row in the Dimension.Employee table.	○	○

Answer Area

Statements	Yes	No
`Dimension.GetDirectReports` is a scalar T-SQL function.	○	○
The `Dimension.GetDirectReports` function will run only once when the query runs.	○	○
The output rows will include at least one row for each row in the Dimension.Employee table.	○	○

114) You have a Fabric tenant named Tenant1 that contains a lakehouse named Lakehouse1.

You need to add data to Lakehouse1 from a CSV file in an Azure Storage account outside of Fabric. The solution must minimize development effort.

What should you use to add the data?

A. copy job

B. shortcut

C. pipeline

D. Dataflow Gen2

114) You have a Fabric tenant named Tenant1 that contains a lakehouse named Lakehouse1.

You need to add data to Lakehouse1 from a CSV file in an Azure Storage account outside of Fabric. The solution must minimize development effort.

What should you use to add the data?

A. copy job

B. shortcut

C. pipeline

D. Dataflow Gen2

Answer: *B*

115) You have a Fabric warehouse named DW1 that loads data by using a data pipeline named Pipeline1. Pipeline1 uses a Copy data activity with a dynamic SQL source. Pipeline1 is scheduled to run every 15 minutes.
You discover that Pipeline1 keeps failing.
You need to identify which SQL query was executed when the pipeline failed.
What should you do?

A. From Monitoring hub, select the latest failed run of Pipeline1, and then view the output JSON.

B. From Monitoring hub, select the latest failed run of Pipeline1, and then view the input JSON.

C. From Real-time hub, select Fabric events, and then review the details of Microsoft.Fabric.ItemReadFailed.

D. From Real-time hub, select Fabric events, and then review the details of Microsoft. Fabric.ItemUpdateFailed.

115) You have a Fabric warehouse named DW1 that loads data by using a data pipeline named Pipeline1. Pipeline1 uses a Copy data activity with a dynamic SQL source. Pipeline1 is scheduled to run every 15 minutes.
You discover that Pipeline1 keeps failing.
You need to identify which SQL query was executed when the pipeline failed.
What should you do?

A. From Monitoring hub, select the latest failed run of Pipeline1, and then view the output JSON.

B. From Monitoring hub, select the latest failed run of Pipeline1, and then view the input JSON.

C. From Real-time hub, select Fabric events, and then review the details of Microsoft.Fabric.ItemReadFailed.

D. From Real-time hub, select Fabric events, and then review the details of Microsoft. Fabric.ItemUpdateFailed.

Answer: *B*

116) You have a Fabric workspace that contains a lakehouse named Lakehouse1.
In an external data source, you have data files that are 500 GB each. A new file is added every day.
You need to ingest the data into Lakehouse1 without applying any transformations. The solution must meet the following requirements
Trigger the process when a new file is added.
Provide the highest throughput.
Which type of item should you use to ingest the data?

A. Data pipeline

B. Environment

C. KQL queryset

D. Dataflow Gen2

116) You have a Fabric workspace that contains a lakehouse named Lakehouse1.
In an external data source, you have data files that are 500 GB each. A new file is added every day.
You need to ingest the data into Lakehouse1 without applying any transformations. The solution must meet the following requirements
Trigger the process when a new file is added.
Provide the highest throughput.
Which type of item should you use to ingest the data?

A. Data pipeline

B. Environment

C. KQL queryset

D. Dataflow Gen2

Answer: *A*

117) You have a Fabric workspace.
You are debugging a statement and discover the following issues:
Sometimes, the statement fails to return all the expected rows.
The PurchaseDate output column is NOT in the expected format of mmm dd, yy.
You need to resolve the issues. The solution must ensure that the data types of the results are retained. The results can contain blank cells.
How should you complete the statement? To answer, select the appropriate options in the answer area.

Answer Area

```
SELECT
  item_id as ItemId
  [dropdown] as ItemName
```

- ,convert(varchar(20), item_name)
- ,convert(varchar(max), item_name)
- ,try_cast(item_name as varchar(20))

```
  ,item_description as ItemDescription
  [dropdown] as PurchaseDate
```

- ,convert(varchar, purchase_date, 7)
- ,convert(varchar, purchase_date, 109)
- ,convert(varchar, purchase_date, 112)

```
FROM
  Table1
WHERE
  item_type = @itemtype_parameter
```

```
SELECT
  item_id as ItemId
  [dropdown] as ItemName
,item_description as ItemDescription
  [dropdown] as PurchaseDate
FROM
  Table1
WHERE
  item_type = @itemtype_parameter
```

Dropdown options (ItemName):

- ,convert(varchar(20), item_name)
- ,convert(varchar(max), item_name)
- ,try_cast(item_name as varchar(20))

Dropdown options (PurchaseDate):

- ,convert(varchar, purchase_date, 7)
- ,convert(varchar, purchase_date, 109)
- ,convert(varchar, purchase_date, 112)

118) You have a Fabric workspace that contains a semantic model named Modell. You need to monitor the refresh history of Model 1 and visualize the refresh history in a chart. What should you use?

A. a data pipeline
B. a notebook
C. a Dataflow Gen2 dataflow
D. the refresh history from the settings of Model1.

118) You have a Fabric workspace that contains a semantic model named Modell. You need to monitor the refresh history of Model 1 and visualize the refresh history in a chart. What should you use?

A. a data pipeline
B. a notebook
C. a Dataflow Gen2 dataflow
D. the refresh history from the settings of Model1.

Correct Answer: B

119) You are processing streaming data from an external data provider.
You have the following code segment.

```
datatable (Location:string, Company:string, UnitsSold:long)
[
  "New York", "Contoso", 300,
  "New York", "Litware", 1000,
  "New York", "Relecloud", 300,
  "New York", "Fabrikam", 200,
  "Seattle", "Contoso", 300,
  "Seattle", "Litware", 100,
  "Seattle", "Fabrikam", 100,
  "San Francisco", "Relecloud", 500,
  "San Francisco", "Litware", 500,
  "Washington DC", "Litware", 300,
  "Washington DC", "Contoso", 400
]
| sort by Location desc, UnitsSold desc
| extend Rank=row_rank_dense(UnitsSold, prev(Location) != Location)
```

For each of the following statements, select Yes if the statement is true. Otherwise, select No.
NOTE: Each correct selection is worth one point.

Answer Area

Statements	Yes	No
Litware from New York will be displayed at the top of the result set.	○	○
Fabrikam in Seattle will have value = 2 in the Rank column.	○	○
Litware in San Francisco will have the same value in the Rank column as Litware in New York.	○	○

For each of the following statements, select Yes if the statement is true. Otherwise, select No.
NOTE: Each correct selection is worth one point.

Statements	Yes	No
Litware from New York will be displayed at the top of the result set.	☒	○
Fabrikam in Seattle will have value = 2 in the Rank column.	○	☒
Litware in San Francisco will have the same value in the Rank column as Litware in New York.	○	☒

120) You have a Fabric workspace that contains an eventstream named EventStreaml. EventStreaml outputs events to a table named Tablel in a lakehouse. The streaming data is souiced from motorway sensors and represents the speed of cars.
You need to add a transformation to EventStream1 to average the car speeds. The speeds must be grouped by non-overlapping and contiguous time intervals of one minute. Each event must belong to exactly one window.
Which windowing function should you use?

A. sliding
B. session
C. tumbling
D. hopping

120) You have a Fabric workspace that contains an eventstream named EventStreaml. EventStreaml outputs events to a table named Tablel in a lakehouse. The streaming data is souiced from motorway sensors and represents the speed of cars.
You need to add a transformation to EventStream1 to average the car speeds. The speeds must be grouped by non-overlapping and contiguous time intervals of one minute. Each event must belong to exactly one window.
Which windowing function should you use?

A. sliding
B. session
C. tumbling
D. hopping

Correct Answer: C

121) What should you do to optimize the query experience for the business users?

A. Introduce primary keys.
B. Create and update statistics.
C. Run the VACUUM command.
D. Enable V-Order.

121) What should you do to optimize the query experience for the business users?

A. Introduce primary keys.
B. Create and update statistics.
C. Run the VACUUM command.
D. Enable V-Order.

Correct Answer: B

122) You have an Azure subscription that contains a blob storage account named sa1. Sa1 contains two files named Filelxsv and File2.csv.
You have a Fabric tenant that contains the items shown in the following table.

Name	Description
Workspace1	Fabric workspace
Pipeline1	Data pipeline in Workspace1
flh1	Lakehouse in Workspace1

You need to configure Pipeline1 to perform the following actions:
* At 2 PM each day, process Filel.csv and load the file into flhl.
* At 5 PM each day. process File2.csv and load the file into flhl.
The solution must minimize development effort. What should you use?

A. a data pipeline schedule
B. a job definition
C. a data pipeline trigger
D. an activator

122) You have an Azure subscription that contains a blob storage account named sa1. Sa1 contains two files named Filelxsv and File2.csv.
You have a Fabric tenant that contains the items shown in the following table.

Name	Description
Workspace1	Fabric workspace
Pipeline1	Data pipeline in Workspace1
flh1	Lakehouse in Workspace1

You need to configure Pipeline1 to perform the following actions:
* At 2 PM each day, process Filel.csv and load the file into flhl.
* At 5 PM each day. process File2.csv and load the file into flhl.
The solution must minimize development effort. What should you use?

A. a data pipeline schedule
B. a job definition
C. a data pipeline trigger
D. an activator

Correct Answer: A

123) You have a Fabric workspace that contains a lakehouse named Lakehouse1.

You plan to create a data pipeline named Pipeline1 to ingest data into Lakehouse1. You will use a parameter named param1 to pass an external value into Pipeline1. The param1 parameter has a data type of int.

You need to ensure that the pipeline expression returns param1 as an int value.

How should you specify the parameter value?

A. "@pipeline().parameters.param1"

B. "@{pipeline().parameters.param1}"

C. "@{pipeline().parameters.[param1]}"

D. "@@{pipeline().parameters.param1}"

123) You have a Fabric workspace that contains a lakehouse named Lakehouse1.

You plan to create a data pipeline named Pipeline1 to ingest data into Lakehouse1. You will use a parameter named param1 to pass an external value into Pipeline1. The param1 parameter has a data type of int.

You need to ensure that the pipeline expression returns param1 as an int value.

How should you specify the parameter value?

A. "@pipeline().parameters.param1"

B. "@{pipeline().parameters.param1}"

C. "@{pipeline().parameters.[param1]}"

D. "@@{pipeline().parameters.param1}"

Correct Answer: A

124) You have the development groups shown in the following table.

Name	Skillset
Group1	Azure Stream Analytics with Azure Event Hubs
Group2	Microsoft Power BI
Group3	R language

You have the projects shown in the following table.

Name	Development group	Requirement
Project1	Group1	Move data between source and sink environments.
Project2	Group2	Transform data by using by a low-code approach.
Project3	Group3	Transform data.

You need to recommend which Fabric item to use based on each development group's skillset The solution must meet the project requirements and minimize development effort What should you recommend for each group? To answer, select the appropriate options in the answer area.
NOTE: Each correct selection is worth one point.

Answer Area

Group1:

- A data pipeline
- A Dataflow Gen2
- An eventstream
- A notebook

Group2:

- A data pipeline
- A Dataflow Gen2
- An eventstream
- A notebook

Group3:

- A data pipeline
- A Dataflow Gen2
- An eventstream
- A notebook

Answer Area

Group1:
- A data pipeline
- A Dataflow Gen2
- **An eventstream**
- A notebook

Group2:
- **A data pipeline**
- A Dataflow Gen2
- An eventstream
- A notebook

Group3:
- A data pipeline
- A Dataflow Gen2
- An eventstream
- **A notebook**

An eventstream

A data pipeline

A notebook